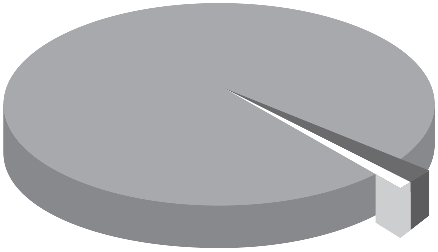
Chapter 2
WATER

2.1 Introduction

Water is the most precious, wonderful and useful gift of nature. It is the most abundant and essential natural resource. It covers nearly 70% of the earth's surface. It is estimated that the hydrosphere contains about 1360 million cubic kilometer (1.3×10^{18} m^3) of water. Of this, about 97% is in the oceans and inland seas, which is not suitable for human consumption because of its high salt content. Of the remaining 3%, 2% is locked in the glaciers and polar ice caps and only 1% is available as fresh water in rivers, lakes, streams, reservoirs and ground water, which is suitable for human consumption.

Water on Earth



Source *www.aid-n.com/earths-water-resources-in-the world/earths water resources-water on earth published Nov 8, 2012.*

2.2 Sources of Water

The chief sources of water are

1. Surface water
2. Ground water
3. Rain water

(1) **Surface water sources** These include lakes, ponds, reservoirs, streams, rivers, seas and oceans.

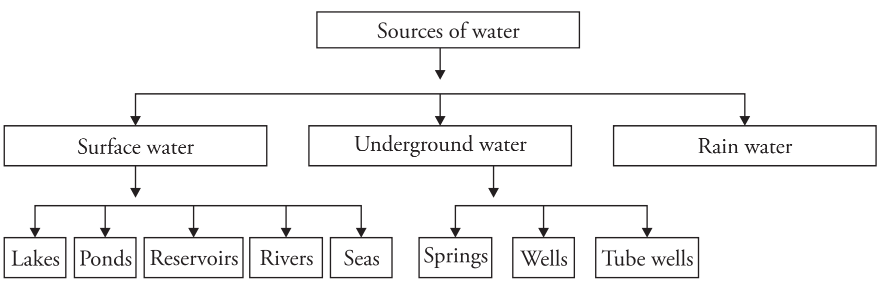
River water It contains dissolved minerals of the soils such as chlorides, sulphates and bicarbonates of sodium, calcium, magnesium and iron. It also contains organic matter derived from dead and decayed plants and animals; besides this, it contains sand and soil in suspension.

Water of lakes, ponds and reservoirs They contain less of dissolved minerals but are rich in organic matter.

Sea water It is the most impure form of natural water. It contains on an average 3.5% of dissolved salts, of which about 2.6% is sodium chloride. Other salts present are sulphates of sodium; bicarbonates of potassium, magnesium and calcium; bromides of potassium and magnesium and a number of other compounds.

(2) **Underground water sources** This is the water accumulated under the ground through seepage. It is obtained from wells, tube wells, springs, etc. Underground water is relatively free from suspended impurities because it is filtered as the water moves down through different layers of soil. The filtration also removes biological contamination. However, underground water is rich in dissolved salts.

(3) **Rain water** It is considered to be the purest form of water. However, it dissolves large amount of gases and suspended solid particles from the atmosphere. Rain water is divided between the two sources, a part of it seeps down to underground water tables and a part of it goes to the surface sources like rivers, ponds, lakes and reservoirs.



2.3 Effect of Water on Rocks and Minerals

Water brings about the weathering of rocks by various chemical processes like

(i) **Dissolution** Certain minerals and salts dissolve in water creating holes in the rock. This leads to the weakening of the rock that may ultimately lead to its collapse. Minerals like halite dissolve directly in water.

(ii) **Hydration** Under humid conditions and in the presence of moisture, the soil-forming minerals in the rocks swell leading to the increase in their volume. This further leads to the disintegration of the rocks in which these minerals are present. Some common examples are as follows:

(a) $\underset{\text{Anhydrite}}{CaSO_4 + 2\,H_2O} \xrightarrow{\text{Hydration}} \underset{\text{Gypsum}}{CaSO_4.\,2H_2O}$

(b) $\underset{\text{Bauxite}}{Al_2O_3 + 3H_2O} \xrightarrow{\text{Hydration}} \underset{\text{hydrated aluminium oxide}}{Al_2O_3.\,3H_2O}$

(c) $\underset{\text{Haematite (Red)}}{2Fe_2O_3 + 3H_2O} \xrightarrow{\text{Hydration}} \underset{\text{Limonite (yellow)}}{2Fe_2O_3.\,3H_2O}$

(d) $\underset{\text{Olivine}}{Mg_2SiO_4} \xrightarrow{\text{Hydration}} \underset{\text{serpentine}}{Mg_2SiO_4.\,x\,H_2O}$

(iii) **Hydrolysis** Water dissociates into H^+ and OH^- ions that combine with minerals to form new compounds. The silicates combine with water, forming clays.

$$\underset{\text{(Orthoclase)}}{KAlSi_3O_8} + H_2O \rightarrow \underset{\text{(Acid silt clay)}}{HAlSi_3O_8} + KOH$$

The acid silt clay further recombines as

$$2HAlSi_3O_8 + 8H_2O \rightarrow \underset{\text{(Hydrated aluminium oxide)}}{Al_2O_3.\,3H_2O} + \underset{\text{(silicic acid)}}{6H_2SiO_3}$$

(vi) **Carbonation** Atmospheric carbondioxide dissolves in water to form carbonic acid.

$$H_2O + CO_2 \rightarrow H_2CO_3$$

This carbonic acid reacts with the rocks and minerals and dissolves them.

$$\underset{\text{Calcite}}{CaCO_3} + H_2O + CO_2 \rightarrow \underset{\text{Calcium bicarbonate (soluble)}}{Ca\,(HCO_3)_2}$$

The removal of $CaCO_3$ that holds the sand particles together leads to disintegration of rocks.

(v) **Oxidation** The oxygen present in soil water and atmosphere leads to oxidation and hydration

$$4FeO \text{ (ferrous oxide)} + O_2 \rightarrow 2Fe_2O_3 \text{ (ferric oxide)}$$

$$4Fe_3O_4 \text{ (magnetite)} + O_2 \xrightarrow{\text{oxidation}} 6Fe_2O_3 \text{ (haematite)}$$

$$2Fe_2O_3 \text{ (haematite)} + 3H_2O \xrightarrow{\text{Hydration}} 2Fe_2O_3.\ 3H_2O \text{ (limonite)}$$

The affected rock becomes reddish brown because of oxidation. The rock weakens and crumbles easily.

2.4 Common Impurities of Water

Water is a very good solvent; therefore, it dissolves a large number of substances in it. The common impurities present in water are as follows

A. ***Dissolved impurities***

- These may be dissolved salts like carbonates, bicarbonates, nitrates, sulphates and chlorides of calcium, magnesium, sodium, potassium, iron, manganese, aluminium, etc.
- Dissolved gases like O_2, CO_2, SO_2, NH_3, N_2 and oxides of nitrogen also fall in this category.

B. ***Colloidal impurities*** These are those impurities whose particle size varies from 1 to 100 nm. It consists of finely divided silica, clay, aluminium hydroxide, ferric hydroxide, organic waste products, coloring matter, etc.

C. ***Suspended impurities*** These are impurities with particle size greater than 100 nm and can be removed by filtration or settling. They may be organic, inorganic or microorganism.

- ***Organic* impurities** consist of animal and vegetable matter, wood pieces, leaves, oil globules, etc.
- ***Inorganic impurities*** include clay, silica, sand, etc.
- ***Microorganisms*** include various types of algae, protozoa, bacteria and fungi.

Suspended impurities give turbidity and color to water. Microorganisms are the main cause of water-born diseases.

Characteristics imparted by impurities in water

Water is a very good solvent and dissolves various impurities in it. These substances present in water affect the characteristics of water in various ways. They alter the physical, chemical and biological characteristics of water. Let us discuss them one by one.

Physical characteristics

- Presence of salts like iron, manganese, industrial effluents, dyes from industries and other salts impart color to water.
- Presence of sand and silt make the water turbid.
- Salts also alter the taste of water. Presence of salts of sodium and potassium makes the water salty. Aluminium, manganese and iron make the water bitter. Presence of sodium bicarbonate gives soapy taste to water.
- Organic matter, decaying vegetation, algae, fungi and sewage impart a characteristic foul odor to water.

Chemical characteristics

Wastes, including organic and inorganic chemicals like dyes, molasses, fertilisers, acids, alkalies from industries insecticides, pesticides, liquors, wastes from tanneries and gases like NH_3 and H_2S produced by anaerobic decomposition, alter the pH, alkalinity, acidity, hardness and chemical characteristics of water and make it unfit for domestic and industrial use.

Biological characteristics

The biological impurities like algae, fungi, pathogens and other microorganisms present in water either naturally or due to discharge of organic waste from domestic sewage, industrial houses or produced by the natural decay of organic matter all adversely affect the biological characteristics of water and make it unfit for domestic use, adversely affecting human health, aquatic flora and fauna.

Table 2.1 Common impurities in water and their effects

Type	Constituents	Effect
1. Suspended impurities	(a) Bacteria	Some causes diseases
	(b) Algae, protozoa	Odor, color, turbidity, diseases
	(c) Silts	Turbidity
2. Dissolved Impurities	(d) Salts	Alkalinity
	(i) Ca and Mg – bicarbonate	Alkalinity, hardness
	– Carbonates	Hardness
	– Sulphate	Hardness
	– Chloride	Hardness
	(ii) Na–bicarbonate	Alkalinity
	– Carbonate	Alkalinity
	– Sulphate	Foaming
	– Fluoride	Dental problems
	– Chloride	Affects taste
	(b) Metals and compounds	
	(i) Oxide of iron	Taste, color, corrosion, hardness
	(ii) Mn	Black and brown color
	(iii) Pb	Poisoning
	(iv) As	Poisoning
	(v) Ba	Effect on heart and nerves

	(vi) Cd	Illness
	(vii) CN	Fatal
	(viii) B	Nervous system
	(ix) Se	Highly toxic
	(x) Ag	Discoloration of skin, eyes
	(xi) Nitrates	Poisoning, color, blue baby syndrome
	(c) Vegetable dyes	Acidity
	(d) Gases–Oxygen	Corrodes metals
	–Hydrogen sulphide	Odor, acidity, corrosiveness
	–Carbon dioxide	acidity, corrosion

2.5 Water Quality Standards

Importance of water quality standards

Water is widely used both in domestic and industrial life. Natural water, no matter how pure, contains some or the other impurities. Water quality standards are important because they help to identify the water quality, problems caused by improper treatment of waste water discharge, run off, addition of fertilisers, chemicals from agricultural areas and so on. The parameters for water quality are decided according to its use. Work in the area of water quality tends to be focused on whether water is treated for human consumption or environmental purpose. It is important to specify water quality standards for the purpose of safety of human beings, drinking characteristics and for the health of ecosystems.

Environmental water quality also called ambient water quality specifies the quality of water in water bodies like lakes, rivers and oceans. Ambient water quality standards vary significantly because of different environmental conditions, ecosystem and intended human uses.

The water quality parameters or characteristics for which analysis is carried out generally fall into three groups:

1. Physical characteristics
2. Chemical characteristics
3. Biological characteristics

(1) **Physical characteristics**

These are characteristics that respond to touch, taste, sight, etc. These include turbidity, temperature, odor, color and taste.

(a) ***Color*** Clean water should be colorless. The presence of color in water indicates the presence of various minerals, decomposed organic matter like leaves, roots, organic and inorganic wastes, wastes from textile mill, paper pulp industries, food-processing industries, domestic wastes, wastes from laundry, dyeing, dairy products, etc.

(b) ***Taste and odor*** Water should be odorless and should have a fairly good taste. There are no specific units to measure these parameters, but generally decaying organic matter imparts odor and bad taste to water. The minimum odor that can be detected is called threshold odor number (TON). The value of TON is determined as follows:

$$TON = \frac{A+B}{A}$$

A = Volume of sample in mL; B = Volume of distilled water (i.e., odor-free water in mL).

(c) ***Temperature*** It is an important water quality parameter. High temperature indicates thermal pollution and disturbs aquatic ecosystem by reducing the dissolved oxygen in water.

(d) ***Electrical conductivity*** It gives an idea about the dissolved solids in water. Greater the amount of dissolved solids, higher will be the conductivity. It can be measured easily with the help of conductivity meter. The average value of conductivity for potable water should be less than 2 μmho/cm (μS/cm).

(2) Chemical characteristics

Some important chemical characteristics are pH, hardness, alkalinity, total dissolved solids, chlorides, fluorides, sulphates, phosphates, nitrates, metal, etc.

(a) ***pH*** The pH of normal drinking water is 6.5–8.5. It can be measured with the help of pH meter using a combined electrode (a glass electrode and a calomel electrode as reference electrode). pH scale ranges from 0 to 14. pH 7 indicates neutral solution, less than 7 is acidic, whereas greater than 7 signifies alkaline or basic water.

(b) ***Hardness*** It expresses the concentration of calcium and magnesium ions in water in terms of equivalent of $CaCO_3$. The maximum acceptance limit is 500 ppm.

(c) ***Total dissolved solids (TDS)*** The maximum permissible limit is 500 mg/L. TDS includes both organic and inorganic dissolved impurities. It can be measured by evaporating a sample to dryness and then weighing the residue.

(d) ***Total solids*** This includes both the dissolved solids as well as suspended impurities.

(e) ***Dissolved oxygen*** It is an important water quality parameter. Higher the amount of DO better is the quality of water. Normal water contains 4.7 mg/L of DO. Lesser amount of DO in water indicates pollution in water. Wrinkler or iodometric methods using membrane electrode is used for measuring DO in water.

(f) ***Chlorides*** Its amount in water should be less than 250 ppm. High percentage of chloride in water harms metallic pipes as well as agriculture crops.

(g) **Fluorides** Maximum permissible limit is 1.5 ppm. The amount of fluoride in water sample can be determined using an ion meter. Excess of fluoride causes discoloration of teeth, bone fluorosis and skeletal abnormalities.

(h) ***Sulphates*** Permissible limit is 250 ppm. These are generally found associated with calcium, magnesium and sodium ions. It leads to scale formation in boilers, causes boiler corrosion and imparts odor to water.

(i) ***Nitrates*** Its concentration in drinking water should not exceed 45 mg/L. Excessive nitrates in drinking water causes 'methemoglobinemia' or blue baby syndrome in infants. Nitrates dissolve in water because of leaching of fertilisers from soil and nitrification of organic matter.

(3) **Biological characteristics**

Water should be free from all types of bacteria, viruses, protozoa and algae. The coliform count in any sample of 100 mL should be zero.

(4) **Bacteriological standards**

(i) ***Water entering the distribution system*** Coliform count in any sample of 100 mL should be zero. A sample of water entering the distribution system that does not confirm to this standard calls for an immediate investigation into both the efficacy of the purification process and the method of sampling.

(ii) ***Water in the distribution system*** It shall satisfy these three criteria:

(a) *E. coli* count in 100 mL of any sample should be zero.

(b) Coliform organism should not be more than 10 per 100 mL of any sample.

(c) Coliform organism should not be present in 100 mL of any two consecutive samples or more than 5% of the samples collected for the year.

(5) ***Virological standards*** 0.5 mg/L of free residual chlorine for 1 h is sufficient to inactivate virus, even in water that was originally polluted. This free residual chlorine should be present in all disinfected supplies in area suspected of infectious hepatitis to inactivate virus and also bacteria. For water supply in such areas, 0.2 mg/L of free residual chlorine for half an hour should be insisted.

The Indian standard for drinking water prescribed by the Bureau of Indian Standards and Indian Council of Medical Research are shown in Table 2.2.

Table 2.2 BIS 105000: Indian standard of drinking water specification

S.No	Substance or characteristic	Acceptable limit	Cause of rejection
	Essential characteristics		
1	Color (units on platinum – cobalt scale)	5	25
2	Odor	Unobjectionable	Unobjectionable
3	Taste	Agreeable	Agreeable
4	Turbidity (JTU, Max)	5	10
5	pH value	7.0–8.5	6.5–9.2
6	Total hardness (as $CaCO_3$) mg/L	200	600
7	Iron (as Fe in mg/L)	0.1	1.0
8	Chlorides (as Cl in mg/L)	200	1000
9	Residual, free chloride, mg/L	0.2	-
	Desirable characteristics		
10	Total dissolved solids mg/L	500	1500
11	Calcium (as Ca), mg/L	75	200
12	Copper (as Cu), mg/L	0.05	1.5
13	Manganese (as Mn), mg/L	0.05	0.5
14	Sulphate (as SO_4), mg/L	200	400
15	Nitrate (as NO_3), mg/L	45	45

16	Fluoride (as F), mg/L	1.0	1.5
17	Phenolic compounds (as C_6H_5OH), mg/L	0.001	0.002
18	Mercury (as Hg), in mg/L	0.001	No relaxation
19	Cadmium (as Cd), in mg/L	0.01	No relaxation
20	Selenium (as Se), in mg/L	0.01	No relaxation
21	Arsenic (as As), in mg/L	0.05	No relaxation
22	Cyanide (as CN), in mg/L	0.05	No relaxation
23	Lead (as Pb), in mg/L	0.1	No relaxation
24	Zinc (as Zn), in mg/L	5	15
25	Anionic detergents (as MBAS), in mg/L	0.2	1.0
26	Chromium (as Cr^{6+}), mg/L	0.05	No relaxation
27	Polynuclear aromatic hydrocarbons (as PAH), mg/L	0.2	0.2
28	Mineral oil, mg/L	0.01	0.3
29	Pesticides, mg/L	Absent	0.001
30	Radioactive materials (i) Gross alpha emitters pCi/L (ii) Gross beta emitters pCi/L	 3 30	 3 30
31	Alkalinity, mg/L	200	600
32	Aluminium (as Al), mg/L	0.03	0.2
33	Boron, mg/L	1	1

Note: The values indicated under the column 'acceptable' are the limits upto which water is generally acceptable to consumers. Values in excess of those mentioned under 'acceptable' render the water not acceptable but still may be tolerated in the absence of alternative and better source upto the limits indicated under column 'cause of rejection' above which the supply will have to be rejected.

2.6 Hardness of Water

Definition It is defined as the characteristic of water that prevents lathering of soap. Originally, it was defined as the soap-consuming capacity of water.

Causes Hardness is due to the presence of certain soluble salts of calcium and magnesium in water. A sample of hard water, when treated with soap (sodium or potassium salts of higher fatty acids like oleic, palmitic or stearic acid), does not form lather or foam but forms a white precipitate or scum instead. This is because of the formation of insoluble salts of calcium and magnesium.

Chemical reactions involved are

$$\underset{\text{Soap (Sodium stearate)}}{2C_{17}H_{35}COONa} + \underset{\text{(from hard water)}}{CaCl_2} \rightarrow \underset{\text{Calcium stearate (Insoluble)}}{(C_{17}H_{35}COO)_2Ca \downarrow} + 2NaCl$$

$$\underset{\text{Soap}}{2C_{17}H_{35}COONa} + \underset{\text{(from hard water)}}{MgSO_4} \rightarrow \underset{\substack{\text{Magnesium stearate}\\ \text{(Insoluble)}}}{(C_{17}H_{35}COO)_2Mg \downarrow} + Na_2SO_4$$

Other salts of calcium and magnesium like $CaSO_4$, $Ca(NO_3)_2$ and $MgCl_2$ also react in a similar way forming insoluble precipitates of calcium and magnesium salts of higher fatty acids.

Table 2.3 Difference between hard and soft water

S.No	Hard water	Soft water
1	Water that does not form lather with soap but forms white precipitate	Water that produces lather or foam easily with soap is called 'soft water'
2	It contains soluble salts of calcium magnesium and other heavy metal ions like Al^{3+}, Fe^{3+}, and Mn^{2+} dissolved in it	It does not contain dissolved salts of calcium and magnesium
3	In hard water, the cleaning properties of soap is depressed and lot of soap is wasted in bathing and washing	The cleaning quality of soap is not depressed hence it is good for washing and cleaning
4	Owing to dissolved hardness, boiling point of water is elevated, therefore more fuel and time is required for cooking	Less fuel and time is required for cooking

Types of hardness

Hardness is of two types:

1. Temporary hardness 2. Permanent hardness

(1) **Temporary hardness**

(a) It is also called carbonate hardness or alkaline hardness. It is called alkaline hardness because it is due to the presence of bicarbonate, carbonate and hydroxide and can be determined by titration with HCl using methyl orange as an indicator.

(b) It can be removed by boiling of water. During boiling, the bicarbonates are decomposed forming insoluble carbonates or hydroxides that are deposited at the bottom of the vessel.

$$\underset{\text{Calcium bicarbonate}}{Ca(HCO_3)_2} \xrightarrow{\text{boil}} \underset{\text{Calcium carb-onate (Insoluble)}}{CaCO_3 \downarrow} + H_2O + CO_2 \uparrow$$

$$\underset{\text{Magnesium bicarbonate}}{Mg(HCO_3)_2} \xrightarrow{\text{boil}} \underset{\text{Magnesium hydroxide}}{Mg(OH)_2 \downarrow} + H_2O + 2CO_2 \uparrow$$

(2) **Permanent hardness**

(a) It is also called non-carbonate or non-alkaline hardness.

(b) It is due to the presence of dissolved chlorides, sulphates and nitrates of calcium, magnesium, iron and other heavy metals.

(c) Salts mainly responsible for permanent hardness are

$CaCl_2$, $CaSO_4$, $MgCl_2$, $MgSO_4$, $Ca\ (NO_3)_2$, etc.

(d) It cannot be removed by simple boiling but can be removed by special chemical methods like lime–soda process, zeolite method, etc.

Table 2.4 Comparison between temporary and permanent hardness

S.No	Temporary hardness	Permanent hardness
1	Also called carbonate hardness or alkaline hardness	Also called non-carbonate or non-alkaline hardness
2	It is due to the presence of bicarbonates, carbonates of calcium and magnesium	It is due to the presence of dissolved chlorides, sulphates and nitrates of calcium, magnesium, iron and other heavy metals
3	It can be removed by boiling $Ca(HCO_3)_2 \xrightarrow{\text{boil}} CaCO_3 \downarrow + H_2O + CO_2 \uparrow$ $Mg\ (HCO_3)_2 \xrightarrow{\text{boil}} Mg(OH)_2 \downarrow + H_2O + 2CO_2 \uparrow$	Cannot be removed by simple boiling but can be removed by specific chemical methods, like lime–soda process, zeolite method, etc.

Degree of hardness

Hardness is always calculated in terms of equivalent of $CaCO_3$, although hardness is never present in the form of $CaCO_3$. There are two basic reasons for choosing $CaCO_3$ as standard:

(i) Calculations become easy as its molecular weight is exactly 100 (and equivalent weight is exactly 50).

(ii) It is the most insoluble salt that can be precipitated in water treatment.

Calculation of equivalents of $CaCO_3$

To find out hardness in a given water sample, it is essential to convert hardness due to different salts ($CaCl_2$, $Ca(NO_3)_2$, $CaSO_4$, $MgCl_2$, $MgSO_4$, etc.) in terms of equivalent of $CaCO_3$. This can be done by the formula:

$$\boxed{\text{Equivalent of } CaCO_3 = \frac{\text{Mass of hardness producing substance} \times \text{Chemical equivalent of } CaCO_3}{\text{Chemical equivalent of hardness producing substance}}}$$

- Chemical equivalent of $CaCO_3$ is 50
- Chemical equivalent of salt $= \dfrac{\text{Molecular weight}}{\text{Valency}}$
- Chemical equivalent of acid $= \dfrac{\text{Molecular weight}}{\text{Basicity}}$

Basicity is the number of replaceable hydrogen ions in an acid.

- Chemical equivalent of base = $\dfrac{\text{Molecular weight}}{\text{Acidity}}$

Acidity is the number of replaceable hydroxyl ions in a base.

The ratio = $\dfrac{\text{Chemical equivalent of } CaCO_3}{\text{Chemical equivalent of hardness producing substance}}$, is constant for a particular compound and is called multiplication factor.

Hence, equivalent of $CaCO_3$ = mass of hardness producing substance × multiplication factor.

The formula and multiplication factor for different salts are tabulated in Table 2.5.

Table 2.5 Formula and multiplication factor for different salts

Dissolved salt/ ion	Molar mass	Chemical equivalent	Multiplication factor = $\dfrac{\text{Eq Wt of } CaCO_3}{\text{Eq Wt of hardness producing substance}}$
$Ca(HCO_3)_2$	162	162/2 = 81	$\dfrac{100/2}{162/2} = \dfrac{100}{162}$
$Mg(HCO_3)_2$	146	146/2 = 73	$\dfrac{100/2}{146/2} = \dfrac{100}{146}$
$CaSO_4$	136	136/2 = 68	$\dfrac{100/2}{136/2} = \dfrac{100}{136}$
$CaCl_2$	111	111/2 = 55.5	$\dfrac{100/2}{111/2} = \dfrac{100}{111}$
$MgSO_4$	120	120/2 = 60	$\dfrac{100/2}{120/2} = \dfrac{100}{120}$
$MgCl_2$	95	95/2 = 47.5	$\dfrac{100/2}{95/2} = \dfrac{100}{95}$
$CaCO_3$	100	100/2 = 50	$\dfrac{100/2}{100/2} = \dfrac{100}{100}$
$MgCO_3$	84	84/2 = 42	$\dfrac{100/2}{84/2} = \dfrac{100}{84}$
$Mg(NO_3)_2$	148	148/2 = 74	$\dfrac{100/2}{148/2} = \dfrac{100}{148}$
Ca^{++}	40	40/2 = 20	$\dfrac{100/2}{40/2} = \dfrac{100}{40}$
Mg^{++}	24	24/2 = 12	$\dfrac{100/2}{24/2} = \dfrac{100}{24}$

HCO_3^-	61	61/1 = 61	$\frac{100/2}{61/1} = \frac{100}{2\times61} = \frac{100}{122}$
CO_3^{2-}	60	60/2 = 30	$\frac{100/2}{60/2} = \frac{100}{60}$
OH^-	17	17/1 = 17	$\frac{100/2}{17/1} = \frac{100}{2\times17} = \frac{100}{34}$
H^+	1	1	$\frac{100/2}{1/1} = \frac{100}{2}$
CO_2	44	44/2 = 22	$\frac{100/2}{44/2} = \frac{100}{44}$
HCl	36.5	36.5/1 = 36.5	$\frac{100/2}{1\times36.5} = \frac{100}{2\times36.5} = \frac{100}{73}$
H_2SO_4	98	98/2 = 49	$\frac{100/2}{98/2} = \frac{100}{98}$
$FeSO_4.\ 7H_2O$	278	278/2 = 139	$\frac{100/2}{278/2} = \frac{100}{278}$
$Al_2(SO_4)_3$	342	342/6 = 57	$\frac{100/2}{342/6} = \frac{100\times3}{342} = \frac{50}{57}$
$NaAlO_2$	82	82/1 = 82	$\frac{100/2}{82/1} = \frac{100}{82\times2} = \frac{100}{164}$

Solved Examples

1. A water sample contains 248 mg $CaSO_4$ per liter. Calculate the hardness in terms of $CaCO_3$ equivalent.

Solution

Weight of $CaSO_4$ per liter = 248 mg

Hardness in terms of $CaCO_3$ equivalent = ?

$$\text{Equivalents of } CaCO_3 = \frac{\text{Mass of hardness producing substance} \times \text{Chemical equivalent of } CaCO_3}{\text{Chemical equivalent of hardness producing substance}}$$

$$\frac{248\times100/2}{136/2} = 182.35 \text{ ppm.}$$

2. How many grams of $FeSO_4$ dissolved per liter gives 300 ppm of hardness?

(Fe = 56, S = 32, O = 16, Ca = 40,C =12)

Solution

1 mole of $FeSO_4$ causes hardness equivalent to that caused by 1 mole of $CaCO_3$ because 1 mole of any two substances contain equal number of molecules, that is, Avogadro's number of molecules.

$\therefore$ 1 mole $FeSO_4$ = 1 mole of $CaCO_3$

or, 56 + 32 + 64 = 152 g of $FeSO_4$ = 40 +12 + 48 = 100 g of $CaCO_3$

$\therefore$ 100 ppm of $CaCO_3$ equivalent hardness is caused by 152 ppm of $FeSO_4$ or,

300 ppm of hardness will be caused by $\frac{152}{100} \times 300$ = 456 ppm of $FeSO_4$ = 456 mg/L = 0.456 g/L of $FeSO_4$

[Ans = 0.456 g/L]

Alternative method

Calculation by direct application of formula

$$\text{Equivalents of } CaCO_3 = \frac{\text{Mass of hardness producing substance} \times \text{Chemical equivalent of } CaCO_3}{\text{Chemical equivalent of hardness producing substance}}$$

Let mass of hardness producing substance, that is, $FeSO_4$ = x mg

$$\text{Then } 300 = \frac{x \times 100/2}{152/2}$$

$$\text{or } 300 = \frac{x \times 100}{152} = 456 \text{ ppm} = 456 \text{ mg/L} = 0.456 \text{ g/L}$$

Units of hardness and their interrelationship

(i) ***Parts per million* (ppm)** It is defined as the number of parts by weight of $CaCO_3$ equivalent hardness present in million (10^6) parts by weight of water.

1 ppm = 1 part of $CaCO_3$ equivalent hardness in 10^6 parts of water.

(ii) ***Milligrams per liter* (mg/L)** It is defined as the number of milligrams of $CaCO_3$ equivalent hardness present per liter of water.

1mg/L = 1mg of $CaCO_3$ equivalent hardness in one liter of water

Since weight of 1 liter of water = 1 kg

$$= 1000 \text{ g} = 1000 \times 1000 = 10^6 \text{ mg}$$

1mg/L = 1mg of $CaCO_3$ per 10^6 mg of water.

= 1 part of $CaCO_3$ per 10^6 parts of water.

= 1 ppm.

Thus mathematically, 1ppm = 1mg/L.

(iii) ***Degree Clarke (° Cl)*** It is defined as the number of grains (1/7000 lb) of $CaCO_3$ equivalent hardness per gallon (10 lb) of water.

1° Clarke = 1 grain of $CaCO_3$ equivalent hardness per gallon of water

∵ 1 grain = 1/7000 lb (pounds)

And 1 gallon = 10 lb

1 grain/gallon = 1/7000 lb/10 lb

= 1: 70,000

1° Cl can also be defined as the number of parts of $CaCO_3$ equivalent hardness per 70,000 parts of water.

(iv) ***Degree French (°Fr)*** It is defined as the number of parts of $CaCO_3$ equivalent hardness per 10^5 parts of water. Thus,

1° Fr = 1 part of $CaCO_3$ equivalent hardness per 10^5 parts of water.

Relationship between different units of hardness As,

1 ppm = 1 part of $CaCO_3$ equivalent hardness in 10^6 parts of water.

1mg/L = 1 part of $CaCO_3$ equivalent hardness in 10^6 parts of water.

1° Cl = 1 part of $CaCO_3$ equivalent hardness in 70000 parts of water.

1° Fr = 1 part of $CaCO_3$ equivalent hardness in 10^5 parts of water.

Hence,

10^6 ppm = 10^6 mg/L = 70,000 °Cl = 10^5 °Fr
Dividing by 10^6
1 ppm = 1mg/L = 0.07 °Cl = 0.1 °Fr
Similarly,
1 °Cl = 1.433 °Fr = 14.3 ppm = 14.3 mg/L
1 °Fr = 10 ppm = 10 mg/L = 0.7 °Cl

Solved Examples

1. Three samples P, Q and R were analysed for their salt contents.
   (i) Sample P was found to contain 155 mg of magnesium carbonate per liter.
   (ii) Sample Q was found to contain 800 mg of calcium nitrate and 2.5 mg of silica and 5.1 mg of sodium chloride per liter.
   (iii) Sample R was found to contain 15 g of potassium nitrate and 3 g of calcium carbonate per liter. Find the hardness of all the above three samples in ppm and in grains per gallon.

Solution

Sample	Constituent	Amount mg/L	Equivalents of $CaCO_3$
P	$MgCO_3$	155 mg/L	$\frac{155\times100/2}{84/2} = 184.52$ mg/L
Q	$Ca(NO_3)_2$	800 mg/L	$\frac{800\times100/2}{164/2} = 487.80$ mg/L
R	$CaCO_3$	3 g/L or 3000 mg/L	$\frac{3000\times100/2}{100/2} = 3000$ mg/L

$\therefore$ Hardness of the three samples in ppm

Sample P = 184.52 ppm; Sample Q = 487.80 ppm; Sample R= 3000 ppm.

Hardness of the three samples in grains/gallon or degree Clarke

Since 1 ppm = 0.07 °Cl = 0.07 grains/gallon

Sample P = 184.52 ppm × 0.07 = 12.91 grains/gallon

Sample Q = 487.80 ppm × 0.07 = 34.146 grains/gallon

Sample R = 3000 ppm × 0.07 = 210 grains/gallon.

2. A sample of water on analysis was found to contain $Ca(HCO_3)_2$ = 4 mg/L; $Mg(HCO_3)_2$ = 6 mg/L; $CaSO_4$ = 8 mg/L; $MgSO_4$ = 10 mg/L. Calculate the temporary, permanent and total hardness of water in ppm, °Fr and °Cl [Mol wt of $Ca(HCO_3)_2$=162, $Mg(HCO_3)_2$= 146; $CaSO_4$=136; $MgSO_4$=120].

Solution

Conversion into $CaCO_3$ equivalent

Constituent	Amount	Equivalent of $CaCO_3$
$Ca(HCO_3)_2$	4 mg/L	$\frac{4\times100/2}{162/2} = 2.469$ mg/L
$Mg(HCO_3)_2$	6 mg/L	$\frac{6\times100/2}{146/2} = 4.110$ mg/L
$CaSO_4$	8 mg/L	$\frac{8\times100/2}{136/2} = 5.882$ mg/L
$MgSO_4$	10 mg/L	$\frac{10\times100/2}{120/2} = 8.333$ mg /L

Temporary hardness is due to Ca $(HCO_3)_2$ and Mg $(HCO_3)_2$.

Hence, temporary hardness = 2.469 + 4.110

= 6.579 mg/L or 6.579 ppm

Since 1mg/L = 1ppm = 0.07 °Cl = 0.1 °Fr

= 6.579 × 0.1 °Fr = 0.6579 °Fr

= 6.579 × 0.07 °Cl = 0.4605 °Cl

Permanent hardness is due to salts of $CaSO_4$ and $MgSO_4$, Hence

Permanent hardness = 5.882 + 8.333 = 14.215 mg/L = 14.215 ppm

= 14.215 × 0.1 = 1.42150 °Fr

= 14.215 × 0.07 = 0.995 °Cl

Total hardness = Temporary hardness + Permanent hardness

= 6.579 +14.215 = 20.794 mg/L = 20.794 ppm

= 20.794 × 0.1 = 2.0794 °Fr

= 20.794 × 0.07 = 1.4555 °Cl

Practice Problems

1. A water sample contains $Ca(HCO_3)_2$ = 32.4 mg/L, $Mg(HCO_3)_2$= 29.2 mg/L, $CaSO_4$= 13.5 mg/L. Calculate the temporary and permanent hardness of water.

   [Ans Temporary hardness = 20 + 20 = 40 ppm; permanent hardness = 9.926 ppm]

2. A sample of water on analysis gives the following data:

   $Ca(HCO_3)_2$ = 24.1°French; $CaSO_4$ = 3.2 °French; $MgCl_2$ = 9.5 °French

   SiO_2 = 2.1°French; $MgSO_4$ = 8.2 °French

   Calculate the temporary and permanent hardness on °Clarke and ppm scale.

   [Ans Temporary hardness = 14.88 °French = 10.41 °Clarke = 148.88 ppm;
   permanent hardness = 19.18 °Fr = 13.426 °Cl = 191.8 ppm]

3. A sample of water on analysis has been found to contain the following impurities

   $Mg(HCO_3)_2$ = 14.6 mg/L; $Mg(NO_3)_2$ = 44.4 mg/L; $MgSO_4$ = 36 mg/L;

   $MgCl_2$ = 19.0 mg/L and $CaCO_3$ = 30 mg/L.

   Calculate the temporary and permanent hardness in ppm and °Clarke.

   [Ans Temporary hardness = 40 ppm and 2.8 °Clarke;
   permanent hardness = 80 ppm and 5.6 °Clarke]

2.7 Disadvantages of Hard Water

(1) ***In Industries***

Large quantities of water is used for steam generation in industries like textile, paper, sugar, dyeing, pharmaceutical, etc.

(a) *Steam generation in boilers* The use of hard water in boilers causes problems like scale and sludge formation, boiler corrosion, caustic embrittlement, priming and foaming, etc (discussed in detail in Section 2.9).

(b) *Sugar industries* Salts dissolved in hard water cause problems in crystallisation of sugar.

(c) *Paper industries* The cations (Ca^{2+}, Mg^{2+}, Fe^{2+} etc) present in hard water react with the chemicals involved in different steps of paper manufacturing. This produces various unwanted side products and undesirable effects like loss of gloss and smoothness, change in color of paper, etc.

(d) *Textile industries* During washing of the fabric or yarn with hard water the calcium and magnesium salts stick to the surface causing undesirable changes in color or texture of fabric.

(e) *Dyeing industries* The salts dissolved in hard water react with the functional groups present in the dyes causing precipitation or change in shade.

(f) *Pharmaceutical industry* Use of hard water in the preparation of drugs can lead to the production of undesirable products that may be harmful or even poisonous.

(2) ***In domestic usage***

(a) *Drinking* Taste of soft water is better than hard water. The dissolved calcium in hard water can help to produce strong teeth and healthy bones in children. However, hard water can have bad affect on our appetite and digestive system; sometimes it produces calcium oxalate that causes different urinary problems.

(b) *Cooking* The boiling point of hard water increases because of the presence of various salts; this causes wastage of time and fuel.

(c) *Bathing and washing* As hard water does not form lather or foam with water, it adversely affects the cleaning properties of soap and a lot of it is wasted because of production of sticky precipitates of calcium and magnesium. The production of sticky precipitate continues till all the calcium and magnesium salts present in water are precipitated, after which the soap starts giving lather with water.

Moreover, the sticky precipitate adheres on the cloths giving stains, streaks and spots.

2.8 Water for Industries

Water is a prime requirement for any industry. In fact, industries are generally located in areas where water is abundantly available. Water is used in industries for washing, cooling, steam generation, as a heat-transfer medium, for dilution, as a solvent, for dust suppression, to reduce pollution, etc. The quality of water has a major bearing on the safety and durability of the machinery. It affects the life of the machinery and also the quality of the products formed. Each industry has its own specifications for water, depending upon the use to which the latter is kept. The specifications of water used in major industries is tabulated in Table 2.6.

Table 2.6 Specifications of water for major industrial use

S.No	Name of the industry/unit	Specifications of water to be used
1.	Boiler feed water	Should have zero hardness, it should be free from dissolved gases, acids, alkalies and organic matter. The presence of hardness causes problems like boiler corrosion, scale and sludge formation, priming, foaming and carry over. Acids and gases like CO_2 in water lead to boiler corrosion. Alkalies in water cause caustic embrittlement.
2.	Cooling systems	Should have little dissolved solids, less amount of dissolved gases as dissolved gases cause corrosion, should be free from organic matter, should not promote the growth of slime and algae as the spray nozzles and cooling pipes may be clogged due to the deposition of solids.
3.	Food production industries like confectionary, dairies, distilleries, beverages, sugar industries, etc.	Water should be soft, should not be acidic or alkaline and should not contain dissolved gases, organic matter, pathogens and fungi. Water should confirm to the standards of drinking water. Presence of salts in water modifies the taste of the product, presence of alkalinity neutralises the acids in the fruits and modifies the taste. In sugar industries, the presence of salts causes crystallisation. Presence of pathogens spoil the food products creating bad odor.
4.	Textiles	Should be soft and free from turbidity, color-forming ions like iron, manganese and organic matter. Presence of salts leads to spots and deposit formation on the textiles. Salts can lead to unwanted product formation modifying the color of the dye in the dyeing process. Hard water precipitates basic dyes and decreases the solubility of acidic dyes. These salts produce insoluble precipitates that stain the cloth and cause uneven dyeing.
5.	Laundry	Should be free from hardness-producing substances. Should not contain iron and manganese salts because hard water consumes soap and does not clean properly. Can cause staining on the cloth due to the formation of iron soap.
6.	Paper and pulp industry	Free from hardness and alkalinity, free from silica, lime, magnesia, iron salts, turbidity-producing material and color should be absent because hardness-producing ions react with the chemicals used for processing paper. Silica may produce cracks in the paper and may decompose the resin films.
7.	Pharmaceutical industries	Water should be very pure, highly sterilised and free from all types of salts, pathogens and should confirm to the standards of potable water. Presence of salts in water can lead to the formation of undesirable products, some of which may be poisonous. Presence of pathogens can lead to infection in the patient.

2.9 Boiler Problems with Hard Water

Boiler feed water should correspond with the following composition
(i) Its hardness should be below 0.2 ppm
(ii) Its caustic alkalinity (due to OH^-) should lie in between 0.15 and 0.45 ppm
(iii) Its soda alkalinity (due to Na_2CO_3) should be 0.45–1 ppm

Excess of impurities, if present, in boiler feed water generally cause the following problems
(i) Scale and sludge formation
(ii) Boiler corrosion
(iii) Caustic embrittlement
(iv) Priming and foaming

Scale and sludge formation

Water evaporates continuously inside the boiler and the concentration of dissolved salts increases gradually. When the solution is saturated with respect to the salt concentration, the salts start precipitating out on the inner walls of the boiler. If the precipitate formed is soft, slimy and loose, it is called *sludge* and if the precipitate formed is hard and adhering on the inner walls of the boiler it is termed as *scale*.

SLUDGE

Definition It is soft, slimy and loose precipitate formed on the inner walls of the boiler.

Formation

(a) Sludges are formed by substances which have greater solubility in hot water than in cold water like $MgSO_4$, $MgCl_2$, $MgCO_3$, $CaCl_2$ etc.
(b) They are formed at colder parts of the boiler and get collected at places where the flow rate is slow or at bends, plug opening, glass gauge connection, thereby causing even choking of the pipes.
(c) They are poor conductors of heat.
(d) They can be removed easily by scraping off with wire brush.

Prevention

(i) By using softened water.
(ii) By frequently 'blow-down operation', that is, drawing off a portion of the concentrated water from the boiler and replacing it with fresh water.

Disadvantages of sludge formation

(i) Being poor conductors of heat they tend to waste a portion of heat generated and thus decreases the efficiency of the boiler.
(ii) Excessive sludge formation disturbs working of the boiler.

(iii) When formed along with scale, they get entrapped in scale and both get deposited as scales.

(iv) It settles in regions of poor water circulation such as pipe connection, plug opening, and glass gauge connection thereby choking the pipes.

SCALE

Definition They are deposits firmly sticking to the inner surface of the boiler which cannot be removed mechanically even with the help of hammer and chisel.

Formation of scale Scales are formed mainly due to four reasons:

(i) ***Decomposition of calcium bicarbonate***

$$Ca(HCO_3)_2 \rightarrow \underset{\text{Scale}}{CaCO_3} \downarrow + H_2O + CO_2 \uparrow$$

$CaCO_3$ scale is soft and it is the main cause of scale formation in low- pressure boilers

However in higher pressure boilers $CaCO_3$ is soluble due to the formation of $Ca(OH)_2$

$$CaCO_3 + H_2O \rightarrow \underset{\text{Soluble}}{Ca(OH)_2} + CO_2$$

(ii) ***Deposition of $CaSO_4$*** The solubility of $CaSO_4$ decreases as temperature increases. $CaSO_4$ is soluble in cold water and is completely insoluble in superheated water. Therefore, $CaSO_4$ gets precipitated as hard scale on the hot portion of the boiler.

(iii) ***Hydrolysis of magnesium salts*** At high temperature, dissolved magnesium salts undergo hydrolysis forming a soft scale of magnesium hydroxide.

$$MgCl_2 + 2H_2O \rightarrow \underset{\text{Soft scale}}{Mg(OH)_2} \downarrow + 2HCl$$

(iv) ***Formation of calcium and magnesium silicates***

Very small quantities of SiO_2 present in hard water react with calcium and magnesium forming calcium silicate ($CaSiO_3$) and/or magnesium silicate ($MgSiO_3$). These are hard scales and are extremely difficult to remove.

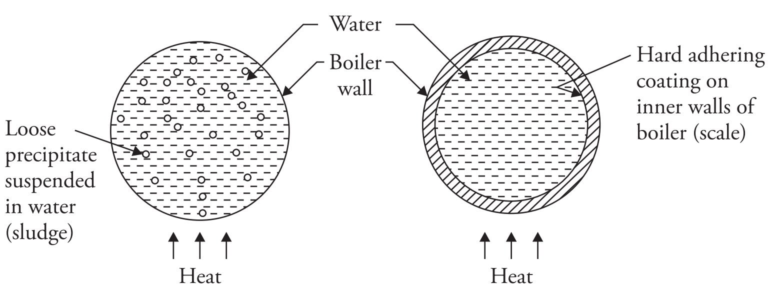


Figure 2.1 *Sludge and scales in boilers*

Table 2.7 Difference between scale and sludge

S.No	Sludge	Scale
1	Sludges are soft, loose and slimy precipitates	Scale are hard deposits
2	Formed at comparatively colder portions of the boiler	Formed generally at heated portion of the boiler
3	Formed by carbonate, bicarbonates of calcium and magnesium along with $Mg(OH)_2$	Formed by substances like $CaSO_4$ and silicates of calcium and magnesium
4	They are non- adherent deposits and can be easily removed	They stick very firmly to the inner surface of the boiler and are very difficult to remove
5	Decrease efficiency of boiler but are less dangerous	Decreases the efficiency of boiler and can even lead to boiler explosion
6	Can be removed by blow-down operation	Cannot be removed by blow-down operation

Disadvantages of scale formation

1. ***Wastage of fuel*** Scales have poor thermal conductivity, therefore the rate of heat transfer from boiler to the water inside is greatly reduced. Hence extra heat is to be supplied to the boiler and this increases the fuel consumption.

   Wastage of fuel depends upon the thickness of the scale. Greater the thickness of the scale larger is the wastage of fuel.
2. ***Lowering boiler safety*** Due to scale formation which is poor conductor of heat, the boiler is to be over heated to provide steady supply of heat. Overheating makes the boiler material softer and weaker. Therefore distortion of boiler tubes takes place and the boiler becomes unsafe to bear the pressure of the steam (in high-pressure boilers).
3. ***Decrease in efficiency*** Deposition of scale in the valves and condensers of the boiler, chokes them partially. This results in decrease in efficiency.
4. ***Danger of explosion*** Due to uneven expansion the thick scales crack, as a result water comes in contact with the overheated inner walls of the boiler resulting in excessive steam formation which can lead to boiler explosion.

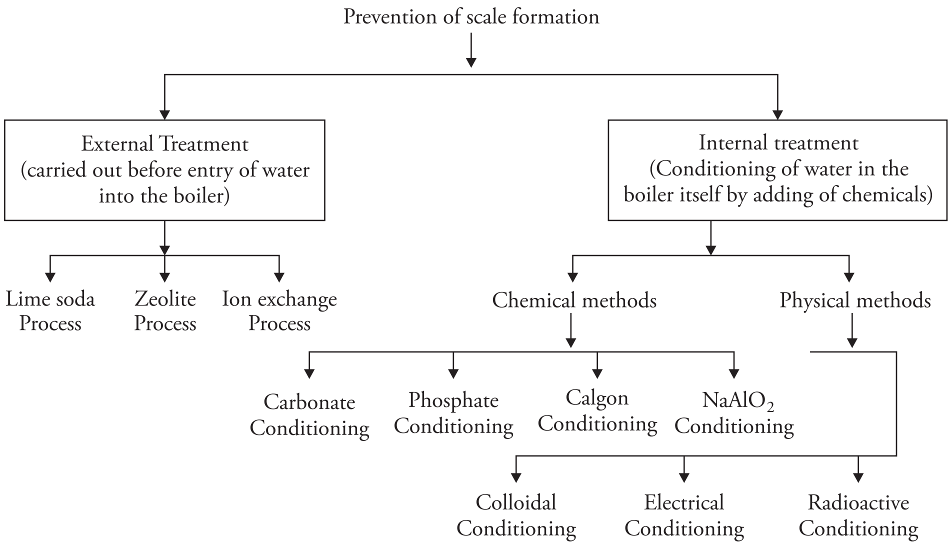
Removal of scale

(i) Loosely adhering scale can be removed either by scraper or wire brush or by blow-down operation.

(ii) Brittle scales can be removed by thermal shocks (i.e heating the boiler and then suddenly cooling with cold water).

(iii) Hard and adherent scales can be removed by dissolving in chemicals, for example, $CaCO_3$ scale can be dissolved by using 5–10% HCl. $CaSO_4$ scale can be removed by adding EDTA since the Ca-EDTA complex is highly soluble in water.

Prevention of scale formation



Internal Treatment

- It is also called *sequestration* or *conditioning*.
- It involves the treatment of boiler water inside the boiler by adding the suitable chemicals which either (a) precipitate the scale forming impurities in the form of sludges, which can be removed by blow-down operation or (b) convert them into compounds, which will stay in dissolved form in water and thus do not cause any harm.

  Internal treatment methods are generally followed by blow-down operation so that accumulated sludge is removed. Important internal conditioning methods are as follows.

Chemical methods

1. **Carbonate conditioning** In low pressure boilers the scale formation can be avoided by adding sodium carbonate to boiler water, so that $CaSO_4$ is converted into calcium carbonate. Consequently deposition of $CaSO_4$ as scale does not take place and calcium is precipitated as loose sludge of $CaCO_3$, which can be removed by blow-down operation.

$$\underset{\text{hard scale}}{CaSO_4} + Na_2CO_3 \rightarrow \underset{\text{loose sludge}}{CaCO_3 \downarrow} + Na_2SO_4$$

2. **Phosphate conditioning** It is used equally in low- and high-pressure boilers. Magnesium chloride which hydrolyses to form $Mg(OH)_2$ scale can be removed by adding appropriate sodium phosphate NaH_2PO_4, Na_2HPO_4, or Na_3PO_4.

   The phosphate reacts with calcium/magnesium salts forming loose sludge of calcium or magnesium phosphate which can be removed by blow-down operation

$$3MgCl_2 + 2Na_3PO_4 \rightarrow Mg_3(PO_4)_2 \downarrow + 6NaCl$$

The choice of phosphate salt depends upon the alkalinity of the boiler feed water. Calcium can be precipitated properly at a pH of 9.5 or above so a phosphate is selected that adjusts pH to optimum value (9.5–10.5)

3. **Calgon conditioning** It involves adding sodium hexametaphosphate (also called calgon) to boiler water to prevent scale and sludge formation. Calgon converts the scale forming impurities into highly soluble complexes.

$$Na_2[Na_4(PO_3)_6] \rightarrow 2Na^+ + [Na_4(PO_3)_6]^{2-}$$

$$2CaSO_4 + [Na_4(PO_3)_6]^{2-} \rightarrow \underset{\text{Soluble complex ion}}{[Ca_2(PO_3)_6]^{2-}} + 2Na_2SO_4$$

$$2Na^+ + [Ca_2(PO_3)_6]^{2-} \rightarrow \underset{\text{Soluble complex}}{Na_2[Ca_2(PO_3)_6]}$$

4. **Sodium metaaluminate ($NaAlO_2$) conditioning** Sodium metaaluminate gets hydrolysed forming NaOH and a gelatinous precipitate of aluminium hydroxide:

$$NaAlO_2 + 2H_2O \rightarrow NaOH + \underset{\text{Gelatinous precipitate}}{Al(OH)_3}$$

The sodium hydroxide so formed precipitates Mg^{2+} as $Mg(OH)_2$

$$MgCl_2 + 2NaOH \rightarrow Mg(OH)_2 + 2NaCl$$

The flocculent precipitate of $Mg(OH)_2$ plus $Al(OH)_3$, produced inside the boiler entraps finely suspended and colloidal impurities (like oil, silica etc.). These loose precipitates are removed by blow-down process.

Physical Methods

5. **Radioactive conditioning** Tablets containing radioactive salts are placed inside the boiler at specific points. The energy radiations emitted out by these prevent the aggregation of scale forming particles.
6. **Electrical conditioning** Sealed glass bulbs, containing mercury connected to a battery are set rotating in the boiler. When water boils, mercury emits discharges which prevents scale forming particles from sticking together to form scale.

7. **Colloidal conditioning** Colloids like tannin, agar-agar gel and kerosene are added to the boiler. They form a coating around scale forming particles and thus prevent them from aggregating together to form scale. This method is effective in case of low-pressure boilers.

BOILER CORROSION

Definition It is defined as the decay or disintegration of boiler material by chemical or electrochemical attack by its environment. Boiler corrosion generally occurs due to three reasons.

1. Acid formation by dissolved salts
2. Dissolved carbon dioxide
3. Dissolved oxygen

(1) **Acid formation by dissolved salts** Magnesium chloride present in the boiler feed water undergoes hydrolysis producing hydrochloric acid

$$MgCl_2 + 2H_2O \rightarrow Mg(OH)_2 \downarrow + 2HCl$$

The acid thus liberated reacts with the boiler material (iron) to form ferrous hydroxide which is then converted to rust

$$Fe + 2HCl \rightarrow FeCl_2 + H_2 \uparrow \quad (1)$$

$$FeCl_2 + 2H_2O \rightarrow Fe(OH)_2 \downarrow + 2HCl \quad (2)$$

The HCl formed in step (2) again attacks boiler material. A chain reaction is set up which causes extensive corrosion.

The $Fe(OH)_2$ formed further reacts with oxygen forming rust

$$\underset{\text{Ferrous hydroxide}}{4Fe(OH)_2 \downarrow} + O_2 \rightarrow \underset{\text{Rust}}{2[Fe_2O_3.2H_2O] \downarrow}$$

Prevention Boiler corrosion by acid formation can be prevented by adding calculated quantity of alkali which neutralises the acid thus formed.

(2) Dissolved carbon dioxide The main source of CO_2 in boilers is by the decomposition of bicarbonates of calcium and magnesium.

$$Mg(HCO_3)_2 \rightarrow Mg(OH)_2 + 2\,CO_2 \uparrow$$

$$Ca(HCO_3)_2 \rightarrow CaCO_3 + H_2O + CO_2 \uparrow$$

Carbon dioxide reacts with water forming carbonic acid

$$H_2O + CO_2 \rightarrow \underset{\text{(Carbonic acid)}}{H_2CO_3}$$

Carbonic acid has a slow corrosive effect on the boiler material.

Removal CO_2 can be removed either by adding ammonium hydroxide or by mechanical de aeration.

$$2NH_4OH + CO_2 \rightarrow (NH_4)_2CO_3 + H_2O$$

(3) **Dissolved oxygen** Water normally contains 8 ppm of oxygen dissolved at room temperature. At high temperature, the dissolved oxygen reacts with iron of the boiler to form ferric oxide (rust):

$$2Fe + 2H_2O + O_2 \rightarrow 2Fe(OH)_2 \downarrow$$

$$\underset{\text{Ferrous hydroxide}}{4Fe(OH)_2} + O_2 \rightarrow \underset{\text{Rust}}{2(Fe_2O_3.2H_2O)} \downarrow$$

Removal (i) By adding chemicals (ii) By mechanical de aeration

Addition of chemicals Dissolved oxygen can be removed by adding a reducing agent like hydrazine, sodium sulphite or sodium sulphide.

$$N_2H_4 + O_2 \rightarrow N_2 + 2H_2O$$

$$2Na_2SO_3 + O_2 \rightarrow 2Na_2SO_4$$

$$Na_2S + 2O_2 \rightarrow Na_2SO_4$$

Aqueous hydrazine is an ideal chemical for removal of dissolved oxygen because it forms nitrogen which is harmless. Pure hydrazine cannot be used as it is explosive.

Excess of hydrazine should not be used as it decomposes to give NH_3.

If Na_2SO_3 or Na_2S are used, Na_2SO_4 is formed, which decomposes at high temperature and produces SO_2 which reacts with water forming corrosive sulphurous acid.

$$SO_2 + H_2O \rightarrow H_2SO_3$$

(iii) ***Mechanical deaeration*** The process consists of spraying water over preheated perforated plates stacked in tower. Removal of dissolved O_2 is ensured by applying high temperature and vacuum.

PRIMING AND FOAMING

PRIMING

Definition When water is heated rapidly inside the boiler or if there is a sudden increase in steam demand then water droplets are carried along the steam. This steam is termed as 'wet steam' and the formation of wet steam is called 'priming'.

Causes

1. Very high velocity of steam formation, when steam is formed at a great speed in the boiler, some droplets of water are carried along with steam.
2. Presence of considerable quantities of dissolved solids.
3. Sudden ebullition/boiling.
4. Faulty boiler design.

Minimisation of Priming

1. Controlling rapid change in steam velocities.
2. Proper boiler design (by putting anti-priming plates or dash plates).
3. Maintaining low water level.
4. Use of water without suspended impurities.
5. By blowing off scales and sludges from time to time.

FOAMING

Definition Formation of small but persistent foam or bubbles which do not break easily.

Causes The main cause of foaming is the presence of oil or alkalies in boiler feed water, which reduces the surface tension of water causing foaming.

Minimisation of foaming

1. By adding anti-foaming agents like castor oil or silicic acid.
2. By adding compounds like sodium aluminates or aluminium sulphate, which removes oil from the boiler. These are hydrolysed to form $Al(OH)_3$, which entraps oil drops.

CAUSTIC EMBRITTLEMENT

It is the phenomenon in which the boiler material becomes brittle due to the accumulation of excess alkali in the boiler. Caustic embrittlement is a form of stress corrosion taking place in boilers operating at high temperature and pressure. It is caused by high alkalinity of water in the boiler, particularly at those places which are under stress such as rivets, joints and bends with the result that the metal plates become brittle.

When water is softened by lime – soda process, the excess sodium carbonate undergoes decomposition in high pressure boilers leading to the formation of NaOH.

$$Na_2CO_3 + H_2O \rightarrow 2NaOH + CO_2$$

This NaOH makes the water alkaline. The alkaline water penetrates into the minute cracks and crevices between the rivets and joints by capillary action. Inside the cracks, the water evaporates and the concentration of NaOH increases on these sites due to poor circulation of water. NaOH attacks the iron in boiler material converting it to sodium ferroate (Na_2FeO_2). A small quantity of sodium ferrite ($NaFeO_2$) is also formed.

$$Fe + 2NaOH \rightarrow Na_2FeO_2 + H_2 \uparrow$$

$$3Na_2FeO_2 + 4H_2O \rightarrow 6NaOH + Fe_3O_4 + H_2 \uparrow$$

$$6Na_2FeO_2 + 6\,H_2O + O_2 \rightarrow 12NaOH + 2Fe_3O_4$$

$$Fe_3O_4 + 4NaOH \rightarrow 2NaFeO_2 + Na_2FeO_2 + 2H_2O$$

It can be seen that NaOH is regenerated in the process and its concentration keeps on increasing maintaining the required environment.

The caustic embrittlement of the boiler may be explained by assuming the formation of a concentration cell as shown below:

Iron at bends, rivets, joints (stressed areas)	Concentrated NaOH solution	Dilute NaOH solution	Iron at plane surface (stress free)

The iron in the anodic area undergoes corrosion and dissolves making the boiler material brittle.

Prevention

1. By adding tannin or lignin to the boiler water. These block the minute cracks and prevent the infiltration of NaOH.
2. Caustic embrittlement can be prevented if Na_2SO_4 is added to the boiler water, because it is useful in blocking the hair cracks. Na_2SO_4/NaOH should be used in the ratio 1:1, 2:1 and 3:1 for boilers working at pressure 10, 20 and above 20 atm, respectively.
3. By using sodium phosphate instead of sodium carbonate for softening water.
4. By adjusting the pH of boiler water between 8–9.
5. Use of excess Na_2CO_3 should be avoided in the lime-soda process.

2.10 Softening Methods: External Treatment Process

External Treatment Process

Important methods employed for this process are

1. Lime-Soda Method
2. Zeolite process and
3. Ion Exchange process

The details of the above three methods are discussed as follows.

Lime soda process

It is an important softening process.

Principle The basic principle of this process is to convert all the soluble hardness causing impurities into insoluble precipitates which can be removed by sedimentation and filtration. It involves the addition of calculated quantities of lime $Ca(OH)_2$ and soda Na_2CO_3. The hardness causing impurities react with these chemicals forming precipitates of $CaCO_3$ and $Mg(OH)_2$. These precipitates are then filtered off to obtain soft water.

The precipitates formed are very fine and do not settle easily causing difficulty in filtration. Therefore small amount of coagulants like alum ($K_2SO_4.Al_2(SO_4)_3.24H_2O$), ferrous sulphate ($FeSO_4$) or sodium aluminate ($NaAlO_2$) are added. They help in the quick setting of precipitates of $CaCO_3$ and $Mg(OH)_2$ by the process of loading.

(The coagulants hydrolyse to form $Al(OH)_3$ which entraps the fine precipitates of $CaCO_3$ and $Mg(OH)_2$. The precipitates become heavy and settle fast. This process is called loading).

Calculation of Lime-Soda Requirement

Calculated quantity of lime and soda are added to hard water to remove temporary and permanent hardness. These chemicals also react with CO_2, H_2S gas, free mineral acids present in water and also with the coagulants added in water.

Table 2.8 Chemical reactions involved in lime soda process and calculation of the requirement of lime and soda for water softening

Constituent	Reaction	Requirement of lime/soda	Explanation
$Ca(HCO_3)_2$, $CaCO_3$ (temporary Ca hardness)	$Ca(HCO_3)_2 + Ca(OH)_2 \rightarrow 2CaCO_3\downarrow + 2H_2O$	L	One equivalent of $Ca(HCO_3)_2$ is removed by one equivalent of lime, hence the requirement is L
$Mg(HCO_3)_2$, $MgCO_3$ (temporary Mg hardness)	$Mg(HCO_3)_2 + 2Ca(OH)_2 \rightarrow 2CaCO_3\downarrow + Mg(OH)_2\downarrow + 2H_2O$ The reaction occurs in two steps $Mg(HCO_3)_2 + Ca(OH)_2 \rightarrow MgCO_3\downarrow + CaCO_3\downarrow + 2H_2O$ $MgCO_3$ is fairly soluble and reacts further $MgCO_3 + Ca(OH)_2 \rightarrow Mg(OH)_2 + CaCO_3$	2L	Removal of one equivalent of $Mg(HCO_3)_2$ requires 2 equivalent of lime hence 2L
Ca^{2+} [$CaCl_2$, $CaSO_4$, $Ca(NO_3)_2$] permanent Ca^{2+} hardness	$Ca^{2+} + Na_2CO_3 \rightarrow CaCO_3 + 2Na^+$	S	One equivalent of Ca^{2+} is removed by one equivalent of soda hence requirement is S
Mg^{2+}[$MgCl_2$, $MgSO_4$, $Mg(NO_3)_2$] permanent Mg^{2+} hardness	$Mg^{2+} + Ca(OH)_2 \rightarrow Mg(OH)_2 + Ca^{2+}$ $Ca^{2+} + Na_2CO_3 \rightarrow CaCO_3 + 2Na^+$	L+S	Lime removes permanent Mg^{2+} hardness but introduces Ca^{2+} hardness which is removed by soda
HCO_3^- (e.g., $NaHCO_3$)	$2HCO_3^- + Ca(OH)_2 \rightarrow CaCO_3 + H_2O + CO_3^{2-}$ or $2NaHCO_3 + Ca(OH)_2 \rightarrow CaCO_3 + H_2O + Na_2CO_3$	L-S	Bicarbonate present in water consumes equivalent amount of lime and equivalent amount of soda is produced which is to be subtracted from total soda requirement

CO_2	$CO_2 + Ca(OH)_2 \rightarrow CaCO_3 + H_2O$	L	One equivalent of CO_2 is removed by one equivalent of lime
H_2S	$H_2S + Ca(OH)_2 \rightarrow CaS + H_2O$	L	Does not produce hardness but consumes equivalent amount of lime and hence is to be considered for calculation
H^+ (Free acids like HCl,H_2SO_4)	$2H^+ + Ca(OH)_2 \rightarrow Ca^{2+} + 2H_2O$ $Ca^{2+} + Na_2CO_3 \rightarrow CaCO_3 + 2Na^+$	L + S	Free acids present in water react with lime introducing Ca^{2+} hardness in water which is in turn removed by soda
(Coagulant) $FeSO_4$	$Fe^{2+} + Ca(OH)_2 \rightarrow Fe(OH)_2 + Ca^{2+}$ $2Fe(OH)_2 + H_2O + O_2 \rightarrow 2Fe(OH)_3$ $Ca^{2+} + Na_2CO_3 \rightarrow CaCO_3 + 2Na^+$	L + S	$FeSO_4$ coagulants added to remove hardness reacts with equivalent amount of lime producing Ca^{2+} which are removed by soda. Hence it is to be considered while calculating lime-soda requirement
$Al_2(SO_4)_3$	$2Al^{3+} + 3Ca(OH)_2 \rightarrow 2Al(OH)_3 + 3Ca^{2+}$ $3Ca^{2+} + 2Na_2CO_3 \rightarrow 3CaCO_3 + 6Na^+$	L + S	$FeSO_4$ coagulants added to remove hardness react with equivalent amount of lime producing Ca^{2+} which are removed by soda. Hence it is to be considered while calculating lime-soda requirement
$NaAlO_2$	$NaAlO_2 + 2H_2O \rightarrow NaOH + Al(OH)_3$	–L	On hydrolysis $NaAlO_2$ gives NaOH which is considered as 1 equivalent of $Ca(OH)_2$ hence the amount of lime formed is to be subtracted from total lime requirement

Lime Requirement

Since 100 parts by mass of $CaCO_3$ are equivalent to 74 parts by mass of $Ca(OH)_2$

∴ Lime required for softening

$$= \frac{74}{100} \text{ (Temp } Ca^{2+} + 2 \times \text{Temp } Mg^{2+} + \text{Permanent } Mg^{2+} + HCO_3^- + CO_2 + H_2S + \text{Free acids} +$$

$$FeSO_4 + Al_2(SO_4)_3 - NaAlO_2 \text{ all in terms of } CaCO_3 \text{ equivalent)}$$

Soda Requirement

100 parts of $CaCO_3$ are equivalent to 106 parts of Na_2CO_3

$$= \frac{106}{100} \text{ (Perm } Ca^{2+} + \text{Permanent } Mg^{2+} - HCO_3^- + \text{Free acids} + FeSO_4 + Al_2(SO_4)_3 \text{ all in}$$

terms of $CaCO_3$ equivalent)

Note The above calculation gives lime and soda requirement for one liter water. The total lime/soda requirement for V liters water is calculated by multiplying volume of water in the above two formulas.

If lime and soda is impure

Say purity of lime = 90%

Purity of soda = 95%

Then lime required to soften water

$$= \frac{74}{100} \text{ (Temp } Ca^{2+} + 2 \times \text{Temp } Mg^{2+} + \text{Permanent } Mg^{2+} + HCO_3^- + CO_2 + H_2S + \text{Free acids} +$$

$$FeSO_4 + Al_2(SO_4)_3 - NaAlO_2 \text{ all in terms of } CaCO_3 \text{ equivalent)} \times \text{vol of water} \times \frac{100}{\%\text{ purity}}$$

$$\text{Lime} = \frac{74}{100} \text{ (Temp } Ca^{2+} + 2 \times \text{Temp } Mg^{2+} + \text{Permanent } Mg^{2+} + HCO_3^- + CO_2 + H_2S + \text{Free acids} +$$

$$FeSO_4 + Al_2(SO_4)_3 - NaAlO_2 \text{ all in terms of } CaCO_3 \text{ equivalent)} \times \text{ Vol of water } \times \frac{100}{90}$$

Similarly, soda required–

$$= \frac{106}{100} \text{ (Perm } Ca^{2+} + \text{Permanent } Mg^{2+} - HCO_3^- + \text{Free acids} + FeSO_4 + Al_2(SO_4)_3 \text{ all in}$$

$$\text{terms of } CaCO_3 \text{ equivalent)} \times \text{ vol of water} \times \frac{100}{\%\text{ purity}}$$

$$= \frac{106}{100} \text{ (Perm } Ca^{2+} + \text{Permanent } Mg^{2+} - HCO_3^- + \text{Free acids} + FeSO_4 + Al_2(SO_4)_3 \text{ all in}$$

$$\text{terms of } CaCO_3 \text{ equivalent)} \times \text{ Vol of water } \times \frac{100}{95}$$

If the treated water contains excess of OH^- and CO_3^{2-} ions, it implies that excess soda and lime has been added.

Reaction involved in giving excess OH^- and CO_3^{2-}

Excess OH^- comes from excess lime added in water

$$Ca(OH)_2 \rightarrow Ca^{2+} + 2OH^-$$

Hence if treated water contains excess OH^- then lime requirement

$$= \frac{74}{100} \text{ (Temp } Ca^{2+} + 2 \times \text{Temp } Mg^{2+} + \text{Permanent } Mg^{2+} + HCO_3^- + CO_2 + H_2S + \text{Free acids} +$$

$$FeSO_4 + Al_2(SO_4)_3 - NaAlO_2 + OH^- \text{ as } CaCO_3 \text{ equivalent).}$$

Similarly excess Na_2CO_3 contributes to both excess OH^- and excess CO_3^{2-} in treated water and hence both are to be added for soda calculation:

$$Na_2CO_3 + H_2O \rightarrow 2\,NaOH + H_2CO_3$$

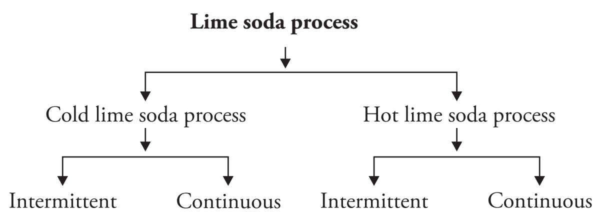
$$= \frac{106}{100} \text{ (Perm } Ca^{2+} + \text{Permanent } Mg^{2+} - HCO_3^- + \text{Free acids} + FeSO_4 + Al_2(SO_4)_3 + OH^- + CO_3^{2-}$$

all in terms of $CaCO_3$ equivalent).

NOTES

1. Substances like KCl, SiO_2, Fe_2O_3 and Na_2SO_4 do not impart any hardness and therefore they do not consume any soda and lime therefore these should not be considered while calculating soda and lime.
2. When impurities are given as $CaCO_3$ or $MgCO_3$ these should be considered as temporary hardness due to bicarbonates.

PROCESS



1. **Cold intermittent lime soda process**

   It is also called batch process. It consists of a tank which has an inlet for raw water and chemicals, outlet for softened water and sludge and a mechanical stirrer (Fig. 2.2). Calculated quantities of lime, soda and coagulant are added through the chemical feed inlet and raw water is added through another inlet simultaneously, and these are mixed thoroughly with the help of stirrer. For accelerating the process some sludge, (from previous operation) is added which acts as nucleus for fresh precipitation. The sludge (precipitate) formed is allowed to settle down, and is then removed through the sludge outlet. The softened water is removed through a floating pipe and sent to the filtering unit. Later when lot of sludge is collected the process has to be stopped to remove the sludge. This problem can be solved by using a set of tanks. Using a set of tanks ensures continuous supply of soft water by planned alternate cycle of reaction and settling.

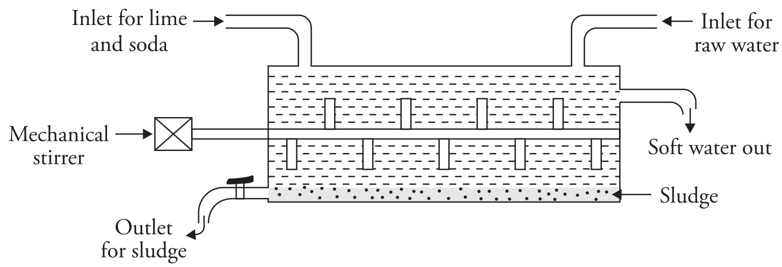


Figure 2.2 *Cold intermittent lime soda process*

2. **Cold continuous lime soda process**

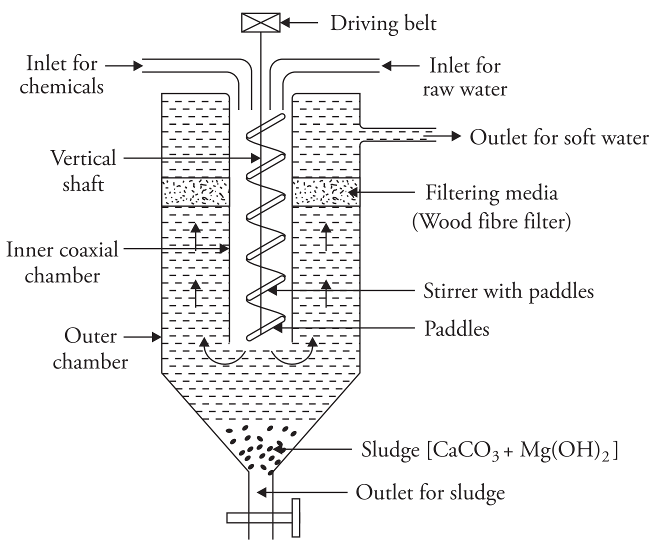


Figure 2.3 *Cold continuous lime soda process*

Apparatus It consists of two concentric vertical co-axial chambers. The inner tank is open from the bottom and has an inlet for chemicals (lime, soda and coagulants) and for hard water at the upper end. It is also provided with a stirrer with paddle plates. The outer chamber has a sludge outlet at the bottom and an outlet for soft water near the top. Just below the soft water outlet is the wood fibre filter for the filtration of soft water (Fig. 2.3).

Process Hard water is first analysed and then calculated quantity of chemicals and raw water are added simultaneously in the inner chamber. Vigorous stirring ensures continuous mixing of raw material and water. Softening of water occurs. The sludge thus formed settles down. Soft water rises up, passes through the wood fibre filter where traces of sludge are removed. Filtered soft water is then collected from the outlet near the top.

Sludge can be removed from the bottom from time to time. The cold lime soda process is not used these days for softening of boiler feed water because the water thus obtained has high residual hardness. This residual hardness harms the tubes in the boilers.

3. **Hot intermittent lime-soda process**

   It is similar to the intermittent type cold lime-soda softener except that the heating coils are installed in it for heating water.

4. **Hot continuous lime-soda process** It is similar to cold lime-soda process except that the raw water and chemicals are heated to a temperature of 80–150 °C.

   The apparatus consists of three parts (Fig. 2.4):

   (a) **Reaction tank** It has three separate inlets, one each for raw water, chemicals and superheated steam. All these are thoroughly mixed inside the reaction tank where the chemical reaction occurs forming insoluble precipitates of $CaCO_3$ and $Mg(OH)_2$ which settle down as sludge.

   (b) **Sedimentation tank** From the reaction tank the contents go to this tank. The sludge settles down and the water rises up and enters the sandfilter.

   (c) **Sand filter** It has layers of fine and coarse sand which acts as filter and ensures complete removal of sludge from the softened water.

Process

All the raw materials are put in through the three inlets in the reaction chamber. Here thorough mixing of raw materials takes place in the presence of superheated steam. Sludges settle down at the conical mouth of vertical outer chamber and 'sand filter' is provided for filtration to ensure complete removal of sludge from softened water giving finally filtered softened water.

Advantages of hot lime-soda process

1. Rate of reaction increases. The process is completed in 15 minutes whereas cold lime-soda process takes several hours for completion.
2. High temperature accelerates the chemical reactions. The rate of settling and filtration also increases. Thus the softening capacity of the hot process is several times higher than the cold process.
3. No coagulant is required.
4. Lesser quantity of chemicals are required.
5. Dissolved gases like CO_2 and H_2S are expelled out.
6. Viscosity of water decreases hence filtration becomes easier.
7. The residual hardness in softened water is 15–30 ppm, which is less than that of water received from cold lime-soda process (upto 60 ppm).
8. The precipitate formed is larger, coarser, heavier and settles rapidly. Most important advantage is that the silica in water is also removed in this process. Dissolved silica in water is absorbed or reacts at the surface of each precipitated particle of $Mg(OH)_2$.

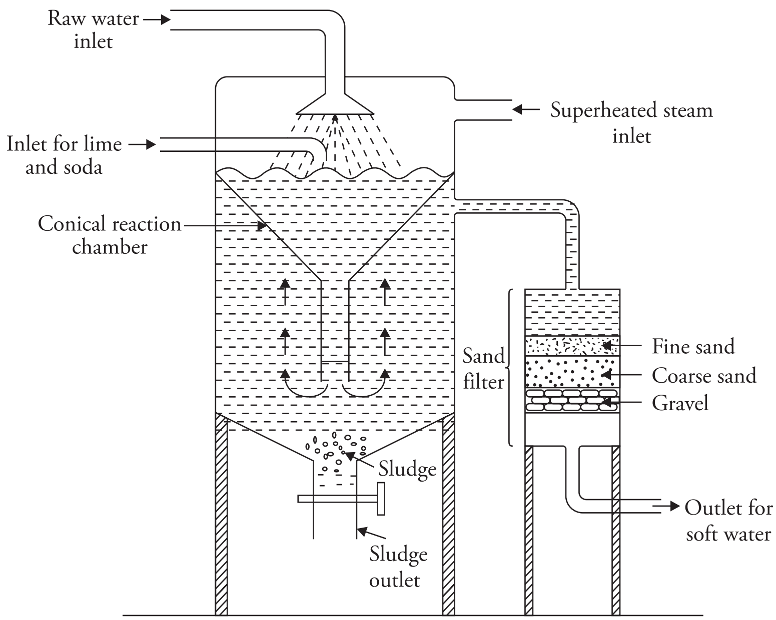


Figure 2.4 *Hot continuous lime soda process*

Merits of lime-soda process

1. Very economical.
2. During softening, the treated water has pH >7; hence it is alkaline and therefore has less corrosion tendencies.
3. Alkaline nature of softened water prevents the growth of bacteria.
4. Fe, Mn and dissolved minerals are also reduced in this process.

Demerits of the process

1. Requires careful and skilled supervision.
2. Sludge disposal is a problem.
3. It can remove hardness only up to 15 ppm which is still not appropriate for boilers.
4. The softened water contains appreciable amount of soluble salts such as sodium sulphate and cannot be used in high pressure boilers.
5. It cannot be used in households as it is not possible to calculate the amount of lime and soda required.

Solved Examples

1. Calculate the amount of lime required for softening 50,000 liters of hard water containing; $CaCO_3$ = 25 ppm; $MgCO_3$ = 144 ppm; $CaCl_2$ = 111 ppm; $MgCl_2$ = 95 ppm; Na_2SO_4 = 15 ppm; Fe_2O_3 = 25 ppm.

Solution

Constituent	Amount	Equivalents of $CaCO_3$	Requirement L/S
$CaCO_3$	25 ppm	$\frac{25\times100/2}{100/2}$ = 25 ppm	L
$MgCO_3$	144 ppm	$\frac{144\times100/2}{84/2}$ = 171.43 ppm	2L
$CaCl_2$	111 ppm	$\frac{111\times100/2}{111/2}$ = 100 ppm	S
$MgCl_2$	95 ppm	$\frac{95\times100/2}{95/2}$ = 100 ppm	L + S
Na_2SO_4	15 ppm	Does not impart hardness	
Fe_2O_3	25 ppm	Does not impart hardness	

Lime required for softening 50,000 liters of hard water

$$= \frac{74}{100}\ [CaCO_3 + 2 \times MgCO_3 + MgCl_2 \text{ as } CaCO_3 \text{ equivalent}] \times \text{Volume of water.}$$

$$= \frac{74}{100}\ [25 + 2 \times 171.43 + 100]\ \text{mg/L} \times 50{,}000\ \text{L}$$

$$= \frac{74}{100}\ [467.86]\ \text{mg/L} \times 50{,}000\ \text{L} = 1{,}73{,}10{,}820\ \text{mg} = 17.311\ \text{kg}\ (1\ \text{kg} = 10^6 \text{mg})$$

Lime Required = 17.311 kg

(Hardness expressed as $CaCO_3$ and $MgCO_3$ is treated as temporary hardness)

2. A water sample contains the following impurities

Ca^{2+} = 20 ppm; Mg^{2+} = 18 ppm; HCO_3^- = 183 ppm; SO_4^{2-} = 24 ppm. Calculate the amount of lime and soda needed for softening.

Solution

Constituent	Amount	Equivalents of $CaCO_3$	Requirement L/S
Ca^{2+}	20 ppm	$\frac{20\times100/2}{40/2} = 50$ mg/L	S
Mg^{2+}	18 ppm	$\frac{18\times100/2}{24/2} = 75$ mg/L	L+S
HCO_3^-	183 ppm	$\frac{183\times100/2}{61/1} = 150$ mg/L	L-S
SO_4^{2-}	24 ppm	–	–

As the volume of water is not given in the question hence calculations may be done for one liter water.

Lime required

$$= \frac{74}{100}\,[Mg^{2+} + HCO_3^- \text{ as } CaCO_3 \text{ equivalent}] \times \text{Volume of water}$$

$$= \frac{74}{100}\,[75 + 150] \times 1$$

$$= \frac{74}{100}\,[225] \times 1 = 166.5 \text{ mg}$$

Lime = 166.5 mg

Soda requirement

$$= \frac{106}{100}\,[Ca^{2+} + Mg^{2+} - HCO_3^- \text{ as } CaCO_3 \text{ equivalent}] \times \text{Volume of water}$$

$$= \frac{106}{100}\,[50 + 75 - 150] \times 1$$

$$= \frac{106}{100}\,[-25] = -26.50 \text{ mg}$$

The answer is negative, which shows that soda formed from bicarbonate is more than the amount of soda required for the removal of various impurities. Hence, the amount of soda to be added will be zero. ***Soda = Nil***

3. Calculate the amount of lime and soda required for softening 18,000 liters of water, which is analysed as follows:

   Temporary hardness = 25 ppm

   Permanent hardness = 20 ppm

   Permanent Mg hardness = 15 ppm

Solution

Temporary hardness is due to $Ca(HCO_3)_2$ and $Mg(HCO_3)_2$ and lime is required to remove both.

For removal of permanent hardness

Due to Mg – Lime + soda is required

Due to Ca – Only soda is required

$\therefore$ to calculate the total lime requirement we add –Temporary hardness + Permanent Mg hardness and to calculate soda requirement only total permanent hardness is considered.

Lime required

$$= \frac{74}{100} \text{ [Temporary hardness + Permanent Mg hardness]} \times \text{Volume of water}$$

$$= \frac{74}{100} [25 + 15] \times 18000$$

$$= \frac{74}{100} [40] \times 18000 = 532800 \text{ mg} = 532.8 \text{ g}$$

Lime = 532.8 g

Soda Requirement

$$\frac{106}{100} \text{ [Permanent hardness]} \times \text{Volume of water}$$

$$\frac{106}{100} [20 \text{ ppm}] \times 18000 = 381600 \text{ mg} = 381.6 \text{ g}$$

Soda = 381.6 g

4. A sample of water on analysis was found to contain the following in mg/L

   $CaSO_4$= 0.45; $MgCO_3$ = 0.50; $CaCO_3$ = 2.50; $MgSO_4$= 0.85; $MgCl_2$ = 0.80; SiO_2 = 2.90; NaCl = 2.80.

Calculate the amount of lime and soda required to soften 25,000 liters of water per day for a year if the purity of lime and soda is 90%.

Solution

Conversion into equivalents of $CaCO_3$

Constituent	Amount(mg/L)	Equivalents of $CaCO_3$	Requirement L/S
$CaSO_4$	0.45	$\frac{0.45\times100/2}{136/2} = 0.3309$	S
$MgCO_3$	0.50	$\frac{0.50\times100/2}{84/2} = 0.595$	2L
$CaCO_3$	2.50	$\frac{2.50\times100/2}{100/2} = 2.5$	L
$MgSO_4$	0.85	$\frac{0.85\times100/2}{120/2} = 0.708$	L + S
$MgCl_2$	0.80	$\frac{0.80\times100/2}{95/2} = 0.842$	L + S
SiO_2	2.90	Does not impart hardness and does not consume lime and soda	
NaCl	2.80	Does not impart hardness and does not consume lime and soda	

Volume of water = 25,000 liters a day for a year.

∴ Total volume of water to be softened = 25000 × 365 days

Purity of lime and soda = 90%

Lime required

$$= \frac{74}{100}\,[2 \times MgCO_3 + CaCO_3 + MgSO_4 + MgCl_2 \text{ as } CaCO_3 \text{ equivalent}] \times \text{Volume of water}$$

$$= \frac{74}{100}\,[2 \times 0.595 + 2.5 + 0.708 + 0.8421] \times 25000 \times 365$$

= 35383100 mg

The above result is for lime of 100% purity. However, lime in question is 90% pure. This means that when 90 mg lime is required 100 mg of lime is to be added to water.

Hence lime actually added = $35383100 \times \frac{100}{90}$ mg = 39314555.56 mg = 39.31 kg

Lime added = 39.31 kg

Similarly soda can also be calculated using the same analogy

Soda Requirement

$$= \frac{106}{100}\,[CaSO_4 + MgSO_4 + MgCl_2 \text{ as } CaCO_3 \text{ equivalent}] \times \text{Volume of water} \times \frac{100}{\%\text{ purity}}$$

$$= \frac{106}{100}\,[0.3309 + 0.708 + 0.842] \times 25000 \times 365 \times \frac{100}{90}$$

$$= \frac{106}{100}\,[1.8809] \times 25000 \times 365 \times \frac{100}{90} = 20214450.28 \text{ mg} = 20.214 \text{ kg.}$$

5. Calculate the amount of lime and soda needed for softening 100000 liters of water containing the following:

   HCl = 8.9 mg/L; $Al_2(SO_4)_3$ = 38.9 mg/L; $MgCl_2$ = 10.2 mg/L; $NaCl$ = 30.38 mg/L

   Purity of lime is 90% and that of soda is 98%. Ten percent chemicals are to be used in excess in order to complete the reaction quickly.

Solution

Conversion into equivalents of $CaCO_3$

Constituent	Amount(mg/L)	Equivalents of $CaCO_3$	Requirement L/S
HCl	8.9 mg/L	$\frac{8.9 \times 100/2}{36.5/1} = 12.191$	L + S
$Al_2(SO_4)_3$	38.9 mg/L	$\frac{38.9 \times 100/2}{342/6} = 34.12$	L + S
$MgCl_2$	10.2 mg/L	$\frac{10.2 \times 100/2}{95/2} = 10.737$	L + S
$NaCl$	30.38 mg/L	Does not impart hardness	

Volume of water = 10^5 liter

Purity of lime = 90%; Purity of soda = 98%

10% chemicals are to be used in excess in order to complete the reaction quickly.

Lime required

$$= \frac{74}{100} [HCl + Al_2(SO_4)_3 + MgCl_2 \text{ as } CaCO_3 \text{ equivalent}] \times \text{Volume of water} \times \frac{100}{\%\ \text{purity}}$$

$$= \frac{74}{100} [12.191 + 34.12 + 10.737] \times 10^5 \times \frac{100}{90}$$

$$= \frac{74}{100} [57.048 \text{ mg/L}] \times 10^5 \times \frac{100}{90} = 4.6906 \times 10^6 \text{ mg}$$

Since, 10% chemicals are used in excess

∴ Lime actually added

$$4.6906 \times 10^6 \times \frac{110}{100} = 5.159 \times 10^6 \text{ mg} = 5.159 \text{ kg}$$

Lime added = 5.159 kg

Soda requirement

$$= \frac{106}{100} [HCl + Al_2(SO_4)_3 + MgCl_2 \text{ as } CaCO_3 \text{ equivalent}] \times \text{Volume of water} \times \frac{100}{\%\ \text{purity}}$$

$$= \frac{106}{100} [12.191 + 34.12 + 10.737] \times 10^5 \times \frac{100}{98}$$

$$= \frac{106}{100} \times 57.048 \times 10^5 \times \frac{100}{98} = 6.1704 \times 10^6 \text{ mg} = 6.1704 \text{ kg}$$

$$\text{Soda added (considering 10\% excess)} = 6.1704 \times \frac{110}{100} = 6.787 \text{ kg}$$

Soda added = 6.787 kg

6. A water works has to supply 1 m^3/s of water. Raw water contains:
$Mg(HCO_3)_2$ = 219 ppm; Mg^{2+} = 36 ppm; HCO_3^- = 18.3 ppm; H^+ = 1.5 ppm. Calculate the cost of treating water per day, if lime (90% pure) and soda (95% pure) cost Rs. 500 per ton and Rs. 7000 per ton, respectively.

Solution

Conversion into equivalents of $CaCO_3$

Constituent	Amount	Equivalents of $CaCO_3$	Requirement L/S
$Mg(HCO_3)_2$	219 ppm	$\frac{219 \times 100/2}{146/2}$ = 150 ppm	2L
Mg^{2+}	36 ppm	$\frac{36 \times 100/2}{24/2}$ = 150 ppm	L+S
HCO_3^-	18.3 ppm	$\frac{18.3 \times 100/2}{61/1}$ = 15 ppm	L - S
H^+	1.5 ppm	$\frac{1.5 \times 100/2}{1/1}$ = 75 ppm	L + S

Purity of lime = 90%; Purity of soda = 95%

Volume of water to be purified in 24 hours

$(1\ m^3/s) \times (60 \times 60 \times 24) = 86400\ m^3/day = 8.64 \times 10^7$ L ($1\ m^3 = 1000$ liters)

Lime required

$$= \frac{74}{100}\ [2 \times Mg(HCO_3)_2 + Mg^{2+} + HCO_3^- + H^+ \text{ as } CaCO_3 \text{ equivalent}] \times \text{Volume of water} \times \frac{100}{\%\ \text{purity}}$$

$$= \frac{74}{100}\ [2 \times 150 + 150 + 15 + 75] \times 8.64 \times 10^7 \times \frac{100}{90}$$

$$= \frac{74}{100}\ [540] \times 8.64 \times 10^7 \times \frac{100}{90} = 3836.16 \times 10^7\ \text{mg} = 38.36 \text{ tons } (1 \text{ ton} = 10^3\ \text{kg} = 10^9\ \text{mg})$$

Lime = 38.36 tons

Soda requirement

$$= \frac{106}{100}\ [Mg^{2+} - HCO_3^- + H^+ \text{ as } CaCO_3 \text{ equivalent}] \times \text{Volume of water} \times \frac{100}{\%\ \text{purity}}$$

$$= \frac{106}{100}\ [150 + 75 - 15] \times 8.64 \times 10^7 \times \frac{100}{95}$$

$$= \frac{106}{100}\,[210] \times 8.64 \times 10^{7} \times \frac{100}{95} = 2024.48 \times 10^{7}\ \text{mg} = 20.24\ \text{tons}$$

Soda = 20.24 tons

Cost of lime = Rs. 500 per ton

Cost of soda = Rs. 7000 per ton

Total cost of treating water per day = (38.36 × 500 + 20.24 × 7000)

= 19180 + 141680 = Rs. 1,60,860

Ans = Rs. 1,60,860

7. Calculate the amount of lime and soda required for softening 30,000 liters of hard water, which is analysed as follows:

Analysis of raw water

Ca^{2+} = 400 ppm; Mg^{2+} = 288 ppm; HCO_3^- = 1586 ppm; dissolved CO_2 = 88 ppm;

Ferrous sulphate = 139 ppm (commercial grade)

Analysis of treated water

CO_3^{2-} = 30 ppm; OH^- = 34 ppm

Purity of lime is 94% and that of soda is 86%.

Solution

Conversion into equivalents of $CaCO_3$

Constituent	**Amount**	**Equivalents of $CaCO_3$**	**Requirement L/S**
Analysis of raw water			
Ca^{2+}	400 ppm	$\frac{400 \times 100/2}{40/2} = 1000$ ppm	S
Mg^{2+}	288 ppm	$\frac{288 \times 100/2}{24/2} = 1200$ ppm	L + S
CO_2	88 ppm	$\frac{88 \times 100/2}{44/2} = 200$ ppm	L
HCO_3^-	1586 ppm	$\frac{1586 \times 100/2}{61/1} = 1300$ ppm	L – S

Ferrous sulphate (commercial grade)	139 ppm	$\frac{139\times100/2}{278/2} = 50$ ppm	L + S
Analysis of treated water			
OH^-	34 ppm	$\frac{34\times100/2}{17/1} = 100$ ppm	L + S
CO_3^{2-}	30 ppm	$\frac{30\times100/2}{60/2} = 50$ ppm	S

Volume of water = 30,000 liters

Purity of lime = 94%; purity of soda = 86%

Lime required

$$= \frac{74}{100}\ [Mg^{2+} + HCO_3^- + CO_2 + FeSO_4.7H_2O \text{ as } CaCO_3 \text{ equivalent}] \times \text{Volume of water} \times \frac{100}{\%\ \text{purity}}$$

$$= \frac{74}{100}\ [1200 + 1300 + 200 + 50\ ] \times 30{,}000 \times \frac{100}{94}$$

$$= \frac{74}{100}\ [2750] \times 30{,}000 \times \frac{100}{94} = 64946809 \text{ mg} = 64.95 \text{ kg}$$

Lime = 64.95 kg

The excess lime added for treatment of water gives OH^- ions in treated water

$$Ca(OH)_2 \leftrightharpoons Ca^{2+} + 2\,OH^-$$

∴ ***Excess lime added***

$$= \frac{74}{100}\ [OH^- \text{ as } CaCO_3 \text{ equivalent}] \times \text{Volume of water} \times \frac{100}{\%\ \text{purity}}$$

$$= \frac{74}{100}\ [100] \times 30{,}000 \times \frac{100}{94} = 2361702.1 \text{ mg} = 2.362 \text{ kg}$$

Total lime = 64.95 + 2.362 = 67.312 kg

Soda Requirement

$$= \frac{106}{100} [Ca^{2+} + Mg^{2+} - HCO_3^- + FeSO_4 .\ 7H_2O \text{ as } CaCO_3 \text{ equivalent}] \times \text{Volume of water} \times \frac{100}{\%\ \text{purity}}$$

$$= \frac{106}{100} [1000 + 1200 - 1300 + 50] \times 30{,}000 \times \frac{100}{86}$$

$$= \frac{106}{100} [950] \times 30{,}000 \times \frac{100}{86} = 35127907 \text{ mg} = 35.128 \text{ kg}$$

Soda = 35.128 kg

The excess soda added during softening process contributes to both CO_3^{2-} and OH^- in treated water.

$$Na_2CO_3 + 2\ H_2O \rightarrow 2NaOH + H_2CO_3$$

Excess soda added

$$= \frac{106}{100} [OH^- + CO_3^{2-} \text{ as } CaCO_3 \text{ equivalent}] \times \text{Volume of water} \times \frac{100}{\%\ \text{purity}}$$

$$= \frac{106}{100} [100 + 50] \times 30{,}000 \times \frac{100}{86} = \frac{106}{100} [150] \times 30{,}000 \times \frac{100}{86}$$

5546511.6 mg = 5.546 kg

Total soda = 35.128 + 5.546 = 40.674 kg

Practice Problems

1. **C**alculate the amount of lime required for softening 50,000 liters of hard water containing

$CaCO_3$ = 20 ppm; $MgCO_3$ =190 ppm; $CaCl_2$ = 120 ppm; $MgCl_2$ = 100 ppm; Na_2SO_4 = 20 ppm; Fe_2O_3 = 30 ppm.

[Ans Lime required = 21.37 kg]

2. A sample of water was analysed and found to contain temporary magnesium hardness = 25 mg/L; permanent magnesium chloride hardness = 15 mg/L; permanent calcium sulphate hardness = 20 mg/L and SiO_2 = 300 mg/L. Calculate the lime and soda required for softening 30,000 liters of hard water.

[Ans Lime = 1.1107 kg; Soda = 0.9699 kg]

3. Calculate the quantities of lime and soda required for softening 3,00,000 liters of water using 164 ppm of sodium aluminate as coagulant. The analytical result of raw water is

   Ca^{2+} = 240 ppm; Mg^{2+} = 96 ppm; HCO_3^- = 732 ppm; CO_2(dissolved) = 44 ppm; NaCl = 60 ppm; Fe_2O_3 = 160 ppm.

   [Ans Lime = 222 kg; soda = 127.2 kg]

4. Calculate the quantity of lime and soda required to soften 25000 gallons of hard water containing $Ca(HCO_3)_2$ = 20 grains; $Mg(HCO_3)_2$ = 15 grains; $CaSO_4$ = 5 grains by lime soda process.

   [Ans Lime = 86.9 lbs (pounds); Soda = 13.93 lbs(pound)]

5. Calculate the quantity of lime and soda required for softening 60,000 liters of hard water. The analysis is as follows:

   Temporary hardness = 55.5 ppm

   Permanent calcium hardness = 42.5 ppm

   Permanent magnesium hardness = 48 ppm

   [Ans Lime = 4.60 kg; Soda = 5.76 kg]

6. Calculate the amount of lime (90% pure) and soda (98% pure) required for the treatment of one million liters of water containing

   $Ca(HCO_3)_2$ = 8.1 ppm; $CaCl_2$ = 33.3 ppm; HCO_3^- = 91.5 ppm; $MgCl_2$ = 38 ppm; $Mg(HCO_3)_2$ = 14.6 ppm. The coagulant $Al_2(SO_4)_3$ was added at a rate of 17.1 mg/L of water.

   [Ans Lime = 127.44 kg; Soda = 10.82 kg]

7. A sample of water on analysis gave the following results:

   H_2SO_4 = 147 mg/L; $MgCl_2$ = 95 mg/L; $MgSO_4$ = 24 mg/L; $CaSO_4$ = 170 mg/L and KCl = 20 mg/L.

   Water is to be supplied to the town of the population of one lakh only. The daily consumption of water is 100 liters per head. Calculate the cost of lime and soda required for softening hard water for the town for one month (November). If lime is 90% pure and Rs. 5 per kg and soda is 92% pure and costs Rs. 8.00 per kg.

   [Ans Cost of lime = Rs 3,33,000; Cost of soda = Rs. 1092260.87]

8. Calculate the amount of lime and soda needed for softening 75,000 liters of hard water. The results of analysis of raw water and treated water are as follows:

   Analysis of raw water

   HCO_3^- = 212 mg/L; H^+ = 3 mg/L; Ca^{2+} = 55.5 mg/L; Mg^{2+} = 42 mg/L; CO_2 = 66 mg/L;

   Analysis of treated water

   CO_3^{2-} = 90 mg/L; OH^- = 45 mg/L.

In this treatment, 20 mg/L of $NaAlO_2$ was used as a coagulant.

[Ans Total Lime = 42.67 kg; Total Soda = 45.5 kg]

Zeolite or Permutit Process

Zeolites are also known as permutit and in Greek it means 'boiling stone'.

They occur naturally as hydrated alumino silicate minerals like $Na_2O.Al_2O_3.\,x\,SiO_2.\,y\,H_2O$ where x = 2–10 and y is 2–6. They are capable of exchanging reversibly sodium ions for hardness producing ions in water.

Zeolites are of two types

1. *Natural Zeolites* Non porous, amorphous and durable, for example, Natrolite $Na_2O.Al_2O_3.4SiO_2.2H_2O$
2. *Synthetic Zeolites* Porous with gel-like structure. They are prepared by heating sodium carbonate (Na_2CO_3), alumina (Al_2O_3) and silica (SiO_2).

Process Hard water is percolated at a specified rate through a bed of zeolite stacked in a cylindrical unit as shown in the diagram (Fig. 2.5). The hardness causing elements (Ca^{2+}, Mg^{2+} etc) are retained by zeolite as CaZe and MgZe, while the outgoing water contains sodium salts.

The chemical reactions involved are

$$Na_2Ze + Ca(HCO_3)_2 \rightarrow CaZe + 2\,NaHCO_3$$

$$Na_2Ze + Mg(HCO_3)_2 \rightarrow MgZe + 2NaHCO_3$$

$$Na_2Ze + CaCl_2\ (\text{or } CaSO_4) \rightarrow CaZe + 2NaCl\ (\text{or } Na_2SO_4)$$

$$Na_2Ze + MgCl_2\ (\text{or } MgSO_4) \rightarrow MgZe + 2NaCl\ (\text{or } Na_2SO_4)$$

After some time, all the sodium ions in zeolite are replaced by Ca and Mg ions giving CaZe/MgZe. Such a zeolite bed is unable to soften water further and is said to be exhausted. It has to be regenerated to enable it to soften water again. It may be noted that the water obtained by zeolite process is used mostly for laundry purpose; it cannot be used in boilers.

Regeneration

When the zeolite bed is exhausted the supply of hard water is stopped and it is regenerated by treating with a concentrated brine (10% NaCl) solution.

$$\underset{\text{(exhausted zeolite)}}{CaZe\ (\text{or } MgZe)} + \underset{\text{(Brine)}}{2NaCl} \rightarrow \underset{\substack{\text{Reclaimed}\\\text{Zeolite}}}{Na_2Ze} + \underset{\text{(Washings)}}{CaCl_2\ (\text{or } MgCl_2)}$$

The washings (containing $CaCl_2$, $MgCl_2$, etc.) are led to drain and the regenerated zeolite bed is used again for softening. ($NaNO_3$, KCl, KNO_3, etc. can also be used for regeneration instead of

NaCl, but NaCl is mostly used because of its low cost and the products of regeneration process ($CaCl_2$ or $MgCl_2$) are highly soluble and can be easily rinsed out of the zeolite bed).

By knowing the amount of NaCl used for regeneration of the exhausted zeolite bed, the hardness of the water sample can be calculated by using the formula.

$$\text{Hardness (H)} = \frac{50 \times m \times V_2 \times 10^3}{58.5 \times V_1}$$

where

V_1 = Total volume of water softened

V_2 = Total volume of NaCl solution used for regeneration

m = Amount of NaCl in g/L present in V_2

Limitations and Disadvantages of Zeolite Process

1. If the water supplied is turbid, the turbidity/suspended matter is to be removed (by coagulation, filtration, etc.) before the water is admitted to the zeolite bed otherwise turbidity will clog the pores of zeolite bed making it inactive.
2. If the water contains ions like Mn^{2+} and Fe^{2+}, they must be removed first because these ions react with zeolite bed to give manganese and iron zeolites which cannot be easily regenerated.
3. Minerals acids, if present in water, destroy the zeolite bed and hence they must be neutralised with soda in advance, before feeding water into the zeolite bed.
4. This method does not remove anions. The bicarbonates present in hard water get converted into $NaHCO_3$ which goes into soft water effluent. If this water is used in boiler, $NaHCO_3$ dissociates as

   $$NaHCO_3 \rightarrow NaOH + CO_2$$

   NaOH leads to caustic embrittlement and CO_2 makes the water acidic and corrosive.
5. The treated water contains more sodium salts than in lime-soda process.
6. Moreover high cost of the plant and material also acts as limiting factor.

Advantages of Zeolite Process

1. Hardness is removed almost completely and the residual hardness is about 10 ppm.
2. The equipment is compact and occupies less space.
3. It is a clean and rapid process. Sludge is not formed as the impurities are not precipitated.
4. It requires less skill for maintenance as well as operation.

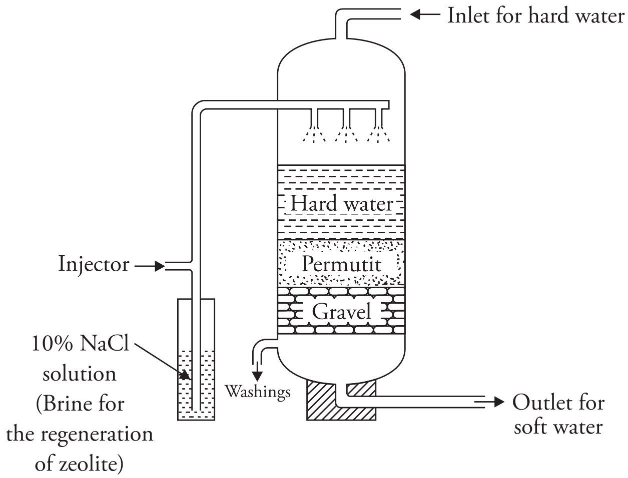


Figure 2.5 *Zeolite process*

Table 2.9 Comparison of Permutit Process with Lime-Soda Process

S.No	Zeolite/Permutit Process	Lime-Soda Process
1	Residual hardnes of water obtained is 10-15 ppm	Water of generally 15-50 ppm residual hardness is obtained
2	Cost of plant and zeolite are higher i.e capital cost is high	Capital cost is lower
3	Operation expenses are low as cheap NaCl is required for regeneration	Operation expenses high as costly chemicals (lime, soda and coagulants) are consumed
4	Occupies less space as it is compact	Plant occupies more space as it depends on the amount of water to be softened
5	It cannot be used for treating acidic water, because the zeolite undergoes disintegration	There is no such limitation
6	The treated water contains large amount of sodium salts than in original raw water. The $NaHCO_3$ in treated water causes caustic embrittlement	Treated water contains lesser % of sodium salts. It is completely free from $NaHCO_3$ as it is removed in the form of insoluble $CaCO_3$ and $Mg(OH)_2$
7	Raw water to be softened, must be free from suspended matter, otherwise the pores of zeolite block and the bed loses its ion exchange capacity	The process is free from such limitation
8	It can be operated under pressure and can be designed for fully automatic operation	This process cannot be operated under pressure.
9	It is clean process as no sludge is formed	Large amount of sludge is formed
10	Control test comprises in checking the hardness of treated water. The process adjusts itself to water of different hardness	In order to meet the changing hardness of incoming water frequent control and adjustment of reagents is needed

Solved Examples

1. A zeolite softener was used to remove the hardness of 95,000 liters of hard water completely. The softener required 475 liters of NaCl solution containing 18 g/liter of NaCl for regeneration. Calculate the hardness of hard water in ppm.

Solution

$$\text{Hardness (H)} = \frac{50 \times m \times V_2 \times 10^3}{58.5 \times V_1}$$

V_1 = Total volume of water softened = 95,000 liters

V_2 = Total volume of NaCl solution used for regeneration = 475 liters

m = Amount of NaCl present in V_2 = 18 g

$$\text{Hardness (H)} = \frac{50 \times 18 \times 475 \times 10^3}{58.5 \times 95{,}000} = 76.92 \text{ ppm}$$

Hardness (H) = 76.92 ppm

2. An exhausted zeolite softener was regenerated by 325 liters of NaCl solution containing 60 g/liter of NaCl. How many liters of hard water of hardness 250 mg/L can be softened by the zeolite softener.

Solution

Given

V_1 = Total volume of water softened = ?

V_2 = Total volume of NaCl solution used for regeneration = 325 liters

m = Amount of NaCl present in V_2 = 60 g/liter

Hardness (H) = 250 mg/liter = 250 ppm

Formula

$$\text{Hardness (H)} = \frac{50 \times m \times V_2 \times 10^3}{58.5 \times V_1}$$

$$250 = \frac{50 \times 60 \times 325 \times 10^3}{58.5 \times V_1}$$

$$V_1 = \frac{50 \times 60 \times 325 \times 10^3}{58.5 \times 250} = 66666.67 \text{ liters}$$

Total volume of hard water softened = 66666.67 liters.

Practice problems

1. A zeolite softener was completely exhausted and was regenerated by passing 100 liters of NaCl solution containing 120 g/liter of NaCl. How many liters of sample of water of hardness 500 ppm can be softened by this softener.

   [Ans 20512.8 liters]

2. A zeolite softener was completely exhausted after softening 145,000 liters of hard water. To regenerate the zeolite softener 540 liters of NaCl solution containing 110 g/liter of NaCl is required. Calculate the hardness of one liter water sample in ppm.

   [Ans 350.13 ppm]

3. 200 liters of NaCl solution containing 85 g/liter of NaCl was required to regenerate a completely exhausted zeolite softener. How many liters of hard water of hardness 600 ppm can be softened by the softener.

   [Ans 24216.5 liters]

4. When a zeolite softener was completely exhausted it was regenerated by passing 200 liters of NaCl solution containing 120 g/liter of NaCl. How many liters of a sample of water of hardness 550 mg/liter can be softened by the zeolite softener before regenerating it again.

   [Ans 37296.037 liters]

Ion Exchange Process – Deionisation or Demineralisation of Water

Ion exchange process is defined as the reversible exchange of ions in the structure of an ion exchanger to ions in solution that is brought in contact with it.

The resins used for the purpose are called ion exchange resins. They are porous, insoluble, cross linked, long chain organic polymers capable of exchanging ions.

Two types of resins are employed for the softening of water.

(i) ***Cation exchange resin***

They are materials capable of exchanging a cation in their structure to the cation in solution. For softening of water, the resins used should be capable of exchanging H^+ ions in their structure to other cations in solution. Commonly used resins are styrene divinyl benzene copolymers, which on sulphonation or carboxylation become capable of exchanging hydrogen ions with the cations in water. They are represented as R^-H^+ (R represents insoluble polymer matrix and H is the exchangeable ion).

Amberlite IR-120 and Dowex-50 are examples of commercially available cation exchange resins.

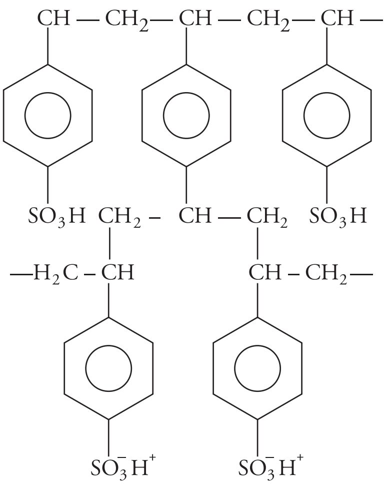


Figure 2.6 *Cation exchange resin*

(ii) ***Anion exchange resin***

They are materials capable of exchanging an anion in their structure (for water softening the exchangeable anion should be OH^- ion) to anion in solution. Anion exchangers employed for water softening are styrene divinyl benzene or amine formaldehyde copolymers, which contain basic functional group such as amino or quaternary ammonium or quaternary phosphonium or tertiary sulphonium groups as an integral part of the resin matrix. These after treatment with dilute NaOH solution become capable of exchanging OH^- ions with the anions in water. They are represented as R^+OH^- (where R^+ is the insoluble polymer matrix and OH^- is the exchangeable ion). Amberlite 400 and Dowex-3 are examples of commercially available anion exchange resins. The structure of a typical anion exchange resin is given in Figure 2.7.

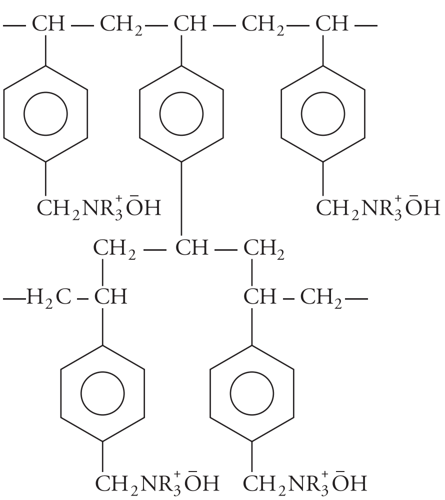


Figure 2.7 *Anion exchange resin*

Process The hard water is first passed through a column containing cation exchange resin. All cations are removed and equivalent amount of H^+ ions are released from this column to water. The water coming out of this chamber has low pH.

The exchange reactions are

$$2RH + Ca^{2+} \rightarrow R_2Ca + 2H^+$$

$$2RH + Mg^{2+} \rightarrow R_2Mg + 2H^+$$

After this the water is passed into second column containing anion exchange resin. All the anions are removed and an equivalent amount of OH^- ions are released.

$$ROH + Cl^- \rightarrow RCl + OH^-$$

$$2ROH + SO_4^{2-} \rightarrow R_2SO_4 + 2OH^-$$

$$2ROH + CO_3^{2-} \rightarrow R_2CO_3 + 2OH^-$$

The H^+ ions released from cation exchange column and OH^- ions released from anion exchange column combine to produce water molecule.

$$H^+ + OH^- \rightarrow H_2O$$

The water coming out of the exchanger is free from all cations (except H^+) and anions (other than OH^-). This ion free water is known as deionised water or demineralised water and has a neutral pH.

Regeneration

After sometime the resin loses all its H^+ and OH^- ions and then its capacity to exchange ions is lost. In such a condition they are said to be exhausted.

The exhausted cation exchange resin is regenerated by passing a solution of HCl or H_2SO_4.

$$R_2Ca + H_2SO_4/2HCl \rightarrow 2RH + CaSO_4/CaCl_2$$

$$R_2Mg + H_2SO_4/2HCl \rightarrow 2RH + MgSO_4/MgCl_2$$

The column is then rinsed with distilled water to remove the salts formed. Ideally, HCl is used for regeneration as $CaCl_2$ and $MgCl_2$ are more soluble.

The anion exchange resin is regenerated by passing a dilute solution of NaOH and then washing with distilled water

$$R_2SO_4 + 2NaOH \rightarrow 2ROH + Na_2SO_4$$

$$RCl + NaOH \rightarrow ROH + NaCl$$

NaCl, Na_2SO_4 are removed by washing with distilled water.

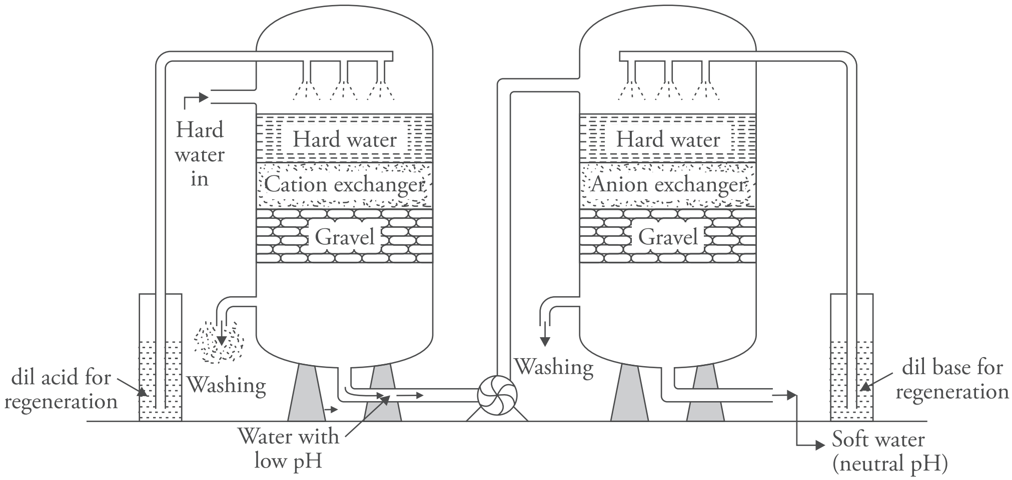


Figure 2.8 *Ion exchange process*

Advantages

1. The process can be used to soften highly acidic or alkaline water.
2. Unlike zeolite process the soft water does not contain sodium ions.
3. It removes all cations and anions other than H^+ and OH^- ions.
4. All ionisable impurities are removed.
5. It produces water of very low residual hardness (about 2 ppm).

Disadvantages

1. The process is expensive, both the equipment and resin are costly.
2. Turbid water decreases the efficiency of the process.

2.11 Municipal Water Supply

Water used for domestic purposes is consumed for various activities like drinking, cooking, bathing, washing, cleaning, etc. It is important that the water consumed for drinking and cooking should be free of impurities. Nearly 80% of the diseases are caused by contaminated water. The common requirements for potable water are

1. It should be clear, colorless, free from solids.
2. It should have good taste and should be odorless.
3. It should be soft.
4. It should be abundant and cheap.
5. Micro-organism and disease producing bacteria should not be present.
6. It should not have objectionable dissolved gases but sufficient dissolved oxygen should be present.
7. It should be free from harmful metallic salts.
8. Radioactive substance should not be present.
9. It should be free from fluorides, chlorides and phenolic compounds.
10. Corrosive metals should not exist.

Water treatment for domestic use employs the following steps:

1. Screening
2. Sedimentation
3. Filtration
4. Disinfection

(i) **Screening**

It is used to remove suspended impurities by passing water through screens having large number of holes when the floating materials are retained by them. Screens are vertical bars with perforations. Water is passed through them where the suspended and floating impurities are removed. To ensure complete removal of such impurities water is first passed through coarse screens and then through fine screens, which consist of woven wire, cloth or perforated plates mounted on a rotating disc or drum partially submerged in the flow. Fine screens should be cleaned mechanically on a continual basis. One of the problems associated with the screening process is the disposal of screenings. In modern water treatment plants, comminutors are installed. The comminutor grinds the coarse material to fine particles, which can be handled satisfactorily at a later stage.

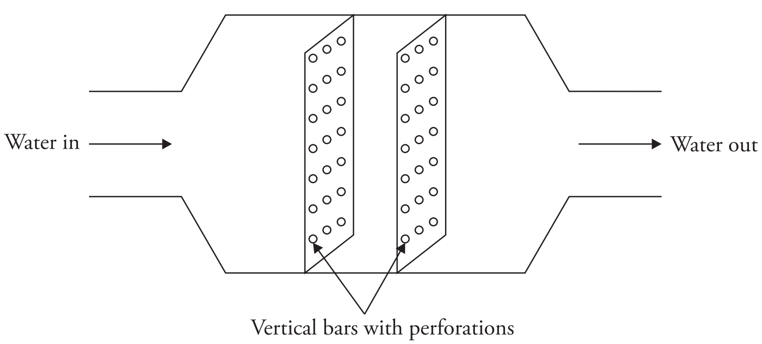


Figure 2.9 *Screening*

(ii) **Sedimentation**

It is the process of removing fine, suspended, colloidal impurities by allowing water to stand undisturbed in big tank about 5 m deep. Most of the suspended particles settle down at the bottom, due to the force of gravity. The clear supernant water is then drawn from the tank with the help of pumps. The retention period in a sedimentation tank ranges from 2–6 hours.

Fine suspended particles like mud particles and colloidal matter present in the water cannot settle by plain sedimentation. They are removed by sedimentation with coagulation.

Sedimentation with coagulation It is the process of removing fine suspended and colloidal impurities by addition of requisite amount of chemicals (coagulants) to water before sedimentation. Coagulant, when added to water, forms an insoluble gelatinous, flocculent precipitate, which traps very fine suspended impurities forming bigger flocs, which settle down.

The coagulants mainly used are alum, ferrous sulphate or copperas ($FeSO_4.7H_2O$), sodium aluminate, etc. They provide positive ions Fe^{3+} or Al^{3+} which neutralise the negative charge on the colloidal clay particles. After losing their charge the tiny particles come nearer to one another and combine to form bigger particles, which settle down, due to the force of gravity. The flocculants or coagulants are generally added in liquid form. Mixers are employed to ensure the proper mixing of the coagulants. General reactions during coagulation are

$$\underset{\text{Coagulant}}{Al_2(SO_4)_3} + \underset{\substack{\text{Calcium}\\\text{bicarbonate}\\\text{present in water}}}{3Ca(HCO_3)_2} \rightarrow \underset{\substack{\text{flocculent}\\\text{precipitate}}}{2Al(OH)_3} + 3CaSO_4 + 6\,CO_2$$

Alum reacts in the presence of alkalinity in water.

$$NaAlO_2 + 2H_2O \rightarrow Al(OH)_3 + NaOH$$

(This gives result in pH range 5.5-8.0. $Al(OH)_3$ floc causes sedimentation whereas NaOH reacts with magnesium salts in water)

$$MgSO_4 + 2NaOH \rightarrow Mg(OH)_2 + Na_2SO_4$$

Similarly, $FeSO_4$ also reacts in slightly alkaline medium

$$FeSO_4 + \underset{\text{from water}}{Mg(HCO_3)_2} \rightarrow Fe(OH)_2\downarrow + MgSO_4 + 2CO_2$$

$$4Fe(OH)_2\downarrow + \underset{\substack{\text{Dissolved}\\\text{oxygen}}}{O_2} + 2H_2O \rightarrow \underset{\substack{\text{Heavy floc which causes}\\\text{quick sedimentation}}}{4Fe(OH)_3}$$

(iii) **Filtration**

It is the process of separating colloidal and suspended impurities from water by passing it through a porous bed made of fine sand and other proper sized granular materials.

Filtration is carried out in a sand filter. It consists of a top thick layer of fine sand, placed over coarse sand layer followed by gravel. Water comes from the top. The suspended particles present in water that are of bigger size than the voids in the sand layer are retained there itself and the water becomes free of them. The sand layer may get choked after sometime and then it is to be replaced with clean sand for further action. This action is analogous to mechanical straining.

Small micro-organisms present in sand voids decompose organic matter and remove some of the biological impurities.

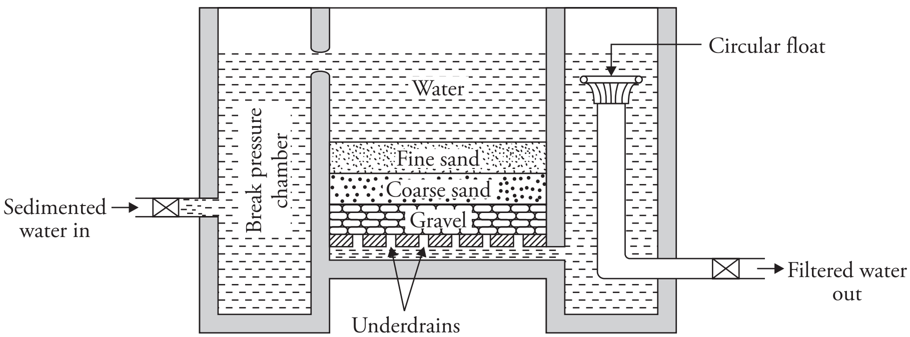


Figure 2.10 *Sand filter*

Advantages

Removes all suspended particles, colloidal impurities and organic matter.

(iv) **Disinfection**

It is the process of removing pathogenic bacteria, viruses and protozoa from water. A large number of methods are utilised, some of them are as follows:

(a) ***Boiling*** It is the simplest and easiest method to disinfect water. But it is useful only for small quantities of water. For proper disinfection, water should be boiled vigorously for 20–25 minutes.

(b) ***By adding bleaching powder*** Bleaching powder ($CaOCl_2$) is a very good disinfectant. About 1 kg of bleaching powder per 100 kiloliters of water is mixed and water is allowed to stand undisturbed for several hours. The chemical action produces hypochlorous acid which is a powerful germicide and kills all the germs.

$$CaOCl_2 + H_2O \rightarrow Ca(OH)_2 + Cl_2$$

$$Cl_2 + H_2O \rightarrow HCl + HOCl$$

$$\text{Germs} + HOCl \rightarrow \text{Germs are killed}$$

Drawbacks

(1) Introduces calcium in water and hence increases its hardness.
(2) Bleaching powder deteriorates during storage.
(3) Excess $CaOCl_2$ gives bad taste and odor to treated water.

(c) ***Direct chlorination*** Chlorine (either gaseous or in concentrated solution form) produces hypochlorous acid, which is a powerful germicide.

$$Cl_2 + H_2O \rightarrow HCl + HOCl$$

$$\text{Germs} + HOCl \rightarrow \text{Germs are killed}$$

Advantage

(1) Economical, requires little space for storage.
(2) Stable, does not deteriorate on storage.
(3) Does not introduce calcium.

Disadvantages

(1) Excess chlorine produces unpleasant taste and odor.
(2) Excess produces irritation on mucous membrane.

(d) ***By adding chloramines ($ClNH_2$)***: Chloramine is formed by reacting chlorine and ammonia in the ratio 2:1

$$Cl_2 + NH_3 \rightarrow \underset{\text{Chloramine}}{ClNH_2} + HCl$$

It is more lasting than chlorine alone, hence it is a better disinfectant

$$ClNH_2 + H_2O \rightarrow \underset{\text{Disinfectant}}{HOCl} + NH_3$$

Advantages

(1) Excess does not produce irritating odor.
(2) It imparts good taste to treated water.

Break Point Chlorination (or free residual chlorination)

When chlorine is added to water and the sample of water is analysed for available chlorine it is found that the available chlorine is less than added. This is because the chlorine added in water is consumed not only for destroying the germs but also for oxidising the organic impurities, reducing substances and coloring matter in water.

If a graph is plotted between the amount of chlorine added and the free residual chlorine a plot as shown in Figure 2.11 is obtained. This graph has four stages:

Stage I: For lower doses of chlorine all the added chlorine is used in oxidising the reducing compounds in water hence there is no residual chlorine in water (part *ab* of the graph).

Stage II: On further addition of chlorine formation of chloro-organic compounds and chloramines takes place. Chloramines act as chlorine reserve and contribute in the chlorine test just as free chlorine does. Hence the graph shows an increase in residual chlorine to a maximum level shown by *bc* in the curve.

Stage III: On further addition of chlorine, the oxidation of chloramines and chloro-organic compounds takes place. The amount of residual chlorine decreases and reaches a minimum shown by point *d* in the graph. This point is termed as the break point. At this point water is completely free of all kinds of impurities and is of potable water quality. Addition of chlorine uptil this point is termed as break point chlorination, which may be defined as the chlorination of water to such an extent that not only the pathogenic bacteria but also the organic matter, reducing impurities, coloring matter and odours from water is oxidised.

Stage IV: After the break point chlorine is not consumed further and the added chlorine is equal to the available chlorine and the graph once again shows an increase in the level of residual chlorine (part *de* in the graph.)

Advantages of break point chlorination:

(1) It removes all compounds that impart colour, bad odour and unpleasant taste to water.

(2) It destroys all organic matter, dissolved ammonia and microorganisms in water and prevents the growth of weeds in water.

Dechlorination: The excess chlorine added in water after break point imparts bad taste and colour to water and hence it has to be removed. It can be removed by passing water over activated charcoal or by the addition of SO_2 or by the use of Na_2SO_3

(a) Use of SO_2

$$H_2O + SO_2 \rightarrow H_2SO_3$$

$$H_2O + Cl_2 \rightarrow HCl + HClO$$

$$HClO + H_2SO_3 \rightarrow HCl + H_2SO_4$$

Net reaction:

$$SO_2 + Cl_2 + 2H_2O \rightarrow H_2SO_4 + 2HCl$$

(b) Use of Na_2SO_3

$$Na_2SO_3 + Cl_2 + H_2O \rightarrow Na_2SO_4 + 2HCl$$

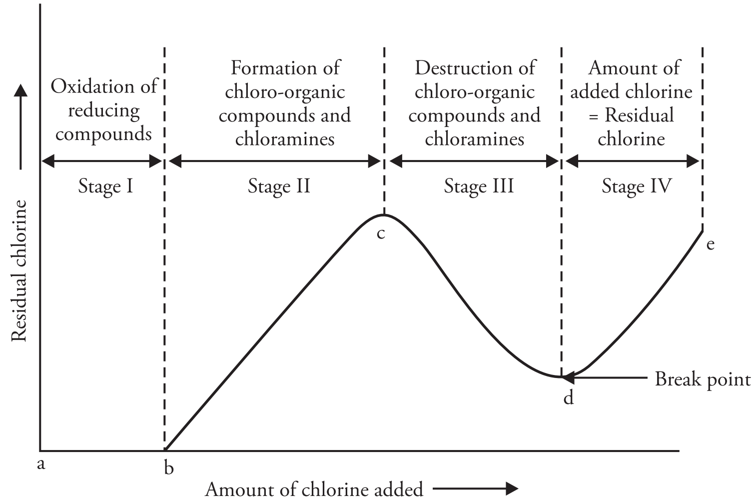


Figure 2.11 *Graphical representation of break point chlorination*

(e) Ozonisation Ozone is produced by passing silent electric discharge through cold and dry oxygen

$$\underset{\text{Oxygen}}{3O_2} \xrightarrow{\text{silent electric discharge}} \underset{\text{Ozone}}{2O_3}$$

Ozone is highly unstable and breaks down liberating nascent oxygen

$O_3 \rightarrow O_2 +$ [Nascent oxygen]

Nascent oxygen is a powerful oxidising agent and kills all the bacteria as well as oxidises the organic matter present in water. For disinfection by ozone, ozone is passed through water in a sterilising tank. The disinfected water is removed from the top. The contact period is about 10–15 minutes and dose is 2–3 ppm.

Advantages

(1) Ozone sterilises, bleaches, decolorises and deodorises water.

(2) Excess of ozone in water is not harmful as it decomposes to give oxygen.

(3) There is no irritation of mucous membrane as is in the case of chlorine treatment.

(4) Ozone improves the taste of water. Highly potable water is thus sterilised with ozone.

Disadvantages Ozonisation is a very expensive process. Moreover ozone is a corrosive agent. It corrodes stainless steel, cast iron, copper, rubber and ebonite.

(f) ***By adding $KMnO_4$*** It is generally used to disinfect surface water systems containing bad taste and odor. When $KMnO_4$ is added to water, it gives purple color to water which finally disappears when the oxidation is completed. It removes H_2S, Fe and Mn. The normal dosage of $KMnO_4$ is about 1–2 mg/L within a period of 4–6 hours. It is generally used in rural areas where supplies are from wells.

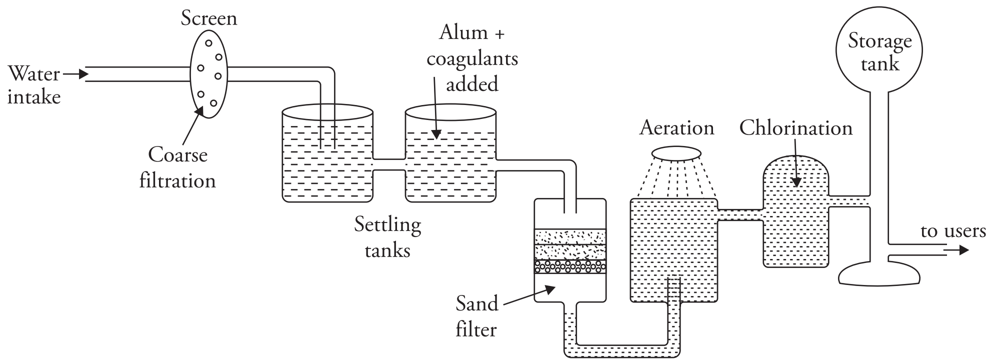


Figure 2.12 *Schematic representation of a municipal water supply system*

2.12 Desalination of Water

Water is a good solvent. It contains dissolved solids in it, depending upon the amount of dissolved solid water may be grouped as follows:

1. Fresh water (having < 1000 ppm of dissolved solids)
2. Brackish water (having > 1000 ppm but < 35000 ppm of dissolved solids)
3. Sea water (having > 35000 ppm of dissolved solids)

Thus water containing high concentration of salts in it is called brackish water. Due to the presence of large quantities of salts, it is salty in taste and is unfit for domestic and industrial purpose. This water can be put to good use by removing the salt content from it. *The process used for the removal of salts from water is called desalination.*

Various techniques employed for desalination are

1. Reverse osmosis
2. Electro dialysis
3. Ultra filtration
4. Flash evaporation

Reverse osmosis

Osmosis is defined as the process of spontaneous flow of solvent particles through a semi permeable membrane from a dilute solution to a concentrated solution (from lower concentration to higher concentration). The diffusion of solvent particles takes place on account of hydrostatic pressure called osmotic pressure, which drives solvent molecules in search of equilibrium.

If an external pressure equal to osmotic pressure is applied on the concentrated solution the process of osmosis will stop and if the external pressure exceeds osmotic pressure, solvent will be squeezed out of the concentrated solution. This process is called reverse osmosis.

Thus in reverse osmosis, pure water is separated from its contaminants rather than removing contaminants from water. This process is also known as super-filtration or hyper filtration. The semi permeable membranes are usually made up of cellulose acetate, polymethacrylate or polyamide polymers.

Advantages This process removes lead, calcium, magnesium, sodium, potassium, aluminium, chloride, nitrate, fluoride, sulphate, boron, most microorganisms and organic chemicals from water.

Disadvantages It requires large volume of water. *It may take as much as 90 liters water to recover 5 liters of useable water.*

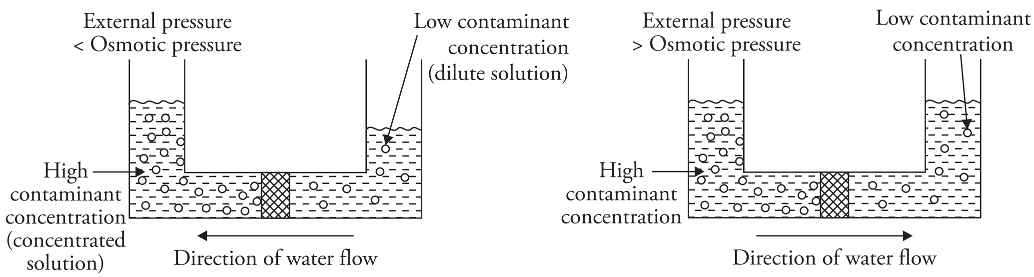


Figure 2.13 *(a) Normal osmosis. (b) Reverse osmosis*

Electrodialysis

It is a method in which the charged ions (salts) are removed from water by passing direct current, using electrodes and thin rigid ion selective semi permeable membranes.

Principle and working Consider a cell which is divided into three compartments using semi permeable membranes. Two electrodes (anode and cathode) are placed in the outer compartments and saline water is introduced in all the three compartments. When electric current is passed through saline water, the sodium ions (Na^+) move towards cathode (negative pole) and the chloride ions (Cl^-) move towards anode (positive pole) through the membrane.

Hence the concentration of brine decreases in the middle compartment and increases in the two side compartments. Desalinated (pure water) is obtained from the middle compartments, while the concentrated brine solution in side compartments is later replaced by fresh sea water.

A = Anion permeable membrane

C = Cation permeable membrane

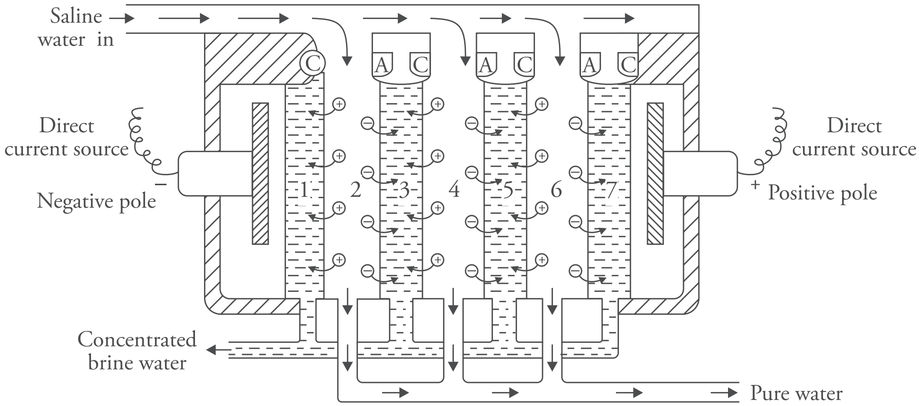


Figure 2.14 *Electrodialysis*

For more efficient separation ion selective membranes are employed. An ion selective membrane has permeability for one kind of ions only. For example, anion selective membrane allows the passage of anion and the cation selective membrane allows the passage of cations.

An electrodialysis cell consists of alternate cation and anion selective membranes. The cathode is placed near the cation selective/permeable membrane and the anode is placed near the anion selective membrane. Saline water is passed under a pressure of about 5–6 kg m^{-2} between membrane pairs and an electric field is applied perpendicular to the direction of water flow. Under the influence of the strong electric field the sodium ions (Na^+) start moving towards cathode through the membrane. As a result, there is depletion of ions in the even numbered compartments and the concentrations of ions in the odd numbered compartments increases. The desalinated water, free of ions is collected separately, whereas concentrated water of the odd numbered compartments is replaced from time to time by fresh lot of saline water.

Advantages

1. It is a compact unit and is cost effective and economical.
2. This technique is useful for the production of fresh water in arid-coastal regions.
3. The method is used when very high purification is not required.

Ultra filtration

Ultra filtration is the process of removing high molecular-weight substances like colloidal materials, organic and inorganic polymeric molecules like proteins by using semi permeable membranes with pore sizes in the range of 0.1 to 0.001 micrometer. Low molecular weight organic molecules and inorganic ions such as sodium, calcium, potassium and magnesium are not removed by this process.

The extent to which dissolved solids and microorganisms are removed is determined by the size of the pores in the membranes. Substances larger than the pore size are fully removed whereas those smaller than the pore size cannot be removed. The size of the molecules retained by the membrane is defined by the molecular weight cut off (MWCO) of the membrane used.

The process of ultra filtration uses hollow fibers of membrane material. The feed water flows inside the hollow fibers. Suspended solids and solutes of high molecular weight are retained inside the hollow membrane (these retained particles are called retentate), while water and low molecular weight solutes pass through the membrane. These particles which pass through the membrane are called permeate.

The membranes commonly used for commercial ultra filtration are 'polysulfone' and cellulose acetate membranes.

Molecular weight cut off The retention properties of ultra filtration membranes are expressed by molecular weight cut-off (MWCO). This value refers to the approximate molecular weight of a dilute solute which is 90% retained by the membrane. For instance, a membrane that removes dissolved solids with molecular weight of 10,000 and higher has a MWCO of 10,000.

Factors affecting ultra filtration

The performance of an ultra filtration system depends on several factors:

(i) **Flow across the membrane surface** Greater the flow velocity of the liquid across the membrane greater will be its permeate rate.

(ii) **Pressure** The permeate rate will be directly proportional to the pressure applied across the membrane surface.

(iii) **Temperature** The permeate rate is directly proportional to the temperature. It increases with the rise in temperature.

Applications

1. Ultra-filtration is used for the removal of suspended solids, colloids, cysts, bacteria and virus from water.
2. Purification of surface water, ground water to make them potable.
3. Removal of pathogens from milk.
4. Ultra filtration finds use in cheese manufacture.
5. It is used for desalting and solvent exchange of proteins.
6. It is used for the recovery of enzymes.
7. It is used in combination with reverse osmosis for pre-treatment of sea water in desalination plants.
8. Used in the treatment of waste water.
9. In medical field the process finds use in blood dialysis and other blood treatments.
10. It is also used in metal industry (oil/ water emulsion separation, paint treatment), filtration of effluents from paper pulp mill, in textile industry, dairy industry (milk, cheese) and food industry (proteins), etc.

Flash Evaporation

In this method sea water is heated to high temperature and then fed into a series of chambers. These chambers are maintained at reduced pressure. Because of sudden drop in pressure, the water gets evaporated very quickly. This is known as flash evaporation. The steam formed is condensed by the incoming sea water passing through coils as shown in Figure 2.15. The concentrated brine then passes to the second chamber which is at still lower pressure. Hence more water evaporates which is also condensed. In the similar manner water passes through a series of 4 to 40 chambers in order of increasingly reduced pressure. As a result more and more water is recovered which is collected and the concentrated brine solution is removed by blow-down process.

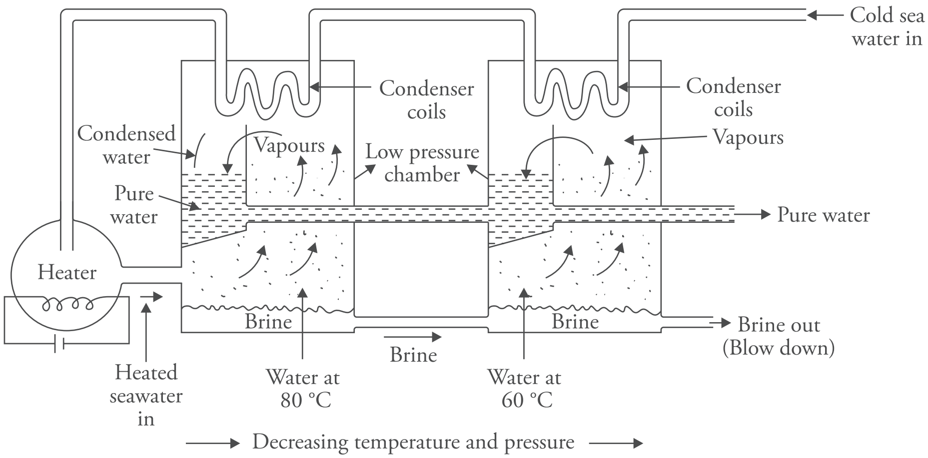


Figure 2.15 *Flash evaporation*

2.13 Defluoridation

Fluorine is an active element found in rocks. Weathering of fluoride bearing minerals from rocks, volcanoes increases the fluoride content in water particularly in ground water. The fluoride concentration in surface water seldom exceeds 0.3 ppm, but depending on the geomorphological conditions the fluoride content in ground water can exceed the acceptable limit of 1.0 ppm leading to chronic health problems in many parts of the country. Within permissible limits fluoride helps to protect teeth enamel and makes it more resistant but excess fluorine in water causes fluorosis which leads to weakening and bending of bones, decay of teeth, etc. Therefore it is mandatory to adopt measures to control fluoride content in potable water. If other options of drinking water are not available then defluoridation (removal of excess fluoride) of water should be adopted. The commonly used defluoridation techniques are as follows:

- Chemical precipitation (Nalgonda technique)
- Adsorption/ion exchange

- Reverse osmosis
- Solar distillation

Fluoride can be removed from water by adsorption on bone charcoal, activated carbon, activated alumina, tamarind gel, etc. Ion exchange method and reverse osmosis have been discussed in the preceding sections. Solar distillation is distillation of water in the presence of sunlight. Here we shall discuss the chemical method to remove fluoride from water.

Nalgonda Technique:

This technique owes its name to the village Nalgonda in Telangana where it was pioneered. This method consists in the removal of fluoride from water by adding chemicals like alum and lime followed by flocculation, sedimentation and filtration. It was first developed by NEERI scientists in India and its first report was published in 1975.

The process involves five steps namely mixing, flocculation, sedimentation, filtration and disinfection. Calculated quantity of alum and lime are added to water and the water is thoroughly mixed so that the chemicals react with the fluoride in water and form suitable precipitates. The chemical reactions involved are as follows:

$$Al_2(SO_4)_3.\ 18H_2O \rightarrow 2Al^{3+} + 3SO_4^{2-} + 18H_2O$$

$$2Al^{3+} + 6H_2O \rightarrow 2Al(OH)_3 + 6H^+$$

$$F^- + Al(OH)_3 \rightarrow Al - F \text{ complex} \downarrow + \text{other products}$$

$$6Ca(OH)_2 + 12H^+ \rightarrow 6Ca^{2+} + 12H_2O$$

Besides this some fluoride is also precipitated with calcium

$$Ca(OH)_2 + 2F^- \rightarrow CaF_2 \downarrow + 2OH^-$$

Flocs containing fluorides, turbidity, bacteria and other impurities are allowed to stand for sometime so that they settle down. Water standing above is decanted carefully and is filtered using sand filters (discussed earlier). The filtered water is collected and disinfected by further addition of bleaching powder and then distributed.

Limitations

Although the Nalgonda technique is being widely used in households and for community defluoridation process it has certain limitations which are as follows:

- The method fails if the fluorine content in water is very high.
- The quantity of alum and lime has to be carefully controlled for proper results.
- Large quantities of sludge is formed which should be disposed off carefully.

2.14 Waste Water Management

Domestic sewage is the waste water from kitchens, bathrooms, lavatories, dishwashers, washing machines and toilets. It generally does not include industrial or agricultural waste water.

Sewage treatment or domestic waste water treatment is the process of removing physical, chemical and biological contaminants from waste water with the objective of producing treated effluent and solid waste (sludge) for discharge or reuse back into the environment.

The main components of sewage treatment process are as follows:

(1) collection, (2) conveyance, (3) treatment and (4) disposal

(1) *Collection of waste water* The waste water from households is transferred to the drain pipes which are connected to the main drain. These pipes are made of either cement or plastic.

(2) *Conveyance or transport of waste water* The domestic waste water from residential or commercial complexes enters the sewer lines laid in the main streets by the municipality. A manhole which is a cemented chamber covered with a cast iron lid is provided for regular inspection and cleaning of the sewer lines. The sewerage network so laid carries the entire sewage of the city to the sewerage treatment plant.

(3) *Treatment of sewage* Sewage treatment include four steps:

  (a) *Pretreatment or preliminary treatment* It involves the removal and size reduction of large objects like stones, twigs, leaves, rags, metals, plastics and other floating impurities. It involves five steps:

    (i) Screening

    (ii) Presedimentation

    (iii) Microstraining

    (iv) Preaeration

    (v) Prechlorination

  (b) *Primary treatment or physico-chemical treatment* It involves the removal of colloidal particles and other settleable impurities.

  (c) *Secondary treatment or biological treatment* It is a biological process to convert organic matter in sewage into bacterial mass followed by removal of the products.

  (d) *Tertiary treatment or advance treatment* Different advanced techniques are employed to remove various nutrients and for disinfecting water.

(a) ***Pretreatment or preliminary treatment*** Preliminary treatment devices are designed to remove weeds, leaves, branches of trees, floating debris as well as gravel, sand, silt (called gritty substances), fats, oils, greases (called FOG), which if not removed can damage or clog the pumps and equipments used in sewage treatment. The steps involved in preliminary treatment process are as follows:

  (i) *Screening* This step involves the removal of large objects by passing water through a series of perforated vertical bars called screens. To ensure complete removal of suspended and free floating impurities water is first passed through coarse screens and then through fine screens. A comminutor is employed to grind the screenings (solid impurities left behind) into small size particles that can be easily handled.

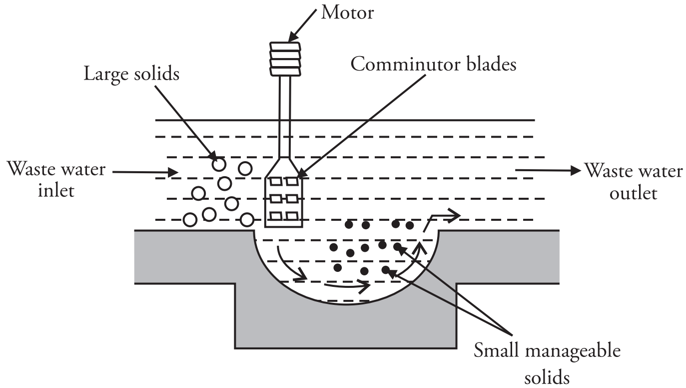


Figure 2.16 *Comminutor*

(ii) *Presedimentation* It is used to remove gravel, sand and silt (grit) from raw water before it enters into the main treatment units, otherwise they can damage the pumps, equipments and other parts of the water treatment plant. Sand traps and cyclone degritter are employed to remove grit from water. In a sand trap the water passes at low velocity so that the gritty substances settle under gravity. In cyclone degritter the grit is removed by centrifugal action.

(iii) *Microstraining* After presedimentation the water is passed through a microstrainer which is a free floating rotating drum consisting of fine stainless steel mesh. By this process small aquatic organisms like algae, fungi and diatoms are removed.

(iv) *Pre-aeration* It is the aeration of waste water before primary treatment. It helps to reduce BOD, assist in the removal of grease and oil in waste water and to freshen up septic water prior to further treatment. For this, waste water is agitated with air in an aerator so that the lighter impurities float and are skimmed off.

(v) *Prechlorination* It is the chlorination of waste water prior to primary treatment. Prechlorination does not disinfect water but prevents further decay of waste water. It is used to control odour, prevent decay of organic matter and reduce the biological oxygen demand of water.

(b) ***Primary treatment*** It involves the removal of colloidal particles and other settleable impurities from the pretreated water. For this the pretreated water is kept in big tanks called primary clarifiers or primary sedimentation tanks where coagulants are added and the faecal solids and colloidal impurities are allowed to settle down. Detention time is typically 3 hours in tanks 10–15 feet deep. Particles too small to be settled out in this tank must be removed by filtration or other methods. Grease and oil rise to the surface and are skimmed off and the sedimented water is decanted.

Primary treatment is capable of removing 80–90% of total dissolved solids, 65% of suspended solids and upto 45% of BOD substances. After this stage a homogenous liquid is obtained which is capable of being treated biologically.

(c) ***Secondary or biological treatment:*** The main objective of this step is to degrade the biological content of sewage derived from human waste, food waste and detergent. Secondary treatment is meant to remove soluble and colloidal organics which remain after primary treatment. It is conceptually similar to the process of decomposition taking place in nature. In this method, organic wastes are degraded microbially in the waste treatment plants. Presence of adequate microbial population and supply of oxygen is required for the effective operation of this process. Two commonly used processes are

- Trickling filter process
- Activated sludge process

Trickling filter process In this process the water to be treated is sprinkled over a bed of rocks covered with a layer of biological growth mass. The rock bed is nearly three meters in depth and is loosely packed to allow circulation of air. As water trickles over the bed the organic matter is consumed and degraded by the microbial growth. As the microorganisms consume the organic matter , the thickness of the slime film increases to a point where it can no longer be supported on the solid media and gets detached from the surface. This process is known as *sloughing*. This sloughed organic matter gets discharged into the effluent. A settling tank which follows the trickling filters removes the detached bacteria and also some suspended solids.

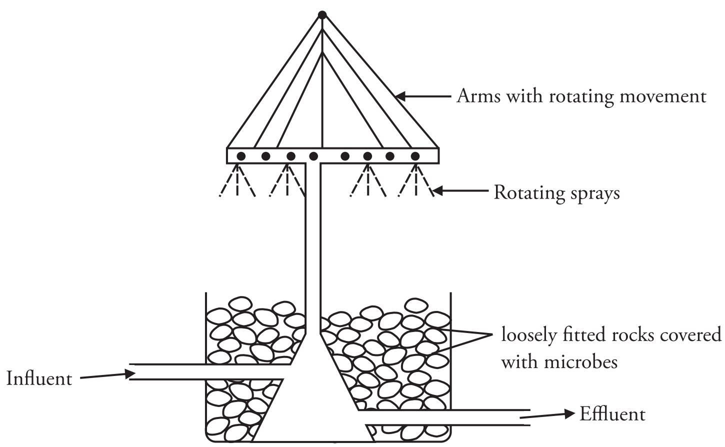


Figure 2.17 *Trickling filter*

Activated sludge process This process is more popular than the trickling filter process. In this method the soluble and unsettleable biodegradable organic compounds are degraded by bacteria in an aeration tank. Waste water after primary treatment enters an aeration tank where the organic matter is brought in contact with recycled sludge (from secondary clarifier), which is heavily laden with bacteria. Air necessary for the survival of bacteria is bubbled in the aeration tank. In the presence of oxygen, these microorganisms feed on the organic matter in waste water and convert it into stabilised low energy contents like sulphates and carbonates.

The microorganisms grow rapidly (because of availability of food/organic matter and air) and form an active mass of microbes and that is why the process is termed as the activated sludge process. This effluent from aeration tank containing the flocculent microbes flows into a settling tank called the secondary clarifier where the sludge settles at the bottom. This sludge containing microbes is called activated sludge or biological floc. It is removed and disposed off after treatment and a small portion is recycled in the aeration tank to act as the seed for the further growth of microbes.

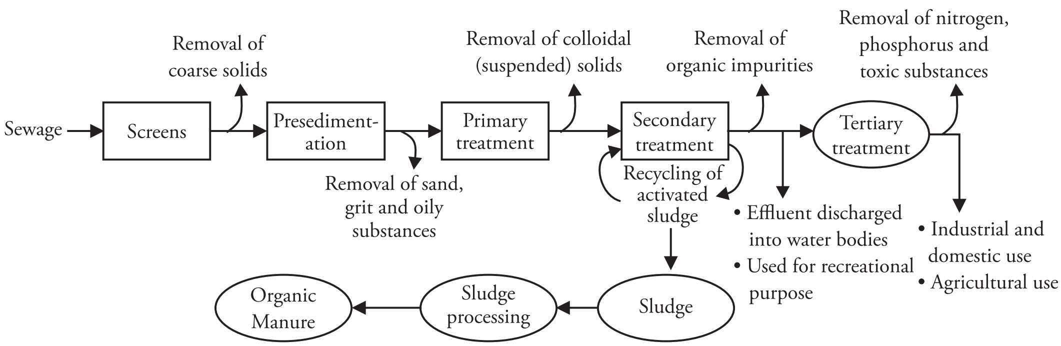


Figure 2.18 *Schematic representation of Sewage/Domestic waste water treatment*

(d) ***Tertiary treatment or advanced treatment*** The primary and secondary treatment is sufficient to meet the waste water effluent standards. If water of very high purity is required then advanced waste water treatment is carried out. The main function of tertiary treatment is to remove the load of nitrogen and phosphorus compounds from the effluents. It also aims at removing toxic substances from water and disinfecting water. It is achieved in a number of ways:

Phosphorus removal It is removed by precipitation. The effluent after secondary treatment is mixed with calcium oxide which reacts with phosphorus compounds in waste water forming calcium phosphate which is removed and used as a fertilizer.

Phosphorus can also be removed by adding polyphosphate accumulating bacteria which feed on phosphorus containing compounds. Later this biomass enriched with phosphorus is separated and used as biofertiliser.

Nitrogen removal A nitrification – denitrification process is used for the removal of nitrogen from waste water. In the first step nitrogen is converted into nitrates with the help of microorganisms and then these nitrates are released into the atmosphere by converting them into nitrogen using denitrifying bacteria.

Toxic substances are removed by special techniques

After this the water is disinfected. (Disinfection of water has been discussed earlier in the preceding sections.)

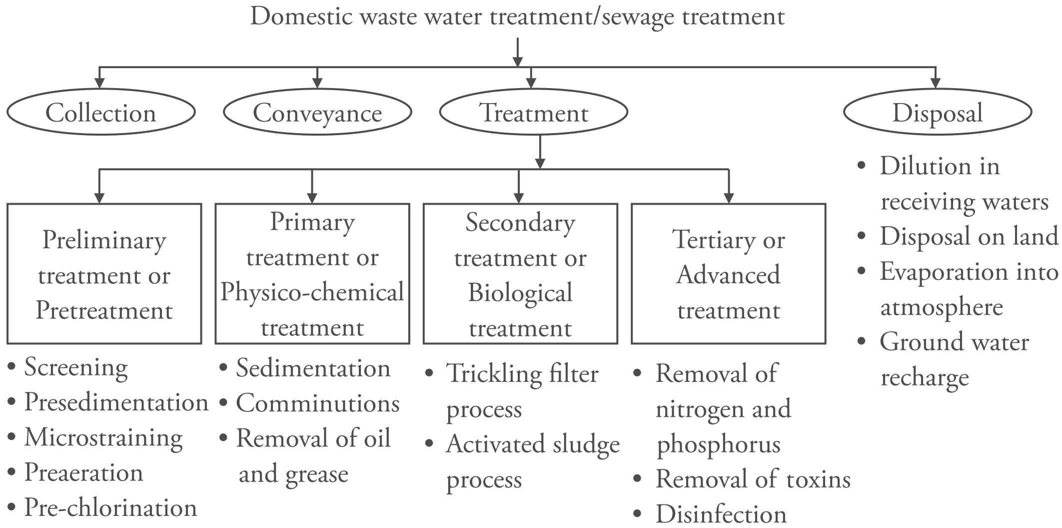


Figure 2.19 *Summary of the Domestic waste water/Sewage treatment process*

2.15 Chemical Analysis of Water

Determination of Hardness

Hardness of water can be determined by the following methods:

1. O. Hehner's method
2. ETDA method
3. Soap titration method

The EDTA method is the most accurate and widely used method for the estimation of hardness in water.

Determination of hardness by O.Hehner's method

(a) Temporary hardness in a water sample is caused by $Ca(HCO_3)_2$ and $Mg(HCO_3)_2$. It is removed on boiling

$$Ca(HCO_3)_2 \xrightarrow{\text{Boiling}} \underset{\substack{\text{Insoluble} \\ \text{ppt}}}{CaCO_3} + H_2O + CO_2$$

$$Mg(HCO_3)_2 \xrightarrow{\text{Boiling}} \underset{\substack{\text{Insoluble} \\ \text{ppt}}}{Mg(OH)_2} + 2CO_2$$

The temporary hardness of the water sample can be determined by titrating the water sample with standard HCl solution before and after boiling.

(b) Permanent hardness caused by chlorides, sulphates and nitrates of calcium and magnesium cannot be removed by boiling. They can however be removed by boiling the water sample with a known excess of standard sodium carbonate. The chloride and sulphates form insoluble carbonates. The residual sodium carbonate is determined by titration with standard HCl solution. The amount of soda used is equivalent to the permanent hardness.

$$CaCl_2 + Na_2CO_3 \rightarrow CaCO_3 + 2\ NaCl$$

$$MgSO_4 + Na_2CO_3 \rightarrow MgCO_3 + Na_2SO_4$$

Procedure

(A) **Determination of temporary hardness** Pipette out 50 mL of the water sample in a conical flask, add 2-3 drops of methyl orange indicator and titrate against N/50 HCl solution. At the end point the color changes to pink and say V_1 mL of HCl is consumed at the end point.

After this take fresh 250 mL of the sample water in 500 mL flask. Gently boil it for half an hour, cool the solution and filter it. Wash the precipitate with distilled water and collect all the washings and filtrate. Now titrate 50 mL of this solution with standard HCl solution using methyl orange as an indicator. Let the volume of HCl consumed be V_2 mL.

Calculation

Volume of N/50 HCl consumed by temporary hardness present in 50 mL of water sample

$= (V_1 - V_2)$ mL

$$\underset{\text{Water sample}}{N_1V_1} = \underset{\text{HCl}}{N_2V_2}$$

$$N_1 \times 50 = N/50 \times (V_1 - V_2)$$

$$N_1 = (V_1 - V_2)/(50 \times 50)$$

Hardness in terms of equivalent of $CaCO_3$

= Normality of water × equivalent weight of $CaCO_3$

$$= \frac{(V_1 - V_2)}{50 \times 50} \times 50 \text{ g/L}$$

$$= \frac{(V_1 - V_2)}{50 \times \cancel{50}} \times \cancel{50} \times 1000 \text{ mg/L}$$

Temporary hardness = $20(V_1 - V_2)$ ppm

(B) **Determination of permanent hardness** Take 250 mL of hard water sample in a conical flask and boil for half an hour to remove temporary hardness. Filter and add 50 mL of N/50 Na_2CO_3 solution. Boil for another 25–30 minutes. All chlorides and sulphates are converted into carbonates. Filter the solution and wash the precipitate several times with hot distilled water, collecting all the washings in the conical flask. Now titrate 50 mL of this solution against standard HCl solution using methyl orange as an indicator.

Let the titre value be V mL

∴ Volume of N/50 Na_2CO_3 solution used for removing permanent hardness in 50 mL water sample = $(50 - V)$ mL

$$\underset{\text{Water sample}}{N_3V_3} = \underset{\text{HCl}}{N_4V_4}$$

$$N_3 \times 50 = N/50 \times (50 - V)$$

$$N_3 = \frac{(50 - V)}{50 \times 50}$$

Hardness in terms of equivalent of $CaCO_3$

= Normality of water × equivalent weight of $CaCO_3$

$$= \frac{(50 - V)}{50 \times 50} \times 50 \text{ g/L} = \frac{(50 - V)}{50 \times 50} \times 50 \times 1000 \text{ mg/L}$$

Permanent hardness = 20 (50-V) ppm.

Solved Examples

1. Calculate the temporary hardness when the following observations were recorded

Observations

Before boiling

Volume of water sample for each titration = 50 mL

Volume of N/50 HCl consumed = 14 mL

After boiling

Volume for each titration = 50 mL

Volume of N/50 HCl consumed = 10 mL.

Solution

Volume of HCl consumed by the temporary hardness present in water = 14 – 10 = 4 mL

Hardness of the water sample $N_1V_1 = N_2V_2$

N_1 = Normality of hardness in water sample; V_1 = volume of water

N_2 = Normality of HCl , V_2 = Volume of HCl consumed

$$N_1 \times 50 = \frac{N}{50} \times 4$$

$$N_1 = \frac{4}{50\times50}$$

Temporary hardness in terms of $CaCO_3$ equivalent in ppm = Normality of water sample × Equivalent weight of $CaCO_3$ × 1000

$$= \frac{4}{50\times50} \times 50 \times 1000$$

$$= 80 \text{ ppm}$$

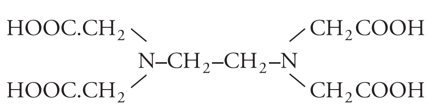
Practice Problems

1. 100 mL of tap water was titrated with N/50 HCl with methyl orange as an indicator.If 28 mL HCl were required, calculate the hardness as parts of $CaCO_3$ per 10,000 parts of water. The hardness is temporary. [Ans 2800 parts per 10000 parts of water]

2. In an experiment to determine the hardness of a sample of water, 25 mL of N/50 Na_2CO_3 solution was added to 100 mL of water sample.After completion of precipitation of the insoluble carbonates the unreacted Na_2CO_3 was titrated against N/50 H_2SO_4 solution when 10 mL of the acid was required. Calculate the degree of hardness and comment on the nature of hardness so determined. [Ans permanent hardness 150 ppm]

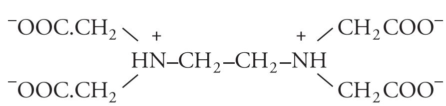
Estimation of hardness by EDTA Method

This is a complexometric titration as ethylene diamine tetra-acetic acid (EDTA) forms a stable complex with Ca^{2+} and Mg^{2+} ions in the pH range 8–10.

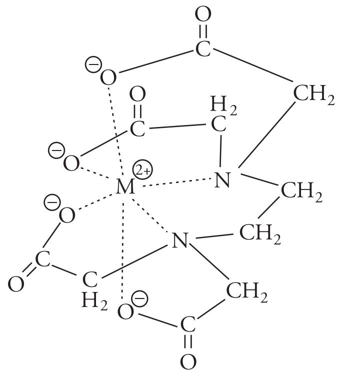


(A)

Its anion is



(B)



(C)

(A) Structure of EDTA

(B) Structure of dianion of EDTA, H_2Y^{2-}

(C) Structure of an M^{2+} -EDTA chelate MY^{2-}

In order to determine hardness, hard water is buffered to a pH value of 10 using NH_4OH-NH_4Cl buffer and a few drops of Eriochrome black-T indicator (EBT) are added. EBT forms a weak unstable wine red complex with metal ions.

$$\underset{\substack{(Ca^{++}\text{ or }Mg^{2+}\\ \text{ions of}\\ \text{hard water)}}}{M^{2+}} + \underset{(\text{Blue})}{\text{EBT}} \xrightarrow{\text{pH}=10} \underset{\substack{\text{Unstable}\\ \text{complex}\\ \text{(wine red)}}}{(\text{M-EBT})\text{ complex}}$$

where M = Ca and Mg ions.

During the course of titration of water sample against EDTA, EDTA combines with free Ca^{2+} or Mg^{2+} ions to give very stable metal–EDTA complex which is colorless.

$$\underset{\text{Wine red}}{(\text{M-EBT})\text{ complex}} + \text{EDTA} \xrightarrow{\text{Titration}} \underset{\substack{\text{Stable complex}\\ \text{(colorless)}}}{(\text{M-EDTA})\text{ complex}} + \underset{(\text{Blue})}{\text{EBT}}$$

Thus at equivalent point, there is a change in color from wine red (due to M-EBT) to blue (due to free EBT).

The various steps involved in this method are as follows:

1. **Preparation of standard hard water** Dissolve 1.0 g of pure, dry $CaCO_3$ in minimum quantity of dilute HCl and evaporate the solution to dryness on a water bath. Dissolve the residue in distilled water to make 1 liter solution. Each mL of this solution contains 1mg of $CaCO_3$ equivalent hardness.
2. **Preparation of EDTA solution**
   0.01 M EDTA is prepared by dissolving 3.7 g of its disodium salt (molar mass = 372.28) in distilled water and making it up to 1 liter.
3. **Preparation of indicator** Dissolve 0.5 g of Eriochrome black – T in 100 mL of alcohol

4. **Preparation of buffer solutions** Add 67.5 g of NH_4Cl to 570 mL of concentrated ammonia solution and then dilute with distilled water to 1 liter.

Procedure

1. **Standardisation of EDTA solution** Fill the burette with EDTA solution after washing and rinsing. Pipette out 50 mL of standard hard water in a conical flask (1 mL of standard hard water = 1 mg of $CaCO_3$ equivalent hardness). Add 10 mL of buffer solution and 2–3 drops of indicator (EBT). Titrate with EDTA solution, till wine red color changes to clear blue. Let the volume of EDTA consumed be V_1 mL.
2. **Determination of total hardness of water** Titrate 50 mL of unknown hard water by the same procedure as done above, let the volume of EDTA consumed this time be V_2 mL.
3. **Determination of permanent hardness of water** Take 250 mL of water sample in a 500 mL beaker and boil it till the volume reduces to about 50 mL. Boiling causes all the bicarbonates to decompose to $CaCO_3$ and $Mg(OH)_2$. Filter and quantitatively collect the filtrate and washings in a 250 mL conical flask and make the volume to 250 mL with distilled water. Titrate 50 mL of this water sample against EDTA as done in step (1). Let the volume of EDTA used be V_3 mL.

CALCULATIONS

Step 1 Standardisation of EDTA solution

V_1 mL of EDTA = 50 mL of standard hard water

(Since each mL of standard hard water contains 1 mg of $CaCO_3$ equivalent hardness)

Therefore, V_1 mL of EDTA = 50 mg of $CaCO_3$ equivalent hardness

$$\text{or } 1 \text{ mL of EDTA} = 50/V_1 \text{ mg of } CaCO_3 \tag{1}$$

Step 2 Determination of total hardness

50 mL of unknown hard water = V_2 mL of EDTA

(Since 1 mL of EDTA has $50/V_1$ mg of $CaCO_3$ hardness – from Eq. (1)

$$= V_2 \times \frac{50}{V_1} \text{ mg of } CaCO_3 \text{ equivalent hardness} \tag{2}$$

The hardness given by Eq. (2) is the hardness in 50 mL water sample ∴ hardness in 1000 mL water sample

$$= V_2 \times \frac{50}{V_1} \times \frac{1}{50} \times 1000$$

$$= 1000 \times \frac{V_2}{V_1} \text{ mg of } CaCO_3 \text{ equivalent hardness}$$

Hence total hardness of water = $1000 \times \frac{V_2}{V_1}$ mg/L or ppm

Step 3 Determination of permanent hardness of water

As 50 mL of boiled water = V_3 mL of EDTA

Again as 1 mL EDTA = $\frac{50}{V_1}$ mg of $CaCO_3$ equivalent hardness [from Eq. (1)]

$\therefore$ V_3 mL of EDTA = $V_3 \times \frac{50}{V_1}$ mg of $CaCO_3$ equivalent hardness (3)

hardness as given by Eq. (3) is the hardness in 50 mL boiled water

$\therefore$ hardness in 1000 mL of boiled water = $V_3 \times \frac{50}{V_1} \times \frac{1}{50} \times 1000 = 1000 \times \frac{V_3}{V_1}$ mg/L

Hence permanent hardness of water = $1000 \times \frac{V_3}{V_1}$ mg/L or ppm

Temporary hardness = Total hardness–Permanent hardness

$$= 1000 \times \frac{V_2}{V_1} - 1000 \times \frac{V_3}{V_1}$$

or temporary hardness = $\frac{1000\ (V_2 - V_3)}{V_1}$ ppm

Volume of EDTA can be equated with $CaCO_3$ equivalent as follows

1000 mL of 1 M EDTA = 100 g of $CaCO_3$

1000 mL of 0.01 M EDTA = 1 g of $CaCO_3$

1 mL of 0.01 M EDTA = 1 mg of $CaCO_3$

1 mL of 0.02 N EDTA = 1 mg of $CaCO_3$

Solved Examples

1. 250 mL of a water sample on EDTA titration with Eriochrome Black –T consumed 13 mL of 0.022 N EDTA till end point is reached. Calculate the hardness of water ?

Solution

1 mL of 0.01 M EDTA = 1 mg of $CaCO_3$ equivalents

13 mL of 0.022 M EDTA = $\frac{1 \times 13 \times 0.022}{1 \times 0.01}$ = 28.6 mg of $CaCO_3$

This amount of hardness is present in 250 mL of water sample.

Hardness present in 1 liter = $\frac{28.6}{250} \times 1000 = 114.4$ ppm

2. 100 mL of a sample of water required 15 mL of 0.01 M EDTA for titration using Eriochrome Black –T as indicator. In another experiment, 100 mL of the same sample was boiled to remove the temporary hardness, the precipitate was removed and the cold solution required 8 mL of 0.01 M EDTA using Eriochrome Black –T as indicator. Calculate the total hardness, permanent hardness and temporary hardness in mg/L of $CaCO_3$ equivalent.

Solution

As we know 1 mL of 0.01 M EDTA = 1 mg of $CaCO_3$ eq. hardness. (1)

Determination of total hardness of water

100 mL of water sample = 15 mL of 0.01 M EDTA = 15 mg of $CaCO_3$ eq. hardness. [From Eq. (1)]

$\therefore$ 1000 mL of water sample = $\frac{15}{100} \times 1000 = 150$ mg/L

Total Hardness = 150 mg/L

Determination of permanent hardness of water

100 mL of boiled hard water = 8 mL of 0.01 M EDTA = 8 mg of $CaCO_3$ eq. hardness

$\therefore$ 1000 mL of water sample = $\frac{8}{100} \times 1000 = 80$ mg/L

Temporary Hardness = Total Hardness – Permanent Hardness

= 150 – 80 = 70 mg/L

3. 0.28 g of $CaCO_3$ was dissolved in HCl and the solution was made to one liter with distilled water. A total of 100 mL of the above solution required 28 mL of EDTA solution on titration.

100 mL of the hard water sample required 35 mL of the same EDTA solution on titration. After boiling 100 mL of this water, cooling, filtering and then titration required 10 mL of EDTA solution. Calculate the temporary and permanent hardness of water.

Solution

Step (i) *Standardisation of EDTA solution*

Given, 1 liter of standard hard water contains 0.28 g $CaCO_3$

or, 1000 mL of standard hard water contains 280 mg $CaCO_3$

or, 1 mL of standard hard water contains 0.28 mg $CaCO_3$ (i)

According to the question

28 mL of EDTA solution ≡ 100 mL of standard hard water

According to eq (i) 1 mL of standard hard water has 0.28 mg of $CaCO_3$

∴ 100 mL of standard hard water has 100 × 0.28 mg of $CaCO_3$

or, 28 mL EDTA solution = 100 × 0.28 = 28 mg of $CaCO_3$

1 mL EDTA solution = $\frac{28}{28}$ = 1 mg of $CaCO_3$

Step (ii) *Determination of total hardness of water*

Given, 100 mL of unknown hard water sample ≡ 35 mL of EDTA ≡ 35×1

= 35 mg of $CaCO_3$ equivalent hardness

∴ 1000 mL (or 1 liter) of unknown hard water sample = $\frac{35}{100} \times 1000$

= 350 mg of $CaCO_3$ equivalent hardness

Hence **Total Hardness = 350 ppm**

Step (iii) *Determination of permanent hardness of water*

Given, 100 mL of boiled water ≡ 10 mL of EDTA ≡ 10×1 = 10 mg of $CaCO_3$ eq. hardness.

∴ 1000 mL (or 1 liter) of unknown boiled water = $\frac{10}{100} \times 1000$ = 100 mg of $CaCO_3$ eq. hardness.

Permanent Hardness = 100 ppm

Step (iv) *Determination of temporary hardness of water*

Temporary Hardness = Total Hardness – Permanent Hardness

350 – 100 = 250 ppm

Temporary Hardness = 250 ppm

4. 25 mL of standard hard water consumes 12 mL of standard EDTA solution. 25 mL of hard water sample consumes 8 mL of EDTA solution. After boiling the sample, 25 mL of the boiled and cooled hard water consumes 6 mL of standard EDTA solution. Calculate the total, permanent and temporary hardness.

Solution

It is assumed that 1 mL of standard hard water has 1 mg of $CaCO_3$ equivalent hardness.

Step (i) *Standardisation of EDTA*

25 mL of standard hard water = 12 mL of EDTA = 25 mg of $CaCO_3$ eq. hardness.

$$1 \text{ mL of EDTA} = \frac{25}{12} \text{ mg of } CaCO_3 \text{ eq. hardness.} \tag{1}$$

Step (ii) *Total hardness*

$$25 \text{ mL of hard water sample} = 8 \text{ mL of EDTA} = 8 \times \frac{25}{12} \text{ mg of } CaCO_3 \text{ eq. hardness.}$$

[From Eq. (1)]

$$1000 \text{ mL of hard water sample} = 8 \times \frac{25}{12} \times \frac{1000}{25} = 666.67 \text{ ppm}$$

Hence **Total Hardness = 666.67 ppm**

Step (iii) *Determination of permanent hardness of water*

$$\text{Given, 25 mL of boiled water} = 6 \text{ mL of EDTA} = \frac{6 \times 25}{12} \text{ mg of } CaCO_3 \text{ eq. hardness}$$

$$\therefore \ 1000 \text{ mL (or 1 liter) of unknown boiled water} = \frac{6 \times 25 \times 1000}{12 \times 25} = 500 \text{ ppm.}$$

Permanent Hardness = 500 ppm

Step (iv) *Determination of temporary hardness of water:*

Temporary Hardness = Total Hardness – Permanent Hardness

666.67 – 500 = 166.67 ppm

Temporary Hardness = 166.67 ppm

5. Calculate the hardness of water sample whose 10 mL required 20 mL of EDTA. 20 mL of $CaCl_2$ solution whose strength is equivalent to 1.5 g of $CaCO_3$ per liter required 30 mL of EDTA solution.

Solution

1 liter of standard hard water = 1.5 g = 1500 mg of $CaCO_3$

or 1 mL of standard hard water = 1.5 mg of $CaCO_3$ (1)

Step (i) *Standardisation of EDTA*

20 mL of standard hard water = 30 mL of EDTA

30 mL of EDTA = 20 × 1.5 mg of $CaCO_3$ equivalent hardness [From Eq (1)]

$$1 \text{ mL of EDTA} = \frac{20\times1.5}{30} \text{ mg of } CaCO_3 \text{ equivalent hardness.} \quad (2)$$

Step (ii) *Total hardness*

10 mL of sample hard water = 20 mL of EDTA

$$= \frac{20\times20\times1.5}{30} \text{ mg of } CaCO_3 \text{ equivalent hardness} \quad \text{[From Eq. (2)]}$$

$$1000 \text{ mL of sample hard water} = \frac{20\times20\times1.5}{30} \times \frac{1000}{10} = 2000 \text{ ppm}$$

Total hardness = 2000 ppm

Practice Problems

1. A standard hard water contains 15 g of $CaCO_3$ per liter. 20 mL of this water sample required 25 mL of EDTA solution, 100 mL of sample water required 18 mL of EDTA solution. The sample after boiling required 12 mL EDTA solution. Calculate total, permanent and temporary hardness of the given water sample in ppm.

   [Ans Total hardness = 2160 ppm; Permanent hardness = 1440 ppm; Temporary hardness = 720 ppm]

2. The EDTA was used to find out permanent and temporary types of hardness in a given hard water sample.Following observations were recorded:
   (i) 22 mL of EDTA was consumed by 50 mL of standard hard water (containing 1 mg of $CaCO_3$ per mL).
   (ii) 50 mL water sample consumed 27 mL EDTA solution.

(iii) 50 mL water sample after boiling, filtering consumed 20 mL of EDTA solution.

[Ans Total hardness = 1227.27 ppm; permanent hardness = 909 ppm; temporary hardness = 318.2 ppm]

3. 1.0 g of $CaCO_3$ was dissolved in hydrochloric acid and made upto 1000 mL. 50 mL of this solution required 46 mL of EDTA solution for titration. 50 mL of hard water sample required 20 mL of the same EDTA solution. After boiling the sample of hard water, 50 mL of it consumed 12 mL of EDTA solution. Calculate the carbonate and non-carbonate hardness of the water sample.

[Ans Total hardness = 434.78 ppm; permanent hardness (non-carbonate hardness) = 260.86 ppm; temporary hardness (carbonate hardness)= 173.92 ppm]

4. 0.25 g of $CaCO_3$ was dissolved in HCl and diluted to 250 mL.100 mL of this solution required 20 mL of EDTA solution for titration. 100 mL of a hard water sample required 30 mL of the same EDTA solution for titration. 100 mL of the sample water on boiling, cooling and filtering required 10 mL of EDTA. Calculate the total, permanent and temporary hardness of the water sample.

[Ans Total hardness = 1500 ppm; permanent hardness = 500 ppm; temporary hardness = 1000 ppm]

Soap titration method or Clarkes's Method

It is based on the principle that when soap solution is added to hard water, the hardness causing ions (Ca^{2+}, Mg^{2+} and other heavy metal ions) react with it forming insoluble precipitate of calcium and magnesium. Formation of insoluble precipitate continues till all the hardness causing ions have precipitated. After this the soap gives lather.

$$\underset{\substack{\text{Soap} \\ \text{(Sodium stearate)}}}{2C_{17}H_{35}COONa} + \underset{\substack{Mg^{2+} \\ \text{or } Ca^{2+}}}{M^{2+}} \rightarrow \underset{\text{Insoluble ppt}}{(C_{17}H_{35}COO)_2M} + 2Na^{+}$$

Also, the addition of soap solution to water decreases its surface tension. Even in distilled water, formation of lather takes place only after the lower critical value of surface tension is reached. The volume of soap solution required to form lather with distilled water (of zero hardness) is known as lather factor.

This volume (corresponding to lather factor) is to be subtracted from all titre values. Both total and permanent hardness in a given water sample can be estimated by this method. Temporary hardness can be calculated by subtracting the two.

Procedure

(1) ***Preparation of standard hard water*** Dissolve 1.0 g of pure, dry $CaCO_3$ in minimum quantity of dilute HCl. Boil to dryness to expel excess of acid and CO_2. Dissolve the residue in distilled water to make 1 liter solution. The hardness of this solution is 1 g/ liter or 1000 ppm or 1 mg/mL.

(2) ***Preparation of soap solution*** Dissolve 100 g of pure dry caustic soap in 800 mL of alcohol plus 200 mL of distilled water.

(3) ***Standardisation of soap solution*** The burette is rinsed and filled with soap solution. In a 250 mL glass-stoppered bottle, 50 mL of standard hard water is taken (the hardness of standard hard water is 1 mg/mL). Soap solution is added 0.2 mL at a time. After each addition, the contents of stoppered bottle are vigorously shaken until lather formation starts. Now, soap solution is added at the rate of 0.1 mL at a time till lather formed after vigorous shaking persist for 2 minutes. Let the volume of soap solution used be V_1 mL.

(4) ***Lather factor of distilled water*** 50 mL of soap solution is titrated against distilled water, till lather persists for two minutes. Let the volume used be V mL. This volume is to be subtracted from all titre values.

(5) ***Determination of total hardness of water*** The above procedure is repeated, taking 50 mL of water sample. Let the titre value of soap be V_2 mL.

(6) ***Determination of permanent hardness of water*** 250 mL of water sample is taken in a beaker. It is boiled and the volume is reduced to 50 mL (boiling causes all bicarbonates to decompose into insoluble $CaCO_3$ and $Mg(OH)_2$). The precipitate is filtered and washed with distilled water. The filtrate and washings are collected in a 250 mL conical flask. The volume is then made to 250 mL by adding distilled water. 50 mL of this water sample is then taken and titrated against soap solution by the procedure as done above. Let the titre value of soap solution be V_3 mL.

Calculation

(i) *Standardisation of soap solution*

50 mL of standard hard water = $(V_1 - V)$ mL of soap solution

As standard hard water has 1 g/liter or 1 mg/mL $CaCO_3$ so 50 mL standard hard water contains $50 \times 1 = 50$ mg of $CaCO_3$

Thus V_1–V mL of soap solution = 50 mg of $CaCO_3$ equivalent hardness

$$\text{or 1 mL of soap solution} = \frac{50}{V_1 - V} \text{ mg of } CaCO_3 \text{ equivalent hardness} \qquad (1)$$

(ii) *Determination of total hardness of water*

50 mL of hard water = $(V_2 - V)$ mL of soap solution

$$\because \text{ 1 mL of soap solution has } \frac{50}{V_1 - V} \text{ mg of } CaCO_3 \text{ equivalent hardness from Eq. (1)}$$

Hence $(V_2$–$V)$ mL of soap solution has

$$= (V_2 - V) \times \frac{50}{V_1 - V} \text{ mg of } CaCO_3 \text{ equivalent hardness (2)}$$

$$\text{Hardness in 1000 mL of hard `water} = (V_2 - V) \times \frac{50}{V_1 - V} \times \frac{1000}{50}$$

$$\text{Total hardness} = 1000 \times \frac{V_2 - V}{V_1 - V} \text{ ppm}$$

(iii) *Determination of permanent hardness in wat*er

50 mL of boiled water = V_3–V mL of soap solution

$$= (V_3 - V) \times \frac{50}{V_1 - V} \text{ mg of } CaCO_3 \text{ equivalent hardness} \qquad \text{[from Eq. (1)]}$$

$$\text{Hardness in 1000 mL of boiled water} = (V_3 - V) \times \frac{50}{V_1 - V} \times \frac{1000}{50}$$

$$= 1000 \times \frac{V_3 - V}{V_1 - V} \text{ mg/L of } CaCO_3 \text{ equivalent hardness}$$

$$\text{Permanent hardness} = 1000 \times \frac{V_3 - V}{V_1 - V} \text{ ppm}$$

(iv) *Determination of temporary hardness*

Temporary hardness = Total hardness – Permanent hardness

$$= 1000 \times \frac{V_2 - V}{V_1 - V} - 1000 \times \frac{V_3 - V}{V_1 - V} \text{ ppm}$$

$$= \frac{1000}{V_1 - V} [(V_2 - V) - (V_3 - V)]$$

$$= \frac{1000}{V_1 - V} (V_2 - V_3) \text{ ppm}$$

$$\text{Temporary hardness} = \frac{1000}{V_1 - V} (V_2 - V_3) \text{ ppm}$$

Solved Examples

1. Estimate different types of hardness present in hard water sample from the following data obtained in soap titration method, when 100 mL of water is titrated with soap solution:
   (i) Standard hard water (400 mg/L of $CaCO_3$) = 36.6 mL of soap solution.
   (ii) Total hardness = 18.6 mL of soap solution.
   (iii) Permanent hardness = 6.2 mL of soap solution.
   (iv) Lather factor = 0.6 mL of soap solution.

Solution

Step (i) *Standardisation of soap solution*

Given, 1 liter of standard hard water (SHW) = 400 mg of $CaCO_3$ equivalent hardness

$\therefore$ 100 mL of SHW = 40 mg of $CaCO_3$ equivalent hardness

Volume of soap solution required to precipitate 100 mL of hard water = 36.6 – 0.6 = 36.0 mL

36.0 mL of soap solution = 40 mg of $CaCO_3$ equivalent hardness.

$$1 \text{ mL of soap solution} = \frac{40}{36} \text{ mg of } CaCO_3 \text{ equivalent hardness} \quad (1)$$

Step (ii) *Determination of total hardness of water*

100 mL of hard water = 18.6 – 0.6 = 18 mL soap solution.

$$100 \text{ mL of hard water} = 18 \times \frac{40}{36} = 20 \text{ mg of } CaCO_3 \text{ equivalent hardness}$$

$$\therefore\ 1 \text{ liter of hard water} = \frac{20}{100} \times 1000 = 200 \text{ mg/L}$$

Thus, ***total hardness = 200 ppm***

Step (iii) *Determination of permanent hardness of water*

100 mL of boiled water = 6.2 – 0.6 = 5.6 mL of soap solution

$$100 \text{ mL of boiled water} = 5.6 \times \frac{40}{36} = 6.22 \text{ mg of } CaCO_3 \text{ equivalent hardness}$$

$$\therefore\ 1 \text{ liter of boiled water} = \frac{6.22}{100} \times 1000 = 62.2 \text{ mg/L}$$

Permanent hardness = 62.2 ppm

Step (iv) *Determination of temporary hardness*

Temporary hardness = Total hardness – Permanent hardness

200– 62.2 = 137.8 mg/L = 137.8 ppm

Temporary hardness = 137.8 ppm

Practice problems

1. A standard hard water was prepared by dissolving 0.2 g of pure and dried $CaCO_3$ in one liter of distilled water. The soap solution was consumed for each titration against 50 mL of the water sample. The following observations were obtained
   (i) Volume with standard hard water = 20.5 mL.
   (ii) Volume with hard water sample = 9.0 mL.
   (iii) Volume for permanent hardness = 3.0 mL.
   (iv) Lather factor = 1.0 mL.
   Find out each type of hardness in ppm.
   [Ans Total hardness = 82 ppm; permanent hardness = 20.5 ppm; temporary hardness = 61.5 ppm]
2. A standard hard water solution was prepared by 250 mg of $CaCO_3$ in 1000 mL of distilled water. And 17.2 mL of soap solution was consumed, when titrated with 50 mL of standard hard water. Lather factor for 50 mL of distilled water is 0.6 mL. The 50 mL of given sample of hard water consumed 7.6 mL of soap solution. 6.8 mL of soap solution was consumed when titrated with 50 mL of boiled filtered hard water sample. Find out total, permanent and temporary hardness.
   [Ans Total hardness = 105.42 ppm; permanent hardness = 93.373 ppm; temporary hardness = 12.047 ppm]

Alkalinity

Alkalinity of water is a measure of total content of those substances which increase the hydroxide ions concentration (OH^-) upon dissociation or due to hydrolysis.

Alkalinity of water is attributed to the presence of

(i) Caustic alkalinity (due to OH^- and CO_3^{2-} ions) and

(ii) Temporary hardness (due to HCO_3^- ions)

These can be estimated separately by titration against standard acid, using phenolphthalein and methyl orange as indicator.

The determination is based on the following reactions

$$\left.\begin{array}{l} \left.\begin{array}{ll}\text{(i)} & OH^- + H^+ \rightarrow H_2O \\ \text{(ii)} & CO_3^{2-} + H^+ \rightarrow HCO_3^- \end{array}\right\} P \\ \begin{array}{ll}\text{(iii)} & HCO_3^- + H^+ \rightarrow H_2O + CO_2\end{array} \end{array}\right\} M$$

The titration of the water sample against a standard acid up to phenolphthalein end point (P) marks the completion of reaction (i) and (ii) only. This amount of acid used thus corresponds to hydroxide plus one half of the normal carbonate present.

On the other hand, titration of the water sample against a standard acid to methyl orange end point (M) marks the completion of reaction (i), (ii) and (iii). Hence the total amount of acid used represents the total alkalinity (due to hydroxide, bicarbonate and carbonate ions)

Thus,

$$P = OH^- \text{ and } \tfrac{1}{2}\, CO_3^{2-}$$

$$\text{and } M = OH^-, CO_3^{2-}, HCO_3^-$$

Alkalinity in water may be due to the presence of the following combinations:

(i) OH^- only

(ii) CO_3^{2-} only

(iii) HCO_3^- only

(iv) OH^- and CO_3^{2-} together

(v) CO_3^{2-} and HCO_3^- together

The possibility of OH^- and HCO_3^- together is ruled out, because of the fact that they combine instantaneously to form CO_3^{2-} ions

$$OH^- + HCO_3^- \rightarrow CO_3^{2-} + H_2O$$

That is, $NaOH + NaHCO_3 \rightarrow Na_2CO_3 + H_2O$

Thus, OH^- and HCO_3^- ions cannot exist together in water

On similar analogy, OH^-, CO_3^{2-} and HCO_3^- ions also cannot exist together.

Procedure Pipette out 100 mL of the water sample in a clean titration flask. Add 2–3 drops of phenolphthalein indicator. Run in N/50 H_2SO_4 (from a burette), till the pink color is just discharged. This is the first end point. Let the volume of acid used until phenolphthalein end point = V_1 mL. Then to the same solution, add 2 to 3 drops of methyl orange. Continue titration, till the pink color reappears. This is the second end point. Let *extra volume* of acid used to methyl orange end point = V_2 mL.

Calculations

100 mL of water up to phenolphthalein end point = V_1 mL of N/50 H_2SO_4

$$\therefore\ \underset{\text{Water}}{100 \text{ mL} \times N_p} = \underset{\text{Acid}}{V_1 \text{ mL} \times N/50}$$

$$N_p = \frac{V_1 \text{ mL}}{100 \text{ mL}} \times \frac{N}{50} = \frac{V_1}{5000} \text{ N}$$

∴ Strength of alkalinity up to phenolphthalein end point in terms of $CaCO_3$ equivalent – Strength = Normality × Eq. weight of $CaCO_3$

$$= \frac{V_1}{5000} \times 50 \text{ g/L}$$

$$= \frac{V_1}{5000} \times 50 \times 1000 \text{ mg/L}$$

P = 10 V_1 ppm

Now 100 mL of water up to methyl orange end point

$= (V_1+V_2)$ mL of N/50 H_2SO_4

∴ 100 $N_m = (V_1+V_2)$ mL × N/50

$$\text{or Normality } N_m = \frac{(V_1+V_2) \text{ mL}}{100 \text{ mL}} \times \frac{N}{50} = \frac{(V_1+V_2)}{5000} \text{ N}$$

∴ Strength of total alkalinity in terms of $CaCO_3$ equivalent

$$= \frac{(V_1+V_2)}{5000} \times 50 \text{ g/L} = \frac{(V_1+V_2)}{5000} \times 50 \times 1000 \text{ mg/L}$$

M = 10(V_1+V_2) ppm

(i) When P = 0 both OH^- and CO_3^{2-} are absent and alkalinity in that case is due to HCO_3^- alone.

(ii) When P = M neither CO_3^{2-} nor HCO_3^- ions are present, only OH^- ions are present, Thus alkalinity due to OH^- = P = M.

(iii) When P = 1/2 M or $V_1 = V_2$ only CO_3^{2-} is present, since half of carbonate neutralisation (ie $CO_3^{2-} + H^+ \rightarrow HCO_3^-$) takes place with phenolphthalein, while complete carbonate neutralisation ($CO_3^{2-} + H^+ \rightarrow HCO_3^-$; $HCO_3^- + H^+ \rightarrow H_2O + CO_2$) occurs when methyl orange indicator is used. Thus alkalinity due to CO_3^{2-} = 2P.

(iv) When P>1/2 M or $V_1>V_2$. In this case, besides CO_3^{2-}, OH^- ions are also present. Now half of CO_3^{2-} (i.e., $HCO_3^-+H^+ \rightarrow H_2O + CO_2$) is equal to (M–P), so alkalinity due to complete CO_3^{2-} = 2(M–P) and alkalinity due to OH^- = M–2(M–P) = (2P–M).

(v) When P<1/2 M or $V_1<V_2$, in this case, besides CO_3^{2-}, HCO_3^- ions are also present, now alkalinity due to CO_3^{2-} = 2P and alkalinity due to HCO_3^- = (M–2P).

[Note If alkalinity is to be expressed in terms of individual components, then their chemical equivalent should be used instead of 50(for $CaCO_3$)]

Table 2.10 Calculation of alkalinity of water

S.No	Result of titration	OH^- (ppm)	CO_3^{2-} (ppm)	HCO_3^- (ppm)
1	P=0	Nil	Nil	M
2	P=M	P or M	Nil	Nil
3	P=1/2M or $V_1 = V_2$	Nil	2P	Nil
4	P>1/2M or $V_1>V_2$	2P-M	2(M-P)	Nil
5	P< 1/2M or $V_1<V_2$	Nil	2P	M-2P

Solved Examples

1. 50 mL of a sample of water required 5 mL of N/50 H_2SO_4 using methyl orange as indicator but did not give any coloration with phenolphthalein. What type of alkalinity is present? Express the same in ppm.

Solution

As the water sample does not give any coloration with phenolphthalein (P = 0), hence only HCO_3^- ions are present.

Now, 50 mL of water sample upto methyl orange end point = 5 mL of N/50 H_2SO_4

$\therefore$ 50 mL × N_M = 5 mL × N/50

or Normality = 5 mL × N/50 × 1/50 = $\frac{1}{500}$ N

Strength of alkalinity in terms of $CaCO_3$ equivalents

$$= N_M \times \text{Equivalent Weight of } CaCO_3 = \frac{1}{500} \times 50 \text{ g/L} = \frac{1}{500} \times 50 \times 1000 \text{ mg/L} = 100 \text{ mg/L}$$

Alkalinity due to HCO_3^- ions = 100 ppm.

2. 200 mL of water sample on titration with N/50 H_2SO_4 using phenolphthalein as indicator gave the end point when 10 mL of acid were run down. Another lot of 200 mL of the sample also required 10 mL of the acid to obtain methyl orange end point. What type of alkalinity is present in the sample and what is its magnitude?

Solution

200 mL of water upto phenolphthalein end point ≡ 10 mL of N/50 H_2SO_4

$\therefore$ 200 × N_p = 10 × N/50

$$N_p = \frac{10}{200} \times \frac{N}{50}$$

Strength of phenolphthalein alkalinity in terms of $CaCO_3$ equivalents

Strength = N_p × Equivalent Weight of $CaCO_3$

$$\text{Strength} = \frac{10}{200} \times \frac{1}{50} \times 50 \text{ g/L} = \frac{10}{200} \times \frac{1}{50} \times 50 \times 1000 \text{ mg/L} = 50 \text{ mg/L}$$

P = 50 ppm

Similarly, M = 50 ppm

Since P=M, hence only OH^- ions are present which are causing alkalinity. Hence, alkalinity due to OH^- = 50 ppm.

3. 500 mL of a water sample on titration with N/50 H_2SO_4 gave a titre value of 29 mL to phenolphthalein end point and another 500 mL sample on titration with same acid gave a titre value of 58 mL to methyl orange end point. Calculate the alkalinity of the water sample in terms of $CaCO_3$ and comment on the type of alkalinity present.

Solution

500 mL of water upto phenolphthalein end point ≡ 29 mL of N/50 H_2SO_4

$$\therefore\ 500 \times N_p = 29 \times N/50$$

$$N_p = \frac{29}{50 \times 500}$$

Strength of phenolphthalein alkalinity in terms of $CaCO_3$ equivalents

Strength = N_p × Equivalent Weight of $CaCO_3$

$$\text{Strength} = \frac{29}{500} \times \frac{1}{50} \times 50 \text{ g/L} = \frac{29}{500} \times \frac{1}{50} \times 50 \times 1000 \text{ mg/L} = 58 \text{ mg/L}$$

P = 58 ppm

Given 500 mL of water upto methyl orange end point ≡ 58 mL of N/50 H_2SO_4

$$\therefore\ 500 \times N_M = 58 \times N/50$$

$$N_M = \frac{58}{500} \times \frac{N}{50}$$

Strength = N_M × Equivalent Weight of $CaCO_3$

$$\text{Strength} = \frac{58}{500} \times \frac{1}{50} \times 50 \text{ g/l} = \frac{58}{500} \times \frac{1}{50} \times 50 \times 1000 \text{ mg/L} = 116 \text{ mg/L}$$

M = 116 ppm

Since $P = \frac{1}{2}$ M, only CO_3^{2-} alkalinity is present in the water sample.

Hence,

Alkalinity in the water sample due to CO_3^{2-} = 2P = M = 116 ppm

4. A sample of water was alkaline to both phenolphthalein and methyl orange. 100 mL of this water sample required 30 mL of N/50 H_2SO_4 for phenolphthalein end point and another 20 mL for complete neutralisation. Determine the type and extent of alkalinity present.

Solution

100 mL of water upto phenolphthalein end point ≡ 30 mL of N/50 H_2SO_4

$$\therefore\ 100 \times N_p = 30 \times N/50$$

$$N_p = \frac{30}{100} \times \frac{N}{50}$$

Strength of alkalinity upto phenolphthalein end point in terms of $CaCO_3$

Strength = $N_p \times$ Equivalent weight of $CaCO_3$

$$\text{Strength} = \frac{30}{100} \times \frac{1}{50} \times 50 \text{ g/L} = \frac{30}{100} \times \frac{1}{50} \times 50 \times 1000 \text{ mg/L} = 300 \text{ mg/L}$$

P = 300 ppm

Given 100 mL of water upto methyl orange end point ≡ 30+20 = 50 mL of N/50 H_2SO_4

$$\therefore\ 100 \times N_M = 50 \times N/50$$

$$N_M = \frac{50}{500} \times \frac{1}{50}$$

Strength = $N_M \times$ Equivalent Weight of $CaCO_3$

$$\text{Strength} = \frac{50}{100} \times \frac{1}{50} \times 50 \text{ g/L} = \frac{50}{100} \times \frac{1}{50} \times 50 \times 1000 \text{ mg/L} = 500 \text{ mg/L}$$

M = 500 ppm

Since $P > \frac{1}{2}$ M therefore both OH^- and CO_3^{2-} are present.

Alkalinity due to OH^- = 2P –M = 2 × 300 – 500 = 100 ppm

Alkalinity due to CO_3^{2-} = 2(M-P) = 2 (500 – 300) = 400 ppm

∴ the water sample contains

OH^- alkalinity = 100 ppm

CO_3^{2-} alkalinity = 400 ppm

5. A water sample is alkaline to both phenolphthalein as well as methyl orange. 200 mL of a this water sample on titration with N/50 HCl required 9.4 mL of the acid to phenolphthalein end point. When a few drops of methyl orange are added to the same solution and titration is further continued, the yellow color of the solution just turned red after addition of another 21 mL of the acid solution. Elucidate on the type and extent of alkalinity present in the water sample.

Solution

200 mL of water upto phenolphthalein end point ≡ 9.4 mL of N/50 HCl

$\therefore\ 200 \times N_p = 9.4 \times N/50$

$$N_p = \frac{9.4}{200} \times \frac{N}{50}$$

Strength of alkalinity upto phenolphthalein end point in terms of $CaCO_3$

Strength = $N_p \times$ Equivalent Weight of $CaCO_3$

$$\text{Strength} = \frac{9.4}{200} \times \frac{1}{50} \times 50 \text{ g/L} = \frac{9.4}{200} \times \frac{1}{50} \times 50 \times 1000 \text{ mg/L} = 47 \text{ mg/L}$$

P = 47 ppm.

As 200 mL of water upto methyl orange end point ≡ 9.4 + 21 = 30.4 mL of N/50 HCl

$\therefore\ 200 \times N_M = 30.4 \times N/50$

$$N_M = \frac{30.4}{200} \times \frac{1}{50}$$

Strength = $N_M \times$ Equivalent Weight of $CaCO_3$

$$\text{Strength} = \frac{30.4}{200} \times \frac{1}{50} \times 50 \text{ g/L} = \frac{30.4}{200} \times \frac{1}{50} \times 50 \times 1000 \text{ mg/L} = 152 \text{ mg/L}$$

M = 152 ppm

Since $P < \frac{1}{2}M$, hence CO_3^{2-} and HCO_3^- are present.

Alkalinity due to $CO_3^{2-} = 2P = 2 \times 47 = 94$ ppm

Alkalinity due to $HCO_3^- = M - 2P = 152 - 94 = 58$ ppm

$\therefore$ the water sample contains

HCO_3^- alkalinity = 58 ppm

CO_3^{2-} alkalinity = 94 ppm

Practice problems

1. A water sample is alkaline to both phenolphthalein as well as methyl orange. 100 mL of this water sample required 7.5 mL of N/50 HCl for neutralisation to phenolphthalein end point (P). At this stage a few drops of methyl orange were added. The acid required further was 14.5 mL of N/50 HCl for neutralisation to methyl orange end point(M). Calculate alkalinity of water as $CaCO_3$ due to the presence of carbonate and bicarbonate.

[Ans Alkalinity due to CO_3^{2-} = 150 ppm
Alkalinity due to HCO_3^{-} = 70 ppm]

2. 100 mL of a raw water sample on titration with N/50 H_2SO_4 required 12.4 mL of the acid to phenolphthalein end point and 15.2 mL of the acid to methyl orange end point. Determine the type and extent of alkalinity present in the water sample.

[Ans Alkalinity due to OH^- = 96 ppm
Alkalinity due to CO_3^{2-} = 56 ppm]

3. 100 mL of a raw water sample required 25 mL of N/50 H_2SO_4 for neutralisation to phenolphthalein end point. After this methyl orange indicator was added to it and further acid required was 3.0 mL. Calculate the alkalinity of water as $CaCO_3$ in parts per million.

[Ans Alkalinity due to OH^- = 220 ppm
Alkalinity due to CO_3^{2-} = 60 ppm; Total alkalinity = 280 ppm]

4. A water sample was alkaline to both phenolphthalein and methyl orange. 50 mL of the water sample required 18 mL of 0.02 N H_2SO_4 for phenolphthalein end point and another 6 mL for complete neutralisation. Describe the type and amount of alkalinity present.

[Ans Alkalinity due to OH^- = 240 ppm
Alkalinity due to CO_3^{2-} = 240 ppm; Total alkalinity = 480 ppm]

Estimation of Free Chlorine

Chlorine is widely used for the disinfection of water. Chlorination is generally done with the help of bleaching powder or chlorine gas or with chloramines.

Excess of free chlorine present in water, makes the water unfit for drinking purposes as it not only imparts unpleasant taste to water but also adversely affects human metabolism. Hence the amount of free chlorine is estimated in municipal water before supplying it for domestic use.

Principle The estimation of free chlorine is based on the oxidation of KI by free chlorine i.e by iodometric titration.

Thus when water sample containing free chlorine is treated with potassium iodide solution, the free chlorine present in water oxidises potassium iodide and liberates an equivalent amount of iodine.

The liberated iodine is titrated against hypo solution ($Na_2S_2O_3$) using starch as indicator.

$$2KI + Cl_2 \rightarrow 2KCl + I_2$$

$$\underset{}{I_2} + \underset{\text{Sodium thiosulphate}}{2Na_2S_2O_3} \rightarrow \underset{\text{Sodium tetrathionate}}{Na_2S_4O_6} + 2NaI$$

$$I_2 + \text{Starch} \rightarrow \text{Deep blue complex}$$

The end point is the disappearance of the blue color.

Procedure A known volume of sample water is pipetted out in a titration flask (say 50 mL). 10 mL of KI solution is added to it. The flask is then kept in dark for 5 minutes so that all the I_2 is liberated. The solution is then titrated against N/50 sodium thiosulphate solution using starch as the indicator. At the end point the color changes from deep blue to colorless. Let the volume of hypo used for titration be V mL. Then

$$\underset{\text{Water sample}}{N_1V_1} = \underset{\text{Hypo}}{N_2V_2}$$

$$N_1 \times 50 = N/50 \times V$$

$$\text{Normality of free chlorine} = \frac{N}{50 \times 50} \times V$$

Strength of free chlorine = Normality of free chlorine × equivalent weight of chlorine

$$= \frac{V}{2500} \times 35.5 \text{ g/L}$$

$$= \frac{V}{2500} \times 35.5 \times 1000 \text{ mg/L} = 14.2 \text{ V ppm}$$

Dissolved Chlorides

Chlorides are generally present in water as NaCl, $MgCl_2$ and $CaCl_2$. Concentration of chlorides above 250 ppm imparts a bad taste to water making it unfit for drinking purposes.

Principle Chloride ions can be determined by titrating the water sample against standard solution of $AgNO_3$ using potassium dichromate as indicator (Argentometric titration)

When $AgNO_3$ is added to the water sample in the presence of K_2CrO_4, the chlorides present in it are precipitated as AgCl. After the precipitation of all chloride ions, the extra drop of $AgNO_3$ gives a red precipitate of silver chromate. This marks the end point.

$$AgNO_3 + NaCl \rightarrow AgCl \downarrow + NaNO_3$$

$$AgNO_3 + K_2CrO_4 \rightarrow \underset{\text{Red}}{AgCrO_4} \downarrow + 2KNO_3$$

Procedure 50 mL of water sample is taken in a conical flask. 3-4 drops of freshly prepared K_2CrO_4 is added. The solution is titrated against N/50 $AgNO_3$ solution. Appearance of brick red color indicates the end point.

Let the volume of $AgNO_3$ used be V mL

Water Sample		$AgNO_3$ solution
N_1V_1	=	N_2V_2

$$N_{\text{chloride}} \times 50 = \frac{N}{50} \times V$$

$$N_{\text{chloride}} = \frac{V}{50 \times 50}$$

$$\text{Strength of chloride ions} = \frac{V}{2500} \times 35.5 \text{ g/L}$$

$$= \frac{V}{2500} \times 35.5 \times 1000 \text{ mg/L}$$

$$= 14.2 \text{ V ppm}$$

Dissolved oxygen

Oxygen dissolved in water is required by living organisms to carry out their biological processes. Dissolved oxygen is determined by the Winkler's method or iodometric titration.

Principle It is based on the fact that oxygen dissolved in water oxidises potassium iodide and an equivalent amount of iodine is liberated. The liberated iodine is titrated against standard hypo solution using starch as indicator. However since oxygen is in molecular state in water it is not capable of reacting with KI, so an oxygen carrier such as manganese hydroxide is used. $Mn(OH)_2$ is produced by the action of manganous sulphate ($MnSO_4$) with KOH.

$$2KOH + MnSO_4 \rightarrow Mn(OH)_2 + K_2SO_4$$

$$\underset{\text{(Oxygen carrier)}}{2Mn(OH)_2} + O_2 \rightarrow \underset{\text{(Basic manganic oxide)}}{2MnO(OH)_2}$$

$$MnO(OH)_2 + H_2SO_4 \rightarrow MnSO_4 + 2H_2O + [O] \text{ (nascent oxygen)}$$

$$2KI + H_2SO_4 + [O] \rightarrow K_2SO_4 + H_2O + I_2$$

The iodine liberated in equivalent amount is titrated against hypo solution

$$2Na_2S_2O_3 + I_2 \rightarrow 2NaI + Na_2S_4O_6$$

$$\text{Starch} + I_2 \rightarrow \text{Starch- iodine (blue)}$$

Sometimes water contains substances like nitrates and sulphites which also react with KI to liberate iodine; hence the presence of these ions gives wrong results. Therefore it is important to eliminate these ions.

Sodium azide (NaN_3) is added to decompose the nitrate or sulphate present in water.

$$2NaN_3 + H_2SO_4 \rightarrow 2HN_3 + Na_2SO_4$$

$$HNO_2 + HN_3 \rightarrow N_2O \uparrow + N_2 \uparrow + H_2O$$

Reagents

(i) **Manganous sulphate solution** 400 g of $MnSO_4$ is dissolved in 1 L of distilled water.

(ii) **Alkaline azide solution** 150 g KI + 500 g NaOH and 20 g of NaN_3 in one liter of distilled water.

(iii) **Sodium thiosulphate solution (N/100)** 2.48 g of $Na_2S_2O_3$ in 1 L of distilled water.

Procedure A known amount of water sample (250 mL) is taken in a stoppered bottle avoiding contact with air. 2 mL of $MnSO_4$ is added with the help of a pipette, dipping the end well below the water surface. Similarly 2.0 mL of alkaline azide solution is also added. The bottle is stoppered and shaken thoroughly. The brown precipitate so formed is dissolved in minimum quantity of concentrated H_2SO_4. The bottle is shaken well to dissolve the precipitate. 100 mL of this solution is then titrated against hypo solution using freshly prepared starch as an indicator. The disappearance of blue color marks the end point. Let the volume of hypo solution used be V mL.

Calculation

$$\begin{array}{ccc} O_2 \text{ in water} & & \text{Hypo} \\ N_1V_1 & = & N_2V_2 \end{array}$$

$$N_1 \times 100 = \frac{1}{100} \times V$$

$$N_1 = \frac{1}{100 \times 100} \times V$$

Strength of O_2 in water = Normality of O_2 × equivalent weight of O_2

$$= \frac{1}{10,000} \times 8 \text{ g/L}$$

$$= \frac{1}{10,000} \times 8 \times 1000 \text{ mg/L} = 0.8 \times V \text{ ppm}$$

BIOLOGICAL OXYGEN DEMAND (BOD)

BOD measures the extent of pollution of waste water, sewage and industrial effluents. The waste water or polluted water generally contains organic matter. This organic matter may either be biodegradable or non biodegradable. The microorganisms feed on the biodegradable matter in the presence of oxygen (aerobically) to form stable, non objectionable end products.

$$\text{Biodegradable organic matter} + O_2 \xrightarrow{\text{Microorganisms}} CO_2 + H_2O + \text{Other stable products}$$

However, if enough oxygen is not available, anaerobic oxidation (in absence of oxygen) takes place and harmful end products like H_2S, NH_3 and CH_4 are formed.

$$\text{Organic matter} \xrightarrow[\text{(Absence of oxygen)}]{\text{Microorganisms}} CO_2 + CH_4 + \text{Unstable products}$$

The amount of oxygen required by microorganisms to oxidise organic matter aerobically is called Biological Oxygen Demand (BOD). If the BOD of the sample is high, the dissolved oxygen is low and this indicates in greater pollution of the water sample.

Determination The BOD is based on the determination of dissolved oxygen over a period of 5 days. This is known as 5 day BOD test. A known volume of sample of sewage is diluted with a known volume of distilled water (water containing nutrients for bacterial growth). The dissolved oxygen is determined on the first day (D_1). This is kept for 5 days at 20 °C in a stoppered bottle, away from light (incubation period). The dissolved oxygen is measured again after 5 days. The difference (D_1–D_2) indicates the oxygen consumed in five days for oxidising the organic matter present in the water sample.

$$\text{BOD} = (D_1 - D_2) \times \text{dilution factor}$$

$$= (D_1 - D_2) \times \frac{\text{Volume of sample after dilution}}{\text{Volume of sample before dilution}}$$

D_1 = Dissolved oxygen of diluted water sample immediately after preparation

D_2 = Dissolved oxygen of diluted water sample after incubation of 5 days at 20 °C

Chemical Oxygen Demand (COD)

BOD is a measure of biodegradable organic impurities in water. It does not measure the non biodegradable matter. COD is a measure of both biodegradable and non biodegradable organic matter. Therefore, COD reveals the actual organic content present in water. *It is the amount of oxygen required by organic matter in a sample of water for its oxidation by a strong chemical oxidising agents such as* $K_2Cr_2O_7$.

Determination Organic matter is almost completely oxidised by a boiling mixture of chromic acid. A measured quantity (say 50 mL) of the sample is refluxed with a known excess of standard potassium dichromate solution and dilute H_2SO_4 in the presence of $Ag_2SO_4 - HgSO_4$ as a catalyst. The amount of $K_2Cr_2O_7$ before the reaction and excess $K_2Cr_2O_7$ left after the reaction is determined by titrating against a standard ferrous ammonium sulphate solution (Mohr's salt) using ferroin as an indicator.

The difference between the $K_2Cr_2O_7$ originally present and that remaining unreacted gives the amount of $K_2Cr_2O_7$ used for the oxidation of oxidisable impurities in water.

Volume of sample taken = 50 mL

Volume of 0.25 N FAS used in sample titration = V_1 mL

Volume of 0.25 N FAS used in blank titration = V_2 mL

Therefore volume of FAS equivalent to $K_2Cr_2O_7$ used for COD determination = $V_2 - V_1$ mL

Now,

$$\underset{\text{Sample}}{N_1V_1} = \underset{\text{FAS}}{N_2V_2}$$

$$N_1 \times 50 = 0.25 \times (V_2 - V_1)$$

$$N_1 = \frac{0.25(V_2 - V_1)}{50}$$

COD = Normality of water sample × equivalent weight of oxygen

$$= \frac{0.25(V_2 - V_1)}{50} \times 8 \text{ g/L}$$

$$= \frac{0.25(V_2 - V_1)}{50} \times 8 \times 1000 \text{ mg/L}$$

$$= (V_2 - V_1) \times 40 \text{ ppm}$$

Solved Examples

1. 30 cm^3 of industrial effluent consumes 8 cm^3 of 0.05 N potassium dichromate solution. Calculate the COD of the effluent.

Solution

Volume of effluent = 30 cm^3

Volume of 0.05 N potassium dichromate consumed = 8 cm^3

COD = ?

Sample $\quad K_2Cr_2O_7$

$$N_1V_1 = N_2V_2$$

$$N_1 = \frac{0.05 \times 8}{30}$$

COD = Normality of waste water sample × equivalent weight of oxygen × 1000

$$COD = \frac{0.05 \times 8}{30} \times 8 \times 1000 = 106.66 \text{ mg/L}$$

Ans COD of effluent sample = 106.66 mg/L

2. 50 cm^3 of acidified potassium dichromate was mixed with 50 cm^3 of a waste water sample and refluxed. 20.4 cm^3 of ferrous ammonium sulphate solution was required by unreacted acidified potassium dichromate solution. Blank titration was performed to estimate the amount of potassium dichromate at the beginning of the reaction. 50 cm^3 of acidified potassium dichromate requires 40.8 cm^3 of the same 0.2 N ferrous ammonium sulphate solution. Calculate the COD of waste water.

Solution

Volume of waste water = 50 cm^3

Volume of ferrous ammonium sulphate used for blank titration V_1 = 40.8 cm^3

Volume of ferrous ammonium sulphate used for sample titration V_2 = 20.4 cm^3

∴ The volume of ferrous ammonium sulphate equivalent to $K_2Cr_2O_7$ used for oxidation of organic matter $= V_1 - V_2$

$= 40.8 - 20.4 = 20.4$ cm^3

Normality of FAS = 0.2 N

Calculation of COD

$$N_1V_1 = N_2V_2$$

N_1 = Normality of O_2 in waste water; V_1 = volume of waste water

N_2 = Normality of FAS; V_2 = Volume of FAS used

$$N_1 \times 50 = 0.2 \times 20.4$$

$$N_1 = \frac{0.2 \times 20.4}{50}$$

$$COD = \frac{0.2 \times 20.4}{50} \times 8 \text{ g/L}$$

$$\text{COD} = \frac{0.2 \times 20.4}{50} \times 8 \times 1000 \text{ mg/L} = 652.8 \text{ mg/L}$$

Answer: COD = 652.8 mg/L

3. 30 cm^3 of waste water is diluted to 600 cm^3 and equal volumes are filled in two BOD bottles. A blank titration was performed using 100 cm^3 of diluted waste water. It required 10 cm^3 of 0.05 N sodium thiosulphate solution for neutralisation. 100 cm^3 of the incubated sample was titrated again after 5 days. It required 5.0 cm^3 of the same sodium thiosulphate solution. Calculate the BOD of waste water.

Solution

Initial volume of waste water = 30 cm^3

Volume of diluted waste water = 600 cm^3

Initially

Sample $Na_2S_2O_3$

$$N_1V_1 = N_2V_2$$

$$N_1 \times 100 = 0.05 \times 10$$

$$N_1 = \frac{0.05 \times 10}{100}$$

$$\text{Dissolved oxygen } (D_1) = \frac{0.05 \times 10}{100} \times \text{equivalent weight of } O_2 \times 1000$$

$$= \frac{0.05 \times 10}{100} \times 8 \times 1000 = 40 \text{ mg/L}$$

After 5 days

Sample $Na_2S_2O_3$

$$N_3V_3 = N_4V_4$$

$$N_3 \times 100 = 0.05 \times 5$$

$$N_3 = \frac{0.05 \times 5}{100}$$

$$\text{Dissolved oxygen after 5 days } (D_2) = \frac{0.05 \times 5}{100} \times \text{equivalent weight of } O_2 \times 1000$$

$$= \frac{0.05 \times 5}{100} \times 8 \times 1000 = 20 \text{ mg/L}$$

$$\text{BOD} = (D_1 - D_2) \times \frac{\text{Volume of sample after dilution}}{\text{Volume of sample before dilution}}$$

$$= (40\text{–}20) \times \frac{600}{30} = 400 \text{ mg/L}$$

Practice problems

1. 20 cm^3 of an industrial effluent consumed 4 cm^3 of 0.5 N $K_2Cr_2O_7$ solution for oxidation. Calculate the COD of the effluent. [Ans 800 mg/L]
2. 25 cm^3 of sewage water for COD is reacted with 20 cm^3 of potassium dichromate solution. The unreacted potassium dichromate requires 7.0 cm^3 of 0.5 N ferrous ammonium sulphate solution under similar conditions. In blank titration, 12 cm^3 of ferrous ammonium sulphate is used up. Calculate the COD of the sample. [Ans 800 mg/L]
3. 25 cm^3 of waste water sample is diluted to 500 cm^3 and equal volumes are filled into two BOD bottles. In a blank titration 100 cm^3 of waste water requires 8.8 cm^3 of 0.02 N sodium thiosulphate solution. 100 cm^3 of incubated sample after 5 days requires 4.2 cm^3 of the same sodium thiosulphate solution. Calculate the BOD of the waste water. [Ans 147.2 mg/L]

Summary

- Water is the most important natural resource. It comprises of nearly 70% of the earth's surface. Out of this 70% water, 97% is in seas. Only 1% water is available as fresh water in lakes, rivers, streams etc
- Water is a very good solvent. It weathers rocks and minerals by dissolution, hydration, hydrolysis, carbonation and oxidation.
- Various organic, inorganic, gaseous and biological impurities are present in water.
- Hard water does not give lather with soap. Hardness is of two types
  - (i) Temporary hardness – It can be removed by boiling and is due to the presence of bicarbonates of calcium and magnesium.
  - (ii) Permanent hardness – It is due to the presence of chlorides, sulphates and nitrates of calcium and magnesium. It cannot be removed by boiling.
- Degree of hardness is always expressed in terms of $CaCO_3$ as the molecular weight of $CaCO_3$ is 100 and it is highly insoluble.

$$\text{Degree of hardness} = \frac{\text{Weight of hardness producing substance} \times \text{Equivalent weight of } CaCO_3}{\text{Equivalent weight of hardness producing substance}}$$

- Hardness can be expressed in various units like ppm, mg/L, degree Clarke and degree French.

  1ppm = 1 mg/L = 0.07° Cl = 0.1° Fr

- Hard water generally causes four type of problems in boilers
  - (i) Scale and sludge formation
  - (ii) Boiler corrosion
  - (iii) Caustic embrittlement
  - (iv) Priming, foaming and carryover
- Scales and sludges are formed due to deposition of salts in boilers. Their formation can be prevented by various internal conditioning methods like carbonate conditioning, phosphate conditioning, aluminate conditioning and calgon conditioning or by physical methods like radioactive, electrical, colloidal conditioning.
- Hard water can be softened by lime soda process, zeolite process and ion exchange process.
- Water supplied for domestic use should be colorless, odorless, tasty, dissolved salt contents should be less, F^-, SO_4^{2-}, etc. should be as per prescribed limits
- The various steps of domestic (municipal) water supply include screening, sedimentation, filtration and disinfection of water.
- Sea water can be desalinated by various methods like reverse osmosis, electrodialysis, ultrafiltration and flash evaporation.
- Water can be defluorinated using the Nalgonda technique which involves addition of lime and alum followed by flocculation, sedimentation, filtration and disinfection.
- Waste water/sewage treatment includes collection, conveyance, treatment and disposal of treated water.
- After collection and conveyance of waste water it is treated in four steps:
  - Preliminary treatment or pretreatment: Its purpose is to remove the floating impurities from water and prevent further decay of sewage. Its steps are screening, presedimentation, microstraining, preaeration and prechlorination.
  - Primary treatment or physico-chemical treatment: It involves the removal of colloidal particles and other settleable impurities from sewage.
  - Secondary treatment or biological treatment: It is the removal of organic impurities in water with the help of bacterial mass. It is carried out either by trickling filter process or by activated sludge process.
  - Tertiary or advanced treatment: It is carried out for a very small fraction of water. It involves the removal of minerals like nitrogen and phosphorus and water is disinfected here.

- The various methods used for the determination of hardness are – O. Hehner's method, EDTA method and soap titration method.
- Parameters used to check the quality of water are dissolved oxygen, biological oxygen demand and chemical oxygen demand.

Review Questions

1. Define hardness of water. Differentiate between temporary and permanent hardness of water.
2. Explain the effect of water on rocks and minerals.
3. Define the various units of hardness. Write down the relationship between them.
4. What are scales and sludges, how are they formed in boilers? Also discuss the disadvantages of scales in boilers.
5. Write short notes on
   (i) Boiler corrosion
   (ii) Caustic embrittlement
   (iii) Priming, foaming and carry over
   (iv) Impurities in water
   (v) Various conditioning methods for the prevention of scale in boilers.
6. Explain the lime soda process used for softening of hard water. Write down the chemical reactions involved in the process.
7. Explain the cold continuous and hot continuous lime soda process, which is better and why?
8. What is zeolite? How can it be used to soften water? What are the disadvantages of the zeolite process?
9. Explain the ion exchange process used to soften water. Why is it considered as the best method to soften the water?
10. What are the essential requirements of domestic water? Explain the various steps involved in the supply of water in a municipality or for domestic use.
11. Explain the term reverse osmosis. How can it be used to obtain fresh water from sea water?
12. Explain the following processes used to obtain soft water from sea water:
    (i) Electrodialysis
    (ii) Ultrafiltration
    (iii) Flash evaporation
13. What is EDTA? Explain the determination of hardness using EDTA method.
14. How can you find out hardness of a given water sample using the soap titration method?

15. Explain the estimation of the following in water:
    (i) Dissolved oxygen
    (ii) Dissolved chlorides
    (iii) Free chloride
16. What are BOD and COD? How can you estimate BOD and COD of a given water sample?
17. Explain the term reverse osmosis. How can it be used to obtain fresh water from sea water?
18. What are the effects of fluorine in water? Explain the Nalgonda technique for the defluoridation of water.
19. Write short notes on the following:
    (i) Preliminary treatment or pretreatment of water
    (ii) Primary treatment of water
    (iii) Trickling filter process
    (iv) Activated sludge process

Multiple Choice Questions

1. Amount of water available for human consumption is
   (a) 1% (b) 3%
   (c) 97% (d) 5%
2. Water causes weathering of rocks due to the following phenomenon
   (a) Dissolution (b) Hydration
   (c) Hydrolysis (d) All of the above
3. Hardness of water is due to
   (a) K and Na salts (b) Salts of iron
   (c) Salts of Ca and Mg (d) SiO_2
4. Permanent hardness is due to
   (a) Carbonate of Ca (b) Bicarbonate of Mg
   (c) Chloride and sulphates of Ca and Mg (d) None of the above
5. The most commonly used unit to express hardness is
   (a) Degree French (b) Degree Clarke
   (c) ppm (d) grains /gallon
6. Scale in boilers are formed due to
   (a) Deposition of $CaCO_3$ (b) Deposition of $CaSO_4$
   (c) Hydrolysis of magnesium salts (d) All of the above
7. Excess nitrate in drinking water causes
   (a) Anaemia (b) Blue baby syndrome
   (c) Irritation in skin (d) Mouth blisters

8. The maximum permissible limit of fluoride in drinking water is
   (a) 1.5 ppm (b) 5 ppm
   (c) 3 ppm (d) 8 ppm
9. Lime soda process uses
   (a) $Ca(OH)_2$ (b) Na_2CO_3
   (c) Both the above (d) None of the above
10. Which process is not used for the desalination of water
    (a) Reverse osmosis (b) Lime soda process
    (c) Electrodialysis (d) Flash evaporation
11. Residual hardness in ion exchange process is
    (a) 10-15 ppm (b) 30-60 ppm
    (c) 0-2 ppm (d) 15-30 ppm
12. Alkalinity in water is due to
    (a) OH^- (b) CO_3^{2-}
    (c) HCO_3^- (d) All of the above
13. Which is not used for disinfection of water
    (a) Chlorination (b) Electrodialysis
    (c) Ozonisation (d) Addition of $KMnO_4$
14. Dissolved oxygen in water is determined by
    (a) Mohr's method
    (b) Gravimetric method
    (c) Winklers method or iodometric titration
    (d) EDTA method
15. Flash evaporation is a method of getting pure water from
    (a) Sea water (b) Industrial waste water
    (c) Domestic sewage (d) River water
16. Disinfection of water removes
    (a) Salts from water (b) Pathogenic bacteria from water
    (c) Dissolved oxygen from water (d) Hardness from water
17. Acceptable pH range for drinking water is
    (a) 7 to 8.5 (b) 6 to 7
    (c) 6.5 to 9.2 (d) 8 to 10
18. Which of the following is removed from water using Nalgonda technique
    (a) Chlorine (b) Fluorine
    (c) Sulphates (d) Hardness in water

19. The organic impurities from sewage are removed by
    (a) Preliminary treatment
    (b) Primary treatment
    (c) Secondary treatment
    (d) Tertiary treatment

Solution

1 (a)	2 (d)	3 (c)	4 (c)	5 (c)	6 (d)	7 (b)	8 (a)
9 (c)	10 (b)	11 (c)	12 (d)	13 (b)	14 (c)	15 (a)	16 (b)
17 (a)	18 (b)	19 (c)					